



General Instructions

- (i) This booklet contains 200 questions, each provided with a complete, step-by-step solution.
- (ii) It comprises 200 single-correct multiple-choice questions.
- (iii) These questions follow the NEET PG examination pattern.
- (iv) Each correct answer is awarded +4 marks, with -1 for an incorrect answer.
- (v) The actual NEET paper has 180 questions to be attempted in 3 hours 20 minutes.
- (vi) Use of calculators is not permitted.
- (vii) Attempt each question on your own before reviewing the given solution.

1. Anterior interosseous nerve is a branch of?

- (A) Radial nerve
- (B) Median nerve
- (C) Ulnar nerve
- (D) Axillary nerve

Correct Answer: (B) Median nerve

Solution:

Step 1: The anterior interosseous nerve (AIN) is a motor branch that arises from the median nerve in the proximal forearm, just below the elbow.

Step 2: It runs on the anterior surface of the interosseous membrane with the anterior interosseous artery, supplying the deep flexor muscles of the forearm - flexor pollicis longus, the lateral half of flexor digitorum profundus, and pronator quadratus.

Step 3: Note the distinction often confused in exams: the anterior interosseous

artery is a branch of the common interosseous artery (from the ulnar artery), but the anterior interosseous NERVE is purely a branch of the median nerve.

Step 4: The radial, ulnar, and axillary nerves do not give rise to the anterior interosseous nerve, so the correct answer is the median nerve.

Quick Tip: Think of the deep forearm flexor motor branch and its parent trunk that passes through pronator teres.



2. Posterior wall of axilla is formed by?

- (A) Pectoralis major
- (B) Pectoralis minor
- (C) Subscapularis
- (D) Intercostal muscles

Correct Answer: (C) Subscapularis

Solution:

Step 1: The axilla is a pyramidal space between the upper arm and chest wall, with an apex, a base, and four walls (anterior, posterior, medial, and lateral).

Step 2: The posterior (subscapular) wall is formed by three muscles: subscapularis above, with teres major and latissimus dorsi below. So among the options, subscapularis is the correct posterior wall component.

Step 3: The other options belong to different walls. Pectoralis major and pectoralis minor form the anterior (pectoral) wall along with subclavius, and the intercostal muscles (with upper four ribs and serratus anterior) form the medial (thoracic) wall.

Step 4: Therefore, the posterior wall of the axilla is formed by subscapularis.

Quick Tip: The subscapular wall is named after the muscle that caps it superiorly.

3. True about popliteus are all except?

- (A) Flexor of knee
- (B) Intracapsular origin
- (C) Supplied by tibial nerve
- (D) Causes locking of knee

Correct Answer: (D) Causes locking of knee

Solution:

Step 1: The popliteus is a deep muscle of the posterior compartment of the leg. It arises from the lateral surface of the lateral condyle of the femur (origin is intracapsular) and from the outer margin of the lateral meniscus, and inserts on the posterior surface of the tibia above the soleal line.

Step 2: Its key action is to UNLOCK the fully extended knee. It does this by laterally rotating the femur on the fixed tibia, which initiates flexion. It is also an accessory flexor of the knee.

Step 3: So options (a) flexor of knee, (b) intracapsular origin, and (c) supplied by the tibial nerve are all correct statements about popliteus.

Step 4: The locking of the knee is produced by the quadriceps and medial rotation of the femur during terminal extension. Popliteus does the opposite - it unlocks the knee. Hence the false statement (the exception) is that it causes locking of the knee.

Quick Tip: Popliteus is the muscle that reverses the screw-home mechanism.

4. Lower limit of the sacroiliac joint lies up to which level in females?

- (A) 1 to 1 1/2
- (B) 2 to 2 1/2
- (C) 3 to 3 1/2
- (D) 4 to 4 1/2

Correct Answer: (B) 2 to 2 1/2

Solution:

Step 1: The sacroiliac joint is the synovial joint between the auricular surfaces of the sacrum and the ilium. Its vertical extent is described in relation to the sacral segments.

Step 2: In females, the articular (auricular) surface and hence the lower limit of the sacroiliac joint extends down to about the level of the second to two and a half sacral segments.

Step 3: This is shorter than in males, where the joint surface tends to extend further. The smaller, more obstetrically favourable female sacrum accounts for this difference.

Step 4: Therefore the lower limit in females lies up to the 2 to 2 1/2 level, matching option (b).

Quick Tip: The female auricular surface is short - think roughly two to two and a half segments.

5. Which structure lies posterior to the transverse pericardial sinus?

- (A) Aorta
- (B) Pulmonary trunk
- (C) SVC
- (D) Left atrium

Correct Answer: (C) SVC

Solution:

Step 1: The transverse pericardial sinus is a short passage within the pericardial cavity that lies between the reflections of serous pericardium around the arterial ends (aorta and pulmonary trunk) and the venous ends of the developing heart tube.

Step 2: Its boundaries are: anteriorly the ascending aorta and pulmonary trunk, posteriorly the superior vena cava (SVC), and inferiorly the left atrium.

Step 3: So the structure lying posterior to the transverse sinus is the SVC, matching option (c).

Step 4: The aorta and pulmonary trunk lie anterior to it, and the left atrium lies inferior to it, so those options describe other boundaries and are not correct.

Quick Tip: Anterior is the great arteries, inferior is the left atrium - what is left for the back?

6. Superior rectal vein drains into?

- (A) Inferior mesenteric vein
- (B) External iliac vein
- (C) Internal iliac vein
- (D) Internal pudendal vein

Correct Answer: (A) Inferior mesenteric vein

Solution:

Step 1: The rectum has a portosystemic anastomosis. The upper part drains via the superior rectal vein, while the middle and lower parts drain via the middle and inferior rectal veins.

Step 2: The superior rectal vein drains into the inferior mesenteric vein, which belongs to the PORTAL venous system. This makes it the site of portosystemic anastomosis at the anorectal junction.

Step 3: The inferior rectal vein drains into the internal pudendal vein, and the middle rectal vein drains into the internal iliac vein - both of these belong to the SYSTEMIC (caval) system.

Step 4: Therefore the superior rectal vein drains into the inferior mesenteric vein, option (a). The external iliac vein has no role here.

Quick Tip: The superior rectal vein is the portal limb of the anorectal anastomosis.

7. Ligament extending from cervix and vagina to the lateral pelvic wall is?

- (A) Broad ligament
- (B) Pubocervical ligament
- (C) Round ligament
- (D) Transverse cervical ligament

Correct Answer: (D) Transverse cervical ligament

Solution:

Step 1: The transverse cervical ligaments (cardinal ligaments or ligaments of Mackenrodt) are fan-shaped condensations of endopelvic fascia on each side of the cervix, lying above the levator ani and surrounding the uterine vessels.

Step 2: They connect the lateral aspect of the cervix and the upper vaginal wall to the lateral pelvic wall, forming a hammock that is the chief supporting structure for the uterus.

Step 3: This exactly matches the description in the question, so the answer is the transverse cervical ligament, option (d).

Step 4: The distractors differ: the broad ligament is a peritoneal fold, the pubocervical ligament runs from the cervix forward to the pubis, and the round ligament passes from the uterus to the labium majus through the inguinal canal. None of these run laterally from cervix and vagina to the pelvic wall.

Quick Tip: Look for the cardinal ligament of Mackenrodt that anchors the cervix laterally.



8. Killian's dehiscence is seen in?

- (A) Oropharynx
- (B) Nasopharynx
- (C) Cricopharynx
- (D) Vocal cords

Correct Answer: (C) Cricopharynx

Solution:

Step 1: The inferior constrictor muscle of the pharynx has two parts. The upper thyropharyngeus has oblique fibres, and the lower cricopharyngeus has transverse fibres.

Step 2: Between these two parts there is a potential weak area in the muscular wall called Killian's dehiscence. This lies at the cricopharyngeal level, so the correct answer is cricopharynx.

Step 3: Clinically this is the gateway to tears, because perforation during esophagoscopy tends to occur here. It is also the site where the pharyngeal mucosa herniates to form a pharyngeal pouch (Zenker's diverticulum).

Step 4: The oropharynx, nasopharynx and vocal cords are not the site of this dehiscence, so they are incorrect.

Quick Tip: Think of the weak gap between thyropharyngeus and cricopharyngeus.

9. Which valve is present at the opening of the nasolacrimal duct?

- (A) Hasner's valve
- (B) Heister valve
- (C) Spiral valve
- (D) None

Correct Answer: (A) Hasner's valve

Solution:

Step 1: The nasolacrimal duct carries tears from the lacrimal sac down into the nasal cavity. It opens into the inferior meatus of the nose, beneath the inferior turbinate.

Step 2: At this lower opening there is a mucosal fold or flap that acts as a valve, called the valve of Hasner. It prevents air and nasal secretions from being forced back up into the lacrimal system. Hence Hasner's valve is the correct answer.

Step 3: The Heister valve, also called the spiral valve, is found in the cystic duct of the biliary tree, not in the lacrimal apparatus, so those options are incorrect.

Quick Tip: It is the mucosal flap where the duct enters the inferior meatus.

10. Scutum is present in which part of the middle ear?

- (A) Roof
- (B) Lateral wall
- (C) Medial wall
- (D) Floor

Correct Answer: (B) Lateral wall

Solution:

Step 1: The scutum is a thin, shield-shaped plate of bone in the middle ear. It forms the upper part of the lateral (outer) wall of the tympanic cavity.

Step 2: The lateral wall of the middle ear is largely formed by the tympanic membrane, with the scutum bounding the attic (epitympanum) above the membrane. Hence the answer is lateral wall.

Step 3: The scutum is an important landmark, because its erosion on imaging is an early sign of an acquired cholesteatoma in the attic region.

Step 4: The roof (tegmen tympani), the medial wall (with the promontory and oval window) and the floor (jugular wall) do not contain the scutum, so they are incorrect.

Quick Tip: It is the shield-shaped bony spur above the tympanic membrane.

11. Optic cup is derived from?

- (A) Neural ectoderm
- (B) Surface ectoderm
- (C) Mesoderm
- (D) Neural crest

Correct Answer: (A) Neural ectoderm

Solution:

Step 1: The eye develops from an outpouching of the forebrain (diencephalon). This outpouching is the optic vesicle, which is made of neuroectoderm.

Step 2: The optic vesicle invaginates to form a double-layered cup, the optic cup. Because it is a direct continuation of the optic vesicle, the optic cup is derived from neural ectoderm. Hence the answer is neural ectoderm.

Step 3: The two layers of the optic cup go on to form the retina, the outer layer becoming the retinal pigment epithelium and the inner layer the neural retina.

Step 4: The lens forms from surface ectoderm, the sclera and choroid largely from mesoderm and neural crest, so those options do not match the optic cup.

Quick Tip: It is an invagination of the optic vesicle from the forebrain.

12. The 2nd part of the duodenum is derived from?

- (A) Foregut
- (B) Midgut
- (C) Both foregut and midgut
- (D) Hindgut

Correct Answer: (C) Both foregut and midgut

Solution:

Step 1: The duodenum forms from the caudal part of the foregut and the cranial part of the midgut. The junction between these two gut segments lies at the opening of the bile duct, which is in the second part of the duodenum.

Step 2: Because the dividing line falls within the second part, this segment

receives a contribution from both the foregut and the midgut. Hence the answer is both foregut and midgut.

Step 3: This dual origin is reflected in the blood supply. The part above the major duodenal papilla is supplied by the coeliac trunk (foregut), while the part below is supplied by the superior mesenteric artery (midgut).

Step 4: So pure foregut, pure midgut, or hindgut are all incorrect for the second part of the duodenum.

Quick Tip: The foregut-midgut junction lies at the major duodenal papilla.

13. Innervated structures of joints are all except?

- (A) Synovium
- (B) Capsule
- (C) Articular cartilage
- (D) Ligaments

Correct Answer: (C) Articular cartilage

Solution:

Step 1: Most components of a synovial joint carry a nerve supply. The fibrous capsule and the ligaments are richly supplied with pain and proprioceptive fibres, and the synovial membrane also receives a nerve supply.

Step 2: Articular cartilage, however, is hyaline cartilage and has no nerve supply, no blood supply and no lymphatic drainage. Hence it is the structure that is not innervated, so it is the exception and the correct answer.

Step 3: Because cartilage is avascular and lacks a perichondrium at the joint surface, it cannot repair or regenerate itself well, which is why cartilage injuries heal poorly.

Step 4: Synovium, capsule and ligaments are all innervated, so they cannot be the exception.

Quick Tip: Which joint tissue is avascular and aneural hyaline cartilage?

14. Mismatch of ventilation/perfusion ratio is seen in?

- (A) Apex
- (B) Base
- (C) Both
- (D) None

Correct Answer: (C) Both

Solution:

Step 1: The overall ventilation-perfusion (V/Q) ratio of the lung is about 0.8, since alveolar ventilation is roughly 4.2 L/min and cardiac output is about 5.0 L/min. Ideally every alveolus would have a V/Q of 0.8, but this is not the case even in normal lungs.

Step 2: Because of gravity, the apex is both underventilated and underperfused, while the base is both overventilated and overperfused. Gravity affects perfusion more than ventilation, so the apex is relatively underperfused and the base relatively overperfused.

Step 3: As a result the V/Q ratio is high at the apex (about 3.0) and low at the base (about 0.6). Both ends therefore deviate from the ideal 0.8, so the mismatch is seen at both apex and base. Hence the answer is both.

Step 4: At the apex ventilation exceeds perfusion, giving high PaO₂ and low PaCO₂; at the base perfusion exceeds ventilation, giving low PaO₂ and high

PaCO₂. Since neither region alone explains the mismatch, options apex only and base only are incorrect.

Quick Tip: V/Q is about 3.0 at the apex and 0.6 at the base, both off the ideal 0.8.

15. Capacitance vessels have in their wall:

- (A) More elastic tissue and less muscle
- (B) Less elastic tissue and more muscle
- (C) More elastic tissue and more muscle
- (D) Less elastic tissue and less muscle

Correct Answer: (D) Less elastic tissue and less muscle

Solution:

Step 1: Capacitance vessels are the veins. They are called capacitance vessels because at any given moment they hold roughly 60 to 70 percent of the total blood volume and act as a reservoir.

Step 2: A vein wall has the same three tunicae as an artery (intima, media, adventitia), but the wall is much thinner. The tunica media in particular has only a small amount of smooth muscle, and the elastic tissue is scanty and is not organized into distinct internal and external elastic laminae.

Step 3: So compared with arteries, the venous wall has both less elastic tissue and less muscle, which makes the veins easily distensible and able to store large volumes at low pressure. This is exactly why they serve as capacitance vessels.

Step 4: The distractors describe arterial or arteriolar walls. Conducting arteries are elastin rich (more elastic tissue), while muscular arteries and arterioles are muscle rich (more muscle). Neither pattern fits a vein, so options a, b and c are wrong and the answer is option d.

Quick Tip: Capacitance vessels are veins, which act as the body's low-pressure blood reservoir.

16. Umami taste is evoked by:

- (A) Glucose
- (B) Glutamic acid
- (C) Quinine
- (D) Sodium chloride

Correct Answer: (B) Glutamic acid

Solution:

Step 1: There are five basic tastes: sweet, sour, salty, bitter and umami. Umami is the savoury or meaty taste and is the fifth, most recently characterized taste.

Step 2: Umami is triggered by glutamate. The taste sensors are glutamate receptors, and the prototypic stimulus is glutamic acid (and its salt monosodium glutamate, MSG). The naturally occurring umami substances are MSG, disodium guanosine monophosphate (GMP) and disodium inosine monophosphate (IMP).

Step 3: Matching the other options to their tastes shows why they are wrong. Glucose gives a sweet taste, quinine gives a bitter taste, and sodium chloride gives a salty taste. None of these produce umami.

Step 4: Therefore the substance that evokes umami is glutamic acid, option b.

Quick Tip: Umami is the savoury taste linked to monosodium glutamate (MSG).

17. Which of the following increases appetite?

- (A) CART
- (B) α -MSH
- (C) AGRP
- (D) Insulin

Correct Answer: (C) AGRP

Solution:

Step 1: Appetite is regulated by the arcuate nucleus of the hypothalamus, which contains two opposing groups of neurons. Orexigenic neurons increase appetite, while anorexigenic neurons decrease it.

Step 2: AGRP (agouti-related peptide), co-released with neuropeptide Y, is the orexigenic signal. It increases food intake, both by directly stimulating feeding pathways and by blocking the melanocortin (MC4) receptor that normally suppresses appetite. So AGRP increases appetite.

Step 3: The other options are all anorexigenic. CART (cocaine and amphetamine regulated transcript) and α -MSH are released by the POMC neurons and reduce appetite. Insulin, like leptin, signals energy sufficiency to the hypothalamus and also reduces appetite.

Step 4: Hence the only appetite-increasing agent in the list is AGRP, option c. (The option is printed as AGPP, an OCR error for AGRP)

Quick Tip: Look for the orexigenic peptide that blocks the MC4 melanocortin receptor.

18. S1 and S2 nerve roots are checked by which reflex?

- (A) Knee jerk
- (B) Patellar reflex
- (C) Calcaneal reflex
- (D) None

Correct Answer: (C) Calcaneal reflex

Solution:

Step 1: Deep tendon reflexes are used clinically to test specific spinal nerve root levels. Each reflex maps to a defined root.

Step 2: The ankle jerk, also called the calcaneal reflex or Achilles reflex, is elicited by tapping the Achilles tendon and tests the S1 and S2 nerve roots through the tibial nerve and gastrocnemius-soleus contraction.

Step 3: The other options test a different level. The knee jerk and the patellar reflex are the same reflex (the patellar tendon jerk) and they test the L2, L3 and L4 roots, mainly L3-L4, not S1-S2.

Step 4: Therefore S1-S2 is checked by the calcaneal (ankle) reflex, option c.

Quick Tip: S1-S2 maps to the ankle jerk, also known as the Achilles reflex.

19. Implantation occurs on which menstrual cycle day?

- (A) 5-7 days
- (B) 20-22 days
- (C) 14-18 days
- (D) 26-28 days

Correct Answer: (B) 20-22 days

Solution:

Step 1: The question gives the day of the menstrual cycle, not the day after fertilization. To answer it you must add the events of the cycle together.

Step 2: In a standard 28 day cycle, ovulation occurs on day 14, and fertilization takes place within about 24 hours of ovulation, so around day 15.

Step 3: Implantation of the blastocyst happens about 6 to 7 days after fertilization. Adding 6 to 7 days to day 14-15 brings you to roughly day 20 to 22 of the menstrual cycle.

Step 4: Therefore implantation corresponds to menstrual cycle days 20-22, option b. Option a (5-7 days) is the post-fertilization interval, not the cycle day, and the other options do not match the timing.

Quick Tip: Ovulation is day 14; add 6-7 days of pre-implantation development.



20. FSH and LH both are inhibited by:

- (A) Cortisol
- (B) Aldosterone
- (C) Estrogen
- (D) Progesterone

Correct Answer: (C) Estrogen

Solution:

Step 1: The pituitary gonadotropins FSH and LH are controlled by negative feedback from the ovarian hormones. The key is to find the hormone that suppresses both gonadotropins.

Step 2: Estrogen is secreted in response to both LH and FSH stimulation, and through negative feedback it in turn inhibits both LH and FSH. It acts at both the hypothalamus and the pituitary, making it the one hormone that suppresses both

gonadotropins.

Step 3: Comparing the more selective feedback signals: progesterone mainly inhibits LH, while inhibin (from granulosa cells) selectively inhibits FSH and acts only at the pituitary. So progesterone does not suppress both, which is why option d is not the best answer.

Step 4: Cortisol and aldosterone are adrenal corticosteroids and are not the physiological feedback regulators of FSH and LH, so options a and b are wrong. The hormone that inhibits both gonadotropins is estrogen, option c. (Note: at the mid-cycle peak, sustained high estrogen switches to positive feedback to trigger the LH surge.)

Quick Tip: Find the ovarian hormone that feeds back on both gonadotropins, unlike inhibin (FSH only) or progesterone (LH only).

21. Maximum amount of K^+ ion is seen in which GI secretion?

- (A) Saliva
- (B) Colonic
- (C) Gastric
- (D) Jejunal

Correct Answer: (A) Saliva

Solution:

Step 1: Each gastrointestinal secretion has a characteristic electrolyte profile. The question asks which one has the highest concentration of potassium (K^+).

Step 2: Saliva is the GI secretion richest in potassium. Salivary potassium concentration is markedly higher than that of plasma, because the salivary ducts reabsorb sodium and secrete potassium as the saliva passes through them.

Step 3: The other secretions have lower potassium content. Gastric, jejunal and colonic fluids contain potassium, but their concentrations are lower than that of saliva (colonic fluid is also potassium rich, yet saliva has the highest concentration among the options offered).

Step 4: Hence the maximum potassium content among these GI secretions is found in saliva, option a.

Quick Tip: Salivary ducts secrete K^+ and reabsorb Na^+ , giving saliva a high potassium content.



22. Maximum water reabsorption in the gastrointestinal tract occurs in?

- (A) Stomach
- (B) Jejunum
- (C) Ileum
- (D) Colon

Correct Answer: (B) Jejunum

Solution:

Step 1: About 9 litres of water enters the gastrointestinal tract each day. This comes from ingested water (about 2.0 L), saliva (about 1.5 L), gastric juice (about 2.5 L), bile (about 0.5 L), pancreatic juice (about 1.5 L) and small intestinal secretions (about 1.0 L).

Step 2: Of this volume, roughly 7.7 L (about 85 percent) is reabsorbed in the small intestine and only about 1.0 to 1.5 L (5 to 10 percent) in the large intestine (colon). Around 0.2 L is finally lost in feces.

Step 3: Within the small intestine, the bulk of this water is reclaimed in the jejunum, which has a very large absorptive surface and a leaky epithelium that allows rapid water movement following solute absorption. Hence the jejunum is

the single site of maximum water reabsorption.

Step 4: The colon (option d) absorbs only the residual fraction, the stomach (option a) absorbs negligible water, and the ileum (option c) absorbs less than the jejunum, so the correct answer is the jejunum.

Quick Tip: Think of the proximal small bowel with its leaky, high-surface villous epithelium.



23. Most diffusible ion across the cell membrane is?

- (A) Na^+
- (B) K^+
- (C) Cl^-
- (D) None

Correct Answer: (C) Cl^-

Solution:

Step 1: The core of the plasma membrane is a hydrophobic lipid bilayer. It strongly resists the passage of charged, water-soluble particles such as ions, glucose and urea, while letting fat-soluble molecules like oxygen, carbon dioxide and alcohol cross easily.

Step 2: The ease with which an ion crosses is described by its permeability coefficient. Among the listed ions, chloride (Cl^-) has the highest membrane permeability and crosses the bilayer most readily compared with sodium and potassium.

Step 3: Sodium (option a) is largely kept out and pumped back by active transport, and potassium (option b), though more permeable than sodium, is still less diffusible than chloride. Therefore chloride is the most diffusible ion of the group.

Step 4: Because Cl^- moves most freely, option c is correct.

Quick Tip: Compare the cations with the chief diffusible anion of the body.



24. Which of the following is present in the cornea?

- (A) Hyaluronic acid
- (B) Chondroitin sulfate
- (C) Dermatan sulfate
- (D) Heparan sulfate

Correct Answer: (B) Chondroitin sulfate

Solution:

Step 1: Glycosaminoglycans (GAGs) are long unbranched sugar chains attached to core proteins, and each type has a characteristic tissue distribution.

Step 2: The cornea is rich in keratan sulfate and chondroitin sulfate. These GAGs are arranged with the precisely spaced collagen fibrils that keep the cornea transparent. Hence chondroitin sulfate is the GAG found in the cornea among the options.

Step 3: Hyaluronic acid (option a) is abundant in synovial fluid, vitreous humor and loose connective tissue. Dermatan sulfate (option c) is typical of skin, blood vessels and heart valves. Heparan sulfate (option d) is found in basement membranes and cell surfaces.

Step 4: Matching each GAG to its tissue, the cornea corresponds to chondroitin sulfate, so option b is correct.

Quick Tip: Pair each GAG with its tissue; the clear cornea uses keratan and this

sulfated GAG.



25. The extrinsic system of coagulation is activated by?

- (A) Factor XI
- (B) Factor X
- (C) Factor XII
- (D) Factor III

Correct Answer: (D) Factor III

Solution:

Step 1: The extrinsic pathway of coagulation is triggered when blood vessels are damaged and tissue is exposed. The trigger molecule is tissue factor, also called tissue thromboplastin or Factor III.

Step 2: Factor III combines with Factor VII and calcium to activate Factor X, which starts the common pathway leading to thrombin and fibrin. So the activator of the extrinsic system is Factor III.

Step 3: Factor XII (option c) and Factor XI (option a) belong to the intrinsic pathway, which is set off by contact with a negatively charged surface. Factor X (option b) is the point where the two pathways converge, not the initiator of the extrinsic limb.

Step 4: Therefore the extrinsic pathway is activated by tissue factor, Factor III, making option d correct.

Quick Tip: The pathway from outside the vessel begins with tissue factor.



26. NADH-CoQ reductase is inhibited by?

- (A) Rotenone
- (B) Carbon monoxide
- (C) Antimycin
- (D) Atractyloside

Correct Answer: (A) Rotenone

Solution:

Step 1: NADH-CoQ reductase is Complex I of the electron transport chain. It carries electrons from NADH to coenzyme Q.

Step 2: Complex I is blocked by rotenone (an insecticide), along with barbiturates such as amobarbital, piericidin A, chlorpromazine and guanethidine. These agents stop electron flow from the iron-sulfur protein to coenzyme Q. So rotenone is the correct inhibitor.

Step 3: Carbon monoxide (option b) inhibits Complex IV (cytochrome c oxidase). Antimycin (option c) inhibits Complex III. Atractyloside (option d) inhibits the ADP-ATP translocase, not Complex I.

Step 4: Since rotenone is the classic Complex I (NADH-CoQ reductase) inhibitor, option a is correct.

Quick Tip: Which insecticide is the classic Complex I blocker?

27. UDP-glucose is NOT used in?

- (A) Uronic acid pathway
- (B) Glycogen synthesis
- (C) Galactose metabolism
- (D) HMP shunt

Correct Answer: (D) HMP shunt

Solution:

Step 1: UDP-glucose is the activated form of glucose. It is made from glucose-6-phosphate, which is converted to glucose-1-phosphate and then to UDP-glucose.

Step 2: Its main fate is glycogen synthesis (option b), where it donates glucose units to the growing glycogen chain. It also feeds the uronic acid pathway to make UDP-glucuronate (option a) and takes part in galactose metabolism (option c) and in the glycosylation of proteins, lipids and proteoglycans.

Step 3: The HMP shunt (hexose monophosphate pathway) starts directly from glucose-6-phosphate and uses no UDP-glucose. Its job is to generate NADPH and ribose-5-phosphate.

Step 4: Since UDP-glucose plays no part in the HMP shunt, option d is the correct answer to a question asking where it is NOT used.

Quick Tip: Which pathway runs straight from glucose-6-phosphate to make NADPH?

28. Which of the following steps is specific for gluconeogenesis?

- (A) Pyruvate to acetyl CoA
- (B) Oxaloacetate to citrate
- (C) Oxaloacetate to phosphoenolpyruvate (PEP)
- (D) Malate to oxaloacetate

Correct Answer: (C) Oxaloacetate to phosphoenolpyruvate (PEP)

Solution:

Step 1: Gluconeogenesis is the making of glucose from non-carbohydrate sources. It largely reverses glycolysis but must bypass the three irreversible glycolytic steps using its own unique enzymes.

Step 2: One of these unique bypass reactions is the conversion of oxaloacetate to phosphoenolpyruvate (PEP), catalysed by PEP carboxykinase. This step belongs to gluconeogenesis and has no role in glycolysis or the citric acid cycle, so it is specific.

Step 3: Pyruvate to acetyl CoA (option a) is an irreversible step of pyruvate oxidation, not gluconeogenesis. Oxaloacetate to citrate (option b) and malate to oxaloacetate (option d) are steps of the citric acid cycle, which are shared with other pathways.

Step 4: Therefore the step specific for gluconeogenesis is oxaloacetate to PEP, making option c correct.

Quick Tip: Look for the PEP carboxykinase bypass reaction.

29. Which of the following is increased in lipoprotein lipase deficiency?

- (A) VLDL
- (B) LDL
- (C) HDL
- (D) Chylomicrons

Correct Answer: (D) Chylomicrons

Solution:

Step 1: Lipoprotein lipase (LPL) is the enzyme on the capillary endothelium that hydrolyses triglycerides carried by chylomicrons and VLDL, releasing free fatty acids for tissue uptake. Apo CII, present on these particles, is the obligatory activator of LPL.

Step 2: When LPL (or its activator Apo CII) is deficient, the triglyceride-rich chylomicrons cannot be cleared from plasma. This is Type I hyperlipoproteinemia (familial chylomicronemia), in which the fasting plasma shows a creamy

supernatant layer of chylomicrons.

Step 3: Therefore chylomicrons are the lipoprotein fraction that accumulates. Clinical features include eruptive xanthomas, hepatosplenomegaly and recurrent abdominal pain or pancreatitis. Management is restriction of dietary fat with medium chain triglyceride (MCT) supplementation, since MCTs are absorbed directly into the portal blood and do not need chylomicrons.

Step 4: VLDL, LDL and HDL are wrong because their metabolism does not depend chiefly on chylomicron clearance in this defect. LDL elevation is seen in Type II (LDL receptor defect), and VLDL excess characterises Type IV.

Quick Tip: LPL clears triglyceride-rich particles; without it, the dietary fat carrier accumulates (Type I).



30. Tay-Sachs disease is due to deficiency of which enzyme?

- (A) Hexosaminidase A
- (B) Hexosaminidase B
- (C) Sphingomyelinase
- (D) Alpha-galactosidase

Correct Answer: (A) Hexosaminidase A

Solution:

Step 1: Tay-Sachs disease is a GM2 gangliosidosis, an autosomal recessive lysosomal storage disorder. The missing enzyme is hexosaminidase A, which is needed to degrade the GM2 ganglioside.

Step 2: With hexosaminidase A absent, GM2 ganglioside accumulates in neurons, especially in the brain and retina. This produces progressive neurodegeneration, an exaggerated startle response, hypotonia and the classic cherry-red spot on the macula. There is no hepatosplenomegaly, which helps separate it from Niemann-

Pick.

Step 3: The distractors map to other disorders. Sphingomyelinase deficiency causes Niemann-Pick disease, alpha-galactosidase A deficiency causes Fabry disease, and hexosaminidase A plus B deficiency together (deficiency of the beta subunit) causes Sandhoff disease.

Quick Tip: GM2 gangliosidosis with a cherry-red spot and no organomegaly.



31. The rate limiting step in the urea cycle is catalyzed by which enzyme?

- (A) Arginase
- (B) Argininosuccinase
- (C) Carbamoyl-phosphate synthase
- (D) Ornithine transcarbamylase

Correct Answer: (C) Carbamoyl-phosphate synthase

Solution:

Step 1: The urea cycle disposes of ammonia as urea in five steps. The first committed step is the formation of carbamoyl phosphate from ammonia and carbon dioxide, using 2 ATP, catalyzed by carbamoyl phosphate synthase I (CPS-I) in the mitochondria.

Step 2: CPS-I is the rate limiting enzyme of the urea cycle. It is an allosteric enzyme that is activated by N-acetylglutamate, so flux through the whole cycle is controlled at this point.

Step 3: Do not confuse CPS-I with the cytosolic CPS-II, which uses glutamine as the nitrogen donor and works in pyrimidine synthesis, not urea formation.

Step 4: The other enzymes act later and are not rate limiting. Ornithine transcarbamylase makes citrulline, argininosuccinase (argininosuccinate lyase)

splits argininosuccinate into arginine and fumarate, and arginase finally cleaves arginine into urea and ornithine.

Quick Tip: First, allosteric, mitochondrial step activated by N-acetylglutamate.



32. Which of the following is a neutral amino acid?

- (A) Aspartate
- (B) Arginine
- (C) Glycine
- (D) Histidine

Correct Answer: (C) Glycine

Solution:

Step 1: Amino acids are grouped by the charge on their side chain. A neutral (uncharged) amino acid has a side chain that carries no net charge at physiological pH.

Step 2: Glycine has only a hydrogen atom as its side chain, so it has no ionizable group and is a neutral amino acid. This makes glycine the correct answer.

Step 3: The distractors all carry charged side chains. Aspartate is acidic (negatively charged carboxyl side chain). Arginine and histidine are basic, with positively charged side chains at physiological pH (the imidazole of histidine and the guanidinium of arginine).

Quick Tip: Look for the amino acid whose side chain has no ionizable charged group.



33. Which is the first purine nucleotide synthesized in purine biosynthesis?

- (A) AMP
- (B) GMP
- (C) IMP
- (D) UMP

Correct Answer: (C) IMP

Solution:

Step 1: In de novo purine biosynthesis the purine ring is built up step by step on a ribose-5-phosphate backbone (derived from the pentose phosphate pathway), which is first activated to PRPP.

Step 2: The end product of this assembly is inosine monophosphate (IMP), whose base is hypoxanthine. IMP is therefore the first complete purine nucleotide formed.

Step 3: IMP is the common precursor: it is then converted into AMP (via adenylosuccinate) and into GMP (via xanthosine monophosphate). So AMP and GMP form after IMP and cannot be the first.

Step 4: UMP is wrong because it is a pyrimidine nucleotide, not a purine, and so does not belong to the purine pathway at all.

Quick Tip: The ring is assembled on ribose; the first complete purine nucleotide carries hypoxanthine.

34. Which of the following is a nucleoside?

- (A) Adenine
- (B) Uridine
- (C) Thymine
- (D) Guanine

Correct Answer: (B) Uridine

Solution:

Step 1: Distinguish the three terms. A base (e.g. adenine) is a nitrogenous ring alone. A nucleoside is a base plus a sugar (ribose or deoxyribose). A nucleotide is a nucleoside plus one or more phosphate groups.

Step 2: Uridine is uracil joined to ribose, so it is a base plus a sugar with no phosphate. That makes uridine a nucleoside, which is the correct answer.

Step 3: Adenine, thymine and guanine are all free nitrogenous bases, not nucleosides, because they lack the attached sugar. Adding ribose would make them adenosine, thymidine and guanosine respectively.

Quick Tip: Nucleoside = base + sugar, no phosphate; its name often ends in -idine or -osine.

35. Which is the strongest interaction among the following?

- (A) Covalent
- (B) Hydrogen
- (C) Electrostatic
- (D) Van der Waals

Correct Answer: (A) Covalent

Solution:

Step 1: Interactions that stabilize molecular structures are of two broad kinds: covalent bonds, which share electron pairs, and non-covalent bonds (hydrogen bonds, electrostatic or ionic interactions and van der Waals forces).

Step 2: Covalent bonds, such as peptide bonds and disulphide bonds in proteins, involve actual sharing of electrons and have far higher bond energies than any

non-covalent interaction. Hence covalent is the strongest interaction.

Step 3: The order of strength runs covalent then electrostatic (ionic) then hydrogen then van der Waals. Van der Waals forces are the weakest of all, so the remaining options are all weaker than covalent.

Quick Tip: Shared electron pairs (peptide, disulphide bonds) beat all non-covalent forces.



36. Hypertrophy is -

- (A) Increase in cell number
- (B) Increase in cell size
- (C) Decrease in cell number
- (D) Decrease in cell size

Correct Answer: (B) Increase in cell size

Solution:

Step 1: Define the term. Hypertrophy is an increase in the size of cells, which results in an increase in the size of the affected tissue or organ. There is no increase in the number of cells. The enlargement happens without cell division, so a hypertrophied organ simply has bigger cells, not new ones.

Step 2: Mechanism. The increase in cell size is driven by the synthesis of more structural proteins and organelles. Nuclei of hypertrophied cells often have a higher DNA content because the cells arrest in the cell cycle without completing mitosis. The most common stimulus is increased workload.

Step 3: Examples. Permanent (non-dividing) tissues such as the myocardium and skeletal muscle classically undergo hypertrophy, for example left ventricular hypertrophy in systemic hypertension.

Step 4: Why the other options are wrong. Increase in cell number (option a) describes hyperplasia, not hypertrophy. Decrease in cell number (option c) is hypoplasia or involution, and decrease in cell size (option d) is atrophy. Hence the correct answer is increase in cell size.

Quick Tip: Trophy means growth of the same cell, not creation of new cells.

37. Matrix metalloproteinases are -

- (A) Cathepsin
- (B) Zn metalloproteinases
- (C) Cu metalloproteinases
- (D) Cd metalloproteinases

Correct Answer: (B) Zn metalloproteinases

Solution:

Step 1: Identify the enzyme family. Matrix metalloproteinases (MMPs) degrade collagen and other extracellular matrix (ECM) components during tissue remodeling and scar maturation. The outcome of repair depends on the balance between synthesis and degradation of ECM proteins.

Step 2: Why zinc. They are called metalloproteinases because they require a metal ion in their active site for catalytic activity, and that ion is zinc. So MMPs are zinc-dependent enzymes, making Zn metalloproteinases the correct answer.

Step 3: Members. The family includes interstitial collagenases (MMP-1, -2, -3) that cleave fibrillar collagen, gelatinases (MMP-2 and MMP-9) that degrade amorphous collagen and fibronectin, and stromelysins (MMP-3, -10, -11) that degrade proteoglycans, laminin, fibronectin and amorphous collagen.

Step 4: Why the others are wrong. Cathepsins are a separate group of lysosomal proteases (often cysteine or aspartic proteases), not MMPs. Copper (Cu) and

cadmium (Cd) are not the catalytic metal of MMPs. Hence Zn metalloproteinases is correct.

Quick Tip: Metallo- in MMP refers to a zinc ion in the active site.

38. Which of the following interleukin is secreted by T helper 2 cells?

- (A) IL 11
- (B) IL 7
- (C) IL 1
- (D) IL 13

Correct Answer: (D) IL 13

Solution:

Step 1: Recall the Th subsets. CD4+ helper T cells differentiate mainly into Th1 and Th2 subsets, each with a characteristic cytokine profile.

Step 2: Th2 cytokines. T helper 2 (Th2) cells secrete IL-4, IL-5, IL-6, IL-10 and IL-13. These drive B cell antibody responses (especially IgE), eosinophil activity and allergic or anti-parasitic immunity. Among the given options, IL-13 is the Th2 cytokine, so option d is correct.

Step 3: Contrast with Th1. Th1 cells secrete IL-2 and interferon-gamma, which activate macrophages and support cell-mediated immunity. IL-13 is not part of the Th1 set.

Step 4: Why the others are wrong. IL-1 is produced mainly by macrophages and is a pro-inflammatory cytokine, IL-7 is largely from stromal and bone marrow cells supporting lymphocyte development, and IL-11 is a stromal-cell cytokine involved in thrombopoiesis. None of these are Th2 products, leaving IL-13 as the answer.

Quick Tip: Th2 cytokines include IL-4, IL-5, IL-6, IL-10 and IL-13.

39. Leukocyte migration through endothelium is induced by -

- (A) Selectin
- (B) N CAM
- (C) C CAM
- (D) PECAM

Correct Answer: (D) PECAM

Solution:

Step 1: Name the step. Migration of leukocytes across the endothelial wall into tissue is called transmigration or diapedesis, and it occurs mainly in postcapillary venules.

Step 2: Identify the molecule. The key adhesion molecule that mediates transmigration is CD31, also called PECAM-1 (platelet endothelial cell adhesion molecule-1). It belongs to the immunoglobulin superfamily and is present on both leukocytes and endothelial cell junctions, allowing the leukocyte to squeeze between cells. So option d, PECAM, is correct.

Step 3: Place it in the cascade. The full leukocyte recruitment cascade is rolling (selectins), firm adhesion (integrins binding ICAM/VCAM), and finally transmigration (PECAM-1). PECAM acts at the last, transmigration step.

Step 4: Why the others are wrong. Selectins mediate the earlier rolling phase, not transmigration. N-CAM is a neural cell adhesion molecule and C-CAM is not the transmigration mediator. Therefore PECAM is the answer.

Quick Tip: Transmigration (diapedesis) is mediated by PECAM-1 (CD31).

40. Pendred syndrome is due to mutation of -

- (A) Bartillin
- (B) Pendrin
- (C) Fibrillin
- (D) Reticulin

Correct Answer: (B) Pendrin

Solution:

Step 1: Define the syndrome. Pendred syndrome is a genetic disorder causing congenital bilateral sensorineural hearing loss together with goitre and occasional hypothyroidism.

Step 2: Identify the gene and protein. It is caused by mutations in the PDS gene (SLC26A4), which codes for the protein pendrin, an anion transporter of the solute carrier family 26. So the mutated protein is pendrin, option b.

Step 3: Inheritance and location. The gene lies on the long arm of chromosome 7 (7q31) and the condition is inherited in an autosomal recessive pattern. Pendrin dysfunction affects iodide transport in the thyroid and ion handling in the inner ear, explaining the combined thyroid and hearing features.

Step 4: Why the others are wrong. Fibrillin mutations cause Marfan syndrome, reticulin is a connective tissue fiber and not the cause here, and bartillin is not the relevant protein. Therefore pendrin is correct.

Quick Tip: Pendred syndrome = pendrin (SLC26A4); deafness plus goitre.

41. Which is not a feature of Down syndrome?

- (A) Clinodactyly
- (B) Pigmented birth marks
- (C) Hypotonia
- (D) Respiratory tract infections

Correct Answer: (B) Pigmented birth marks

Solution:

Step 1: Recall the features of Down syndrome (trisomy 21). General features include mental retardation and short stature. Craniofacial features include brachycephaly, epicanthic folds, a protruding tongue, small ears, upward slanting palpebral fissures (Mongoloid slant), strabismus, nystagmus and Brushfield spots in the iris.

Step 2: Limb and other features. Limb features include fifth finger clinodactyly, a single palmar (simian) crease and a wide sandal gap between the first and second toes. Hypotonia (floppiness) is a classic neonatal sign, and recurrent respiratory tract infections are common because of immune and structural factors.

Step 3: Match the options. Clinodactyly (a), hypotonia (c) and respiratory tract infections (d) are all recognized features of Down syndrome.

Step 4: Identify the odd one out. Pigmented birth marks are not a characteristic feature of Down syndrome; they are not part of its typical phenotype. Hence option b is the feature that is NOT seen, making it the correct answer.

Quick Tip: Look for the sign that is not part of the trisomy 21 phenotype.

42. Keratinization and pearl formation is characteristic of -

- (A) Squamous cell carcinoma
- (B) Basal cell carcinoma
- (C) Melanoma
- (D) Lymphoma

Correct Answer: (A) Squamous cell carcinoma

Solution:

Step 1: Link the clue to the tumor. Keratin (squamous) pearls are concentric whorls of keratinizing cells. Their presence indicates that the tumor cells are producing keratin, which is a property of squamous epithelium.

Step 2: Histology. Squamous cell carcinoma is characterized histologically by keratinization, the formation of squamous (keratin) pearls and the presence of intercellular bridges. Keratinization may appear as pearls or as individual cells with markedly eosinophilic, dense cytoplasm. So option a is correct.

Step 3: Why the others are wrong. Basal cell carcinoma shows basaloid cells with peripheral palisading and retraction artifact, not keratin pearls. Melanoma shows pigmented atypical melanocytes, and lymphoma is a tumor of lymphoid cells; none of these form keratin pearls.

Step 4: Conclusion. Keratin pearl formation is the hallmark of well-differentiated squamous cell carcinoma, so squamous cell carcinoma is the answer.

Quick Tip: Keratin pearls signal a keratin-producing squamous epithelial tumor.

43. The phenomenon by which cancer cells are able to sustain and proliferate under adverse conditions of hypoxia is?

- (A) Warburg
- (B) Wanton
- (C) Wormian
- (D) Wolf

Correct Answer: (A) Warburg

Solution:

Step 1: The question asks about the metabolic adaptation that lets tumour cells survive low-oxygen conditions. This is the Warburg phenomenon, described by Otto Warburg.

Step 2: Cancer cells preferentially use aerobic glycolysis (converting glucose to lactate) even when oxygen is available. This shift in bioenergetics and metabolism permits the cells to grow despite hypoxia and supports proliferation, progression, invasiveness, and distant metastasis.

Step 3: Because it actively promotes tumour growth, it is called the growth-promoting Warburg phenomenon (or Warburg effect).

Step 4: The other terms are distractors. Wormian refers to small extra bones in skull sutures, Wolffian relates to embryology, and Wanton is not a recognised oncology term. Hence Warburg is correct.

Quick Tip: Named after the scientist who described aerobic glycolysis in tumours.

44. Which of the following is not a large vessel vasculitis?

- (A) Takayasu arteritis
- (B) Cogan syndrome
- (C) Churg-Strauss syndrome
- (D) Giant cell arteritis

Correct Answer: (C) Churg-Strauss syndrome

Solution:

Step 1: Vasculitides are classified by the calibre of the vessel predominantly involved - large, medium, or small.

Step 2: Large vessel vasculitis includes giant cell arteritis (temporal arteritis), Takayasu arteritis, and Cogan syndrome. So options a, b and d all belong to the large vessel group.

Step 3: Churg-Strauss syndrome (eosinophilic granulomatosis with polyangiitis) is an ANCA-associated small vessel vasculitis. It is therefore the odd one out and the correct answer.

Step 4: For context, medium vessel vasculitis includes classical polyarteritis nodosa, Kawasaki disease and Buerger's disease, while other small vessel types include HSP, Wegener's granulomatosis and microscopic polyangiitis.

Quick Tip: One option is an ANCA-associated small vessel vasculitis linked to asthma.

45. True about gastric lymphoma:

- (A) Non-Hodgkin lymphoma is the commonest variety
- (B) Diagnosis is made by biopsy
- (C) H. pylori has a direct relationship
- (D) All of the above

Correct Answer: (D) All of the above

Solution:

Step 1: The stomach is the most common site for extranodal lymphoma. Nearly all gastric lymphomas are B-cell lymphomas of mucosa-associated lymphoid tissue (MALT lymphoma), a type of non-Hodgkin lymphoma. So option a is true.

Step 2: About 80% of cases are associated with chronic gastritis and H. pylori infection. Eradicating H. pylori with antibiotics induces durable remission, confirming a direct causal relationship. So option c is true.

Step 3: Diagnosis is established by endoscopic biopsy, so option b is true.

Step 4: Since options a, b and c are all correct, the best answer is option d, All of the above.

Quick Tip: MALToma is NHL, biopsy-diagnosed, and H. pylori driven - check each option.



46. Maintenance dose is calculated by using the value of?

- (A) Clearance
- (B) Volume of distribution
- (C) Oral bioavailability
- (D) Daily dosage

Correct Answer: (A) Clearance

Solution:

Step 1: The maintenance dose is the dose given repeatedly to keep the drug concentration at steady state. At steady state, the rate of drug administration equals the rate of drug elimination.

Step 2: The rate of elimination depends on clearance (CL) and the target steady-state concentration (C_{ss}). The relationship is **Maintenance dose rate = $CL \times C_{ss}$** . Therefore clearance is the key parameter for the maintenance dose.

Step 3: Volume of distribution, by contrast, is used to calculate the loading dose (Loading dose = $V_d \times$ target concentration). So option b governs the loading

dose, not the maintenance dose.

Step 4: Oral bioavailability only corrects the dose for the oral route and daily dosage is an outcome, not the determining pharmacokinetic value. Hence clearance is correct.

Quick Tip: At steady state, dosing rate replaces elimination rate - which depends on this parameter.



47. A young child weighing 20 kg was given a drug in the dose 100 mg/kg body weight. The plasma concentration of the drug is 2 mg/dl and the clearance is 200 ml/hr. What is the time required to reach steady state plasma concentration?

- (A) 10 hrs
- (B) 20 hrs
- (C) 30 hrs
- (D) 40 hrs

Correct Answer: (B) 20 hrs

Solution:

Step 1: First find the total dose. Total dose = dose per kg x body weight = 100 × 20 = 2000 mg.

Step 2: Calculate the volume of distribution. $V_d = \frac{\text{total dose}}{\text{plasma concentration}}$
 $= \frac{2000 \text{ mg}}{2 \text{ mg/dl}} = 1000 \text{ dl}$. In matching units this corresponds to 1000 dl, used directly below.

Step 3: Calculate the half-life. $t_{1/2} = \frac{0.693 \times V_d}{CL} = \frac{0.693 \times 1000}{200} = 3.5 \text{ hours}$.

Step 4: Steady state is reached after about 5.5 half-lives. Time = 5.5×3.5 = 19.25 hours, which rounds to the closest option, 20 hours.

Quick Tip: Find Vd, then half-life from $0.693 \times Vd / CL$, then multiply by about 5.5.



48. Most common cytochrome P-450 enzyme for drug metabolism detoxification reactions is?

- (A) CYP 3A4
- (B) CYP 1A2
- (C) CYP 2A6
- (D) CYP 2B6

Correct Answer: (A) CYP 3A4

Solution:

Step 1: Cytochrome P-450 (CYP) isoenzymes are grouped into families by capital letters (A, B, C) based on amino acid sequence homology, with individual enzymes given numerals such as CYP1A2, 2A6, 2B6, 2C8 and 3A4/3A5.

Step 2: In humans, only a few isoenzymes from three families carry out metabolism of most drugs.

Step 3: CYP3A4/3A5 carries out the biotransformation of the largest number of drugs - nearly 50%. Hence CYP3A4 is the most common enzyme involved in drug metabolism.

Step 4: Important inducers of CYP3A4/3A5 include rifampin, barbiturates, glucocorticoids, carbamazepine, phenytoin, macrolide antibiotics and pioglitazone. The other listed isoenzymes metabolise far fewer drugs, so CYP3A4 is correct.

Quick Tip: This isoform metabolises nearly 50% of all drugs.

49. A patient on verapamil should not be given a beta blocker as?

- (A) Conduction block
- (B) Bronchospasm
- (C) Neurogenic shock
- (D) Anaphylaxis

Correct Answer: (A) Conduction block

Solution:

Step 1: Verapamil is a non-dihydropyridine calcium channel blocker with strong negative chronotropic and dromotropic effects. It slows AV nodal conduction and can accentuate conduction defects.

Step 2: Beta blockers also depress AV nodal conduction and heart rate. Combining the two produces additive suppression of the AV node and sinus node.

Step 3: This combination risks severe bradycardia, AV (conduction) block and even asystole. Verapamil should be avoided in 2nd and 3rd degree block, sick sinus syndrome and together with a beta blocker.

Step 4: Bronchospasm and anaphylaxis are not the concern with this drug pair, and neurogenic shock is unrelated. The correct answer is conduction block.

Quick Tip: Both agents depress the AV node - their additive effect is the danger.

50. Which of the following is an inverse agonist at the benzodiazepine receptor?

- (A) Phenobarbitone
- (B) Flumazenil
- (C) Beta carboline
- (D) Gabapentin

Correct Answer: (C) Beta carboline

Solution:

Step 1: The GABA-A receptor complex carries a benzodiazepine (BZD) binding site that can be engaged by three functional classes of ligand. Agonists (the classical BZDs) increase the frequency of chloride channel opening and produce sedation and anxiolysis. An antagonist simply blocks the site. An inverse agonist binds the same site but produces the opposite effect, reducing chloride conductance and causing anxiety and convulsions.

Step 2: Beta carbolines are the prototype inverse agonists at the BZD receptor. They reduce GABA-A mediated inhibition, so they are anxiogenic and proconvulsant. This makes option c correct.

Step 3: Flumazenil is a pure antagonist (it reverses BZD overdose but has no intrinsic effect of its own), so it is not an inverse agonist. Phenobarbitone acts at the barbiturate site, not as a BZD inverse agonist. Gabapentin is a gabapentinoid that modulates the alpha-2-delta calcium channel subunit and has no inverse agonist action at the BZD site.

Quick Tip: Look for the anxiogenic and proconvulsant agent that does the opposite of diazepam.

51. A patient on carbidopa plus levodopa for 10 years now has a wearing-off effect. Which drug should be added to restore the action?

- (A) Tolcapone
- (B) Amantadine
- (C) Rasagiline
- (D) Benzhexol

Correct Answer: (A) Tolcapone

Solution:

Step 1: The wearing-off phenomenon is a motor fluctuation in advanced Parkinson disease where the benefit of each levodopa dose fades before the next dose is due. The most direct fix is to prolong the plasma availability of levodopa.

Step 2: Tolcapone is a COMT inhibitor. By blocking catechol-O-methyltransferase it prevents the peripheral breakdown of levodopa to 3-O-methyldopa, raising and sustaining levodopa levels reaching the brain. This smooths out the wearing-off, increases 'on' time and reduces 'off' time, which is exactly what is needed here, so option a is correct.

Step 3: Rasagiline (an MAO-B inhibitor) does help wearing-off too, but tolcapone is the classic adjunct emphasised for prolonging each levodopa dose; the question expects the COMT inhibitor. Amantadine is mainly used for levodopa-induced dyskinesias, not pure wearing-off. Benzhexol (trihexyphenidyl) is an anticholinergic useful for tremor, not for extending levodopa action.

Quick Tip: Block the second peripheral enzyme that destroys levodopa to make each dose last longer.

52. Let A = ACE inhibitor, B = beta blocker, C = calcium channel blocker, and D = diuretic. For an elderly patient with hypertension, the antihypertensive drug of choice is?

- (A) A or D
- (B) A or B
- (C) A or C
- (D) C or D

Correct Answer: (D) C or D

Solution:

Step 1: Antihypertensive therapy is structured by the AB/CD rule. A and B (ACE inhibitor and beta blocker) act mainly by suppressing the renin-angiotensin system and tend to work better in younger patients who have high renin. C and D (calcium channel blocker and diuretic) work independently of renin and are preferred in older patients and in those of African origin, who typically have a low-renin, volume-dependent pattern.

Step 2: In the elderly, renin activity is low, so renin-targeting drugs are less effective as first-line agents. Calcium channel blockers and diuretics give better blood pressure control in this group, which is why C or D is the answer (option d).

Step 3: The other combinations all pair in a renin-dependent agent (A or B), which is the preferred starting choice for the younger high-renin patient, not the elderly. Hence A or D, A or B and A or C are not the best fit here.

Quick Tip: Elderly patients are low-renin, so favour the renin-independent arm of the AB/CD rule.

53. Colistin is obtained from which of the following?

- (A) Bacteria
- (B) Fungi
- (C) Actinomycetes
- (D) Herbs

Correct Answer: (A) Bacteria

Solution:

Step 1: Antibiotics are classified by their natural source. Fungi yield penicillin, cephalosporin and griseofulvin. Bacteria (especially *Bacillus* species) yield the polypeptide antibiotics. Actinomycetes (*Streptomyces* and similar) yield aminoglycosides, tetracyclines, chloramphenicol, macrolides and polyenes.

Step 2: Colistin is also called polymyxin E. It is a polypeptide antibiotic produced by a bacterium, *Bacillus* (*Paenibacillus*) *polymyxa*. So colistin is obtained from bacteria, making option a correct.

Step 3: It is not from fungi (those give the beta-lactams and griseofulvin) and not from actinomycetes (those give aminoglycosides, tetracyclines and macrolides). It is certainly not a herbal product. The other bacterially derived polypeptides in the same family are polymyxin B, bacitracin and tyrothricin.

Quick Tip: Colistin is polymyxin E, a polypeptide from a *Bacillus* species.

54. Which cephalosporin characteristically causes thrombocytopenia?

- (A) Cefixime
- (B) Ceftazidime
- (C) Cefoperazone
- (D) Cefdinir

Correct Answer: (B) Ceftazidime

Solution:

Step 1: Cephalosporins share a class of adverse effects (hypersensitivity, diarrhea, superinfection) but several individual agents have signature toxicities. Recognising the drug tied to a specific effect is the key to this question.

Step 2: Ceftazidime is the cephalosporin classically linked with haematological suppression, producing neutropenia and thrombocytopenia. That makes option b the correct answer.

Step 3: The distractors carry different signatures. Cefoperazone (like ceftriaxone, cefamandole and moxalactam) is associated with hypoprothrombinaemia and bleeding from its methylthiotetrazole side chain and biliary excretion, plus a disulfiram-like reaction with alcohol, but not specifically thrombocytopenia. Cefixime and cefdinir are oral third-generation agents whose main issue is gastrointestinal upset, not platelet suppression.

Quick Tip: Pair low platelets and low neutrophils with the antipseudomonal third-generation agent.

55. Irreversible hearing loss is caused by which of the following?

- (A) Gentamicin
- (B) Clarithromycin
- (C) Both of the above
- (D) None of the above

Correct Answer: (A) Gentamicin

Solution:

Step 1: Ototoxicity differs in reversibility depending on the drug. The crucial distinction here is between a permanent and a temporary effect on hearing.

Step 2: Gentamicin, the most commonly used aminoglycoside, damages the cochlear and vestibular hair cells. It causes vestibular toxicity and irreversible (permanent) hearing loss, because the destroyed hair cells do not regenerate. So option a is correct.

Step 3: Clarithromycin, a macrolide, can cause hearing loss too, but it is reversible and resolves on stopping the drug. Therefore 'Both' is wrong (only gentamicin gives irreversible loss), and 'None' is clearly incorrect because gentamicin does cause it.

Quick Tip: Aminoglycoside hair-cell damage does not recover; macrolide hearing loss does.



56. Regarding nicotinic acid (niacin), which of the following is correct?

- (A) Increases HDL
- (B) Increases triglyceride synthesis
- (C) Used for type II hyperlipoproteinemia
- (D) Decreases hydrolysis of VLDL

Correct Answer: (A) Increases HDL

Solution:

Step 1: Nicotinic acid (niacin) acts on lipid metabolism mainly by inhibiting hormone-sensitive lipase in adipose tissue. This reduces the flow of free fatty acids to the liver, so the liver makes less triglyceride and less VLDL.

Step 2: Niacin is the most effective drug for raising HDL cholesterol. It reduces the apo-A1 mediated hepatic clearance of HDL, so HDL levels rise. This makes option a the correct statement.

Step 3: The other options are the opposite of niacin's actions. It decreases (not increases) triglyceride synthesis. It increases lipoprotein lipase activity, so it

increases (not decreases) hydrolysis of VLDL triglyceride. It is used in type I, III, IV and V hyperlipoproteinaemias, not specifically type II, so that option is also incorrect.

Quick Tip: Niacin is the single best drug for raising the protective lipoprotein.

57. Prophylactic dose of vitamin K given to newborn infants at delivery is?

- (A) 1 mg
- (B) 5 mg
- (C) 10 mg
- (D) 15 mg

Correct Answer: (A) 1 mg

Solution:

Step 1: Newborns are highly prone to vitamin K deficiency. This is because of low fetal fat stores, low levels of vitamin K in breast milk, a sterile infantile gut that cannot synthesise vitamin K, an immature liver, and poor placental transfer of the vitamin.

Step 2: Vitamin K is needed for hepatic synthesis of clotting factors II, VII, IX and X. A deficiency leads to bleeding, known as vitamin K deficiency bleeding (haemorrhagic disease of the newborn), which may present as intracranial, gastrointestinal or skin bleeding within the first week of life.

Step 3: To prevent this, a single prophylactic dose of vitamin K **1 mg** is given intramuscularly to every newborn at the time of delivery. Hence the correct answer is 1 mg.

Step 4: Doses of 5 mg, 10 mg or 15 mg are far higher than required for prophylaxis and are not the standard recommended dose, so they are incorrect.

Quick Tip: Every newborn gets a single small intramuscular injection right after birth.

58. DMPA (Depot medroxyprogesterone acetate) is given once in?

- (A) 3 months
- (B) 6 months
- (C) 9 months
- (D) 45 days

Correct Answer: (A) 3 months

Solution:

Step 1: DMPA is a long-acting injectable progestin-only contraceptive. It is given intramuscularly as an oily depot preparation, from which the drug is slowly released.

Step 2: The standard dose is 150 mg given at 3-month intervals. The slow release provides reliable suppression of ovulation for about 12 weeks, after which the next injection is due. Hence DMPA is given once in 3 months.

Step 3: For comparison, norethisterone enanthate (NEE) 200 mg is the other injectable progestin and is given every 2 months, while combined long-acting injectables are given once a month. None of these matches 6 or 9 months or 45 days, so those options are wrong.

Step 4: A key side effect of DMPA is disruption of the menstrual bleeding pattern with eventual amenorrhoea, which is more common with DMPA than with other progestins.

Quick Tip: This 150 mg depot progestin injection lasts about one quarter of a year.

59. Drug which decreases efficacy of testosterone?

- (A) Isoniazid
- (B) Ketoconazole
- (C) Rifampicin
- (D) None

Correct Answer: (B) Ketoconazole

Solution:

Step 1: Testosterone is synthesised through a series of steps involving cytochrome P450 enzymes, including those that convert cholesterol into androgens in the testes and adrenal glands.

Step 2: Ketoconazole is an azole antifungal that inhibits several P450 enzymes involved in steroidogenesis (such as 17,20-lyase and 17-alpha-hydroxylase). By blocking these enzymes it reduces testosterone synthesis, thereby decreasing its effect. This anti-androgen action is why ketoconazole has even been used to lower testosterone in conditions like prostate cancer.

Step 3: Hence ketoconazole is the drug that decreases the efficacy of testosterone, making it the correct answer.

Step 4: Isoniazid and rifampicin are antitubercular drugs; rifampicin is a strong enzyme inducer but is not classically the answer for lowering testosterone action here, and isoniazid does not reduce testosterone efficacy. So options a, c and d are incorrect.

Quick Tip: An azole antifungal that blocks P450 steroid-synthesis enzymes and acts as an anti-androgen.

60. Mechanism of action of actinomycin D is?

- (A) Inhibits DNA dependent RNA synthesis
- (B) Activates DNA dependent RNA synthesis
- (C) Inhibits RNA dependent DNA synthesis
- (D) Activates RNA dependent DNA synthesis

Correct Answer: (A) Inhibits DNA dependent RNA synthesis

Solution:

Step 1: Actinomycin D (dactinomycin) is an anticancer antibiotic obtained from microorganisms. This group also includes doxorubicin, daunorubicin, bleomycin, mitomycin and mitoxantrone.

Step 2: Actinomycin D works by intercalating between the two strands of the DNA double helix, specifically binding to guanine-cytosine base pairs. By wedging into the DNA, it interferes with the template function of DNA.

Step 3: This intercalation blocks the movement of RNA polymerase along the DNA, so transcription (DNA-dependent RNA synthesis) is inhibited. Hence the correct answer is that it inhibits DNA-dependent RNA synthesis.

Step 4: It does not activate transcription, nor does it act on RNA-dependent DNA synthesis (reverse transcription), so options b, c and d are wrong. Note that most antitumour antibiotics are cell-cycle non-specific except bleomycin, which acts in the G2 phase.

Quick Tip: It intercalates into DNA and blocks RNA polymerase, halting transcription.

61. Grievous hurt comes under which section of the Indian Penal Code?

- (A) 319
- (B) 320
- (C) 324
- (D) 326

Correct Answer: (B) 320

Solution:

Step 1: The Indian Penal Code defines different categories of hurt. Simple hurt is defined under Section 319, while grievous hurt is defined under Section 320.

Step 2: Section 320 lists the eight types of hurt designated as grievous, including emasculation, permanent loss of sight of either eye, permanent loss of hearing of either ear, loss or destruction of any member or joint, permanent disfiguration of the head or face, fracture or dislocation of a bone or tooth, and any hurt that endangers life or causes the victim to be in severe bodily pain or unable to follow ordinary pursuits for 20 days. Hence grievous hurt falls under Section 320.

Step 3: Section 319 deals with simple hurt, Section 324 deals with voluntarily causing hurt by dangerous weapons or means, and Section 326 deals with voluntarily causing grievous hurt by dangerous weapons. These define offences related to causing hurt, not the definition of grievous hurt itself, so options a, c and d are incorrect.

Quick Tip: Simple hurt is 319; the very next section defines grievous hurt.

62. Which of Gustafson's parameters is the most accurate?

- (A) Attrition
- (B) Periodontosis
- (C) Root resorption
- (D) Transparency of the tooth

Correct Answer: (D) Transparency of the tooth

Solution:

Step 1: Gustafson's method is used to estimate age in persons older than about 21 years, based on six progressive physiological changes in the dental tissues: attrition, periodontosis (paradentosis), secondary dentine formation, cementum apposition, root resorption and transparency of the root.

Step 2: Of these six criteria, transparency of the root dentine is the single most reliable and accurate parameter. It appears after about 30 years of age and increases steadily with age as minerals are deposited in the dentinal tubules, making the dentine progressively transparent.

Step 3: Because this change is steady and consistent with advancing age, transparency of the tooth (root) is the most accurate of Gustafson's parameters. Hence option d is correct.

Step 4: Attrition, periodontosis and root resorption are also used but are more variable. Attrition depends on the functional use of teeth and diet, periodontosis is influenced by oral hygiene, and root resorption may be pathological, so they are less reliable and are incorrect choices.

Quick Tip: The root dentine change that appears after age 30 is the most reliable criterion.

63. Preauricular sulcus is a part of?

- (A) Humerus
- (B) Femur
- (C) Pelvis
- (D) Skull

Correct Answer: (C) Pelvis

Solution:

Step 1: The preauricular sulcus is a groove found on the ilium of the pelvic bone, just below and in front of the auricular surface of the sacroiliac joint.

Step 2: It lies at the site of attachment of the anterior (ventral) sacroiliac ligament. Hence anatomically it is a part of the pelvis, making option c correct.

Step 3: The preauricular sulcus is an important feature in forensic sex determination of skeletal remains. It is more frequent, broader and deeper in the female pelvis, reflecting the adaptations of the female pelvis for childbirth.

Step 4: The humerus, femur and skull do not bear a preauricular sulcus, so options a, b and d are incorrect.

Quick Tip: This groove near the sacroiliac joint helps determine sex and is broader in females.

64. Maximum soft tissue bruising in the neck is seen in -

- (A) Strangulation
- (B) Hanging
- (C) Burking
- (D) Smothering

Correct Answer: (A) Strangulation

Solution:

Step 1: The question compares the amount of soft tissue bruising in the neck across different forms of mechanical asphyxia. The key is to identify which method applies the most violent, sustained external force directly on the neck structures.

Step 2: Manual strangulation, also called throttling, involves gripping the neck with the hands. This is one of the most violent forms of asphyxial death. The fingers and thumb press deeply into the soft tissues, producing extensive bruising of the strap muscles, fracture of the hyoid bone, and injury to the thyroid cartilage.

Step 3: The distractors cause less neck bruising. In hanging the body weight stretches the neck smoothly via a ligature, so deep muscle bruising is usually minimal. Burking combines smothering with chest compression and does not focus violent force on the neck. Smothering blocks the nose and mouth, so injuries are mainly perioral, not in the neck.

Step 4: Because throttling delivers the greatest direct mechanical trauma to neck soft tissue, strangulation shows the maximum soft tissue bruising. Hence option (a) is correct.

Quick Tip: Think of the hands-on method that crushes the neck and breaks the hyoid.

65. Lucid interval may be seen in?

- (A) Intracerebral hemorrhage
- (B) Alcohol intake
- (C) Insanity
- (D) Subdural hemorrhage

Correct Answer: (C) Insanity

Solution:

Step 1: A lucid interval is a period of apparent normal consciousness or normal behaviour that lies between two episodes of abnormality. The term is used in two distinct settings in forensic medicine and psychiatry.

Step 2: In head injury, the classic lucid interval is seen in extradural (epidural)

haemorrhage. The patient is briefly conscious after the impact, then deteriorates as the arterial bleed expands. In psychiatry, a lucid interval is a period of sanity between two phases of insanity.

Step 3: Among the given options, the only valid answer is insanity. During this lucid interval a person can make a valid will, give valid evidence, and is held legally responsible for civil and criminal acts.

Step 4: The distractors do not apply. Intracerebral and subdural haemorrhage typically cause progressive symptoms without the classic lucid interval seen in extradural bleeds, and simple alcohol intake does not produce a true lucid interval. Hence option (c) Insanity is correct.

Quick Tip: Besides extradural bleed, which listed mental state has a sane gap between two abnormal phases?

66. Legal age by which a fetus is capable of independent existence is?

- (A) 240 days
- (B) 230 days
- (C) 220 days
- (D) 210 days

Correct Answer: (D) 210 days

Solution:

Step 1: The concept here is viability, which is the physical ability of a fetus to lead a separate existence after birth, apart from the mother, by virtue of a certain degree of development.

Step 2: A child is considered viable after 210 days, that is about 7 months of intrauterine life. In some cases a fetus may survive after 180 days (6 months), but most of these are immature and rarely survive.

Step 3: The legally accepted figure for viability is therefore 210 days. The other options (240, 230 and 220 days) are higher values that do not match the recognised medico-legal threshold of independent existence.

Step 4: A useful marker of viability is the primary ossification centre of the talus, which appears at the end of the 7th month. Hence option (d) 210 days is correct.

Quick Tip: Viability is about 7 months of intrauterine life.



67. Postmortem blood is collected from which vessel?

- (A) Femoral artery
- (B) Femoral vein
- (C) Cephalic vein
- (D) Brachial artery

Correct Answer: (B) Femoral vein

Solution:

Step 1: At autopsy, blood is often needed for toxicological analysis, blood grouping, alcohol estimation and other tests. The ideal site is a large peripheral vein that can be reached before the body cavity is opened, to avoid contamination.

Step 2: Before autopsy, about 10 to 20 ml of blood is collected from the femoral vein in the groin. This peripheral site reduces the chance of postmortem diffusion artefacts that affect central blood.

Step 3: The jugular vein or subclavian vein may be used as alternatives, but the femoral vein remains the preferred standard site.

Step 4: The arterial options (femoral artery, brachial artery) and the cephalic

vein are not the recommended sites for routine postmortem blood collection. Hence option (b) Femoral vein is correct.

Quick Tip: A large peripheral vein in the groin, sampled before opening the body.

68. Chocolate brown postmortem staining is seen in?

- (A) KCl poisoning
- (B) Opium poisoning
- (C) H₂S poisoning
- (D) Cyanide poisoning

Correct Answer: (A) KCl poisoning

Solution:

Step 1: The colour of postmortem lividity (hypostasis) gives a clue to the cause of death, because it reflects the form of haemoglobin present in the blood after death.

Step 2: Chocolate brown postmortem staining indicates methaemoglobin formation. Poisons that oxidise haemoglobin to methaemoglobin, such as potassium chlorate and related agents, produce this brownish discolouration. Among the options, KCl (potassium salts in this poisoning context) is the answer linked to chocolate brown staining.

Step 3: The distractors give different colours. Cyanide poisoning gives bright cherry red lividity because of cyanmethaemoglobin and retained oxyhaemoglobin. Hydrogen sulphide (H₂S) gives a greenish or bluish-green colour from sulphaemoglobin. Opium does not classically produce a characteristic chocolate brown stain.

Step 4: Because chocolate brown corresponds to methaemoglobinaemia produced in this poisoning, option (a) is correct.

Quick Tip: Chocolate brown means methaemoglobin from an oxidising agent.



69. For phage typing, how many phages of *Staphylococcus aureus* are used?

- (A) 12
- (B) 15
- (C) 20
- (D) 23

Correct Answer: (D) 23

Solution:

Step 1: Bacteriophage typing of *Staphylococcus aureus* is an epidemiological tool based on the susceptibility of the cocci to a standard set of bacteriophages. It helps trace the source of infection in outbreaks.

Step 2: The technique uses the pattern method, in which a standard set of 23 typing phages is applied to the staphylococcal isolate. The strains are distinguished from one another by their individual patterns of susceptibility to lysis.

Step 3: The phage type of a strain is named by the phages that lyse it. For example, if a strain is lysed by phages 83A, 84 and 85, it is designated type 83A/84/85.

Step 4: Since the international basic set contains 23 typing phages, option (d) 23 is correct. The other figures (12, 15, 20) do not match the standard set size.

Quick Tip: Recall the standard international basic set size for Staph phage typing.

70. Infective endocarditis after tooth extraction is probably due to?

- (A) *Streptococcus viridans*
- (B) *Streptococcus pneumoniae*
- (C) *Streptococcus pyogenes*
- (D) *Staphylococcus aureus*

Correct Answer: (A) *Streptococcus viridans*

Solution:

Step 1: Infective endocarditis following dental procedures is linked to the normal flora of the mouth that enters the bloodstream during the procedure.

Step 2: Viridans streptococci are normal residents of the mouth and upper respiratory tract. They cause a transient bacteremia after tooth extraction or other dental work, then settle on damaged, prosthetic or congenitally diseased heart valves and grow into vegetations.

Step 3: In patients with pre-existing cardiac lesions, viridans streptococci (with *Streptococcus sanguis* often responsible) are the classic cause of subacute bacterial endocarditis. Their adherence to biological surfaces explains this predilection.

Step 4: The distractors fit other diseases. *Strep. pneumoniae* causes pneumonia and meningitis, *Strep. pyogenes* causes pharyngitis and rheumatic fever, and *Staph. aureus* typically causes acute endocarditis (often in IV drug users), not the post-dental subacute form. Hence option (a) *Streptococcus viridans* is correct.

Quick Tip: Which mouth commensal seeds damaged valves after dental work?

71. Which *Vibrio* species is the most halophilic (most salt tolerant)?

- (A) *V. cholerae*
- (B) *V. vulnificus*
- (C) *V. alginolyticus*
- (D) *V. parahaemolyticus*

Correct Answer: (C) *V. alginolyticus*

Solution:

Step 1: Halophilic vibrios need added sodium chloride for growth and tolerate high salt concentrations. Among the medically relevant species, the degree of salt tolerance varies, and the question asks for the one that survives in the highest salt content.

Step 2: *Vibrio alginolyticus* is the most halophilic of the vibrios. It can grow in media containing up to 10 percent sodium chloride, which is higher than the other species listed.

Step 3: *Vibrio parahaemolyticus* and *Vibrio vulnificus* are halophilic too but tolerate less salt (roughly up to 8 percent). *Vibrio cholerae* is non-halophilic and grows without added salt, so it is the least salt tolerant of the group.

Step 4: Therefore the most salt tolerant species is *V. alginolyticus*.

Quick Tip: Think of the vibrio that tolerates the highest salt, around 10 percent NaCl.

72. Which of the following is a liquid medium used for the culture of tuberculosis bacilli?

- (A) LJ medium
- (B) Dorset medium
- (C) Loeffler's medium
- (D) MGIT

Correct Answer: (D) MGIT

Solution:

Step 1: The question asks for a liquid culture medium for *Mycobacterium tuberculosis*. Most classical mycobacterial media are solid egg based media.

Step 2: MGIT stands for Mycobacterial Growth Indicator Tube. It is an automated liquid culture system containing about 7 ml of modified Middlebrook 7H9 broth base, and it detects growth by oxygen consumption using a fluorescent sensor. It is the liquid medium among the options.

Step 3: Lowenstein-Jensen (LJ) medium and Dorset egg medium are solid egg based media used for tuberculosis. Loeffler's serum slope is a solid medium used mainly for *Corynebacterium diphtheriae*, not for primary TB culture.

Step 4: Hence the correct answer is MGIT.

Quick Tip: Look for the automated broth based Middlebrook 7H9 system.

73. Indian tick typhus is caused by which organism?

- (A) *R. typhi*
- (B) *R. conorii*
- (C) *R. akari*
- (D) *C. burnetii*

Correct Answer: (B) *R. conorii*

Solution:

Step 1: Indian tick typhus belongs to the spotted fever group of rickettsial diseases, which are tick borne.

Step 2: *Rickettsia conorii* is the agent of the spotted fever group that causes

Indian tick typhus as well as the closely related Mediterranean spotted fever (boutonneuse fever). It is transmitted by the dog tick.

Step 3: The other options do not fit. *R. typhi* causes endemic (murine) typhus and is flea borne. *R. akari* causes rickettsialpox and is mite borne. *Coxiella burnetii* causes Q fever, which is usually acquired by inhalation rather than by tick bite in classic presentation.

Step 4: Therefore Indian tick typhus is caused by *R. conorii*.

Quick Tip: It is a tick borne spotted fever group rickettsia spread by the dog tick.

74. What is the natural reservoir of plague?

- (A) Domestic rat
- (B) Wild rat
- (C) Rat flea
- (D) Man

Correct Answer: (B) Wild rat

Solution:

Step 1: Plague is caused by *Yersinia pestis*. A reservoir is the host in which the organism is maintained in nature over the long term.

Step 2: Wild rodents, especially wild rats, form the natural reservoir of plague. The infection is maintained silently in these wild populations as sylvatic or wild plague.

Step 3: The domestic rat acts as an amplifier host and a link to humans (domestic or urban plague) rather than the long term reservoir. The rat flea (*Xenopsylla cheopis*) is the vector, not the reservoir, and man is an incidental dead end host.

Step 4: Hence the reservoir of plague is the wild rat.

Quick Tip: The flea is the vector and man is incidental, so look at the sylvatic rodent host.

75. Molluscum contagiosum virus belongs to which group of viruses?

- (A) Poxviruses
- (B) Herpesviruses
- (C) Picornaviruses
- (D) Adenovirus

Correct Answer: (A) Poxviruses

Solution:

Step 1: Molluscum contagiosum produces small umbilicated skin papules and is caused by the molluscum contagiosum virus.

Step 2: This virus is a member of the Poxviridae family. Like other poxviruses it is a large, brick shaped, double stranded DNA virus that replicates in the cytoplasm and produces characteristic intracytoplasmic inclusion bodies (molluscum or Henderson-Paterson bodies).

Step 3: The other families do not fit. Herpesviruses are enveloped DNA viruses causing diseases such as herpes and chickenpox. Picornaviruses are small RNA viruses. Adenoviruses are non enveloped DNA viruses causing respiratory and eye infections.

Step 4: Therefore molluscum contagiosum virus belongs to the poxviruses.

Quick Tip: Think of the large brick shaped DNA virus with cytoplasmic inclusion

bodies.

76. Exanthema subitum is caused by which virus?

- (A) HHV-6
- (B) HHV-8
- (C) Parvovirus
- (D) Coxsackievirus

Correct Answer: (A) HHV-6

Solution:

Step 1: Exanthema subitum is another name for roseola infantum or sixth disease, a febrile rash illness of infants and young children.

Step 2: It is caused by human herpesvirus 6 (HHV-6). The typical course is a high fever for a few days that resolves abruptly, followed by the appearance of a rose pink maculopapular rash.

Step 3: The distractors cause other rash illnesses. HHV-8 is linked to Kaposi sarcoma. Parvovirus B19 causes erythema infectiosum (fifth disease) with a slapped cheek appearance. Coxsackievirus causes hand, foot and mouth disease and herpangina.

Step 4: Hence exanthema subitum (roseola infantum) is caused by HHV-6.

Quick Tip: Roseola infantum or sixth disease points to one herpesvirus.

77. Which serum marker appears after successful Hepatitis B vaccination?

- (A) Anti-HBsAg
- (B) Anti-HBeAg
- (C) Anti-HBcAg
- (D) HBsAg

Correct Answer: (A) Anti-HBsAg

Solution:

Step 1: The hepatitis B vaccine contains only the surface antigen (HBsAg) produced by recombinant technology. It does not contain the core or e antigens.

Step 2: After vaccination the body mounts an immune response only to the surface antigen, so the marker that appears is anti-HBs (antibody to HBsAg). A protective titre confirms successful immunization.

Step 3: Anti-HBc would appear only after natural infection because the core antigen is not part of the vaccine, so it helps distinguish vaccination from past infection. Anti-HBe also reflects infection rather than vaccination. HBsAg itself is a marker of active infection or carrier state, not of vaccine response.

Step 4: Therefore the serum marker after hepatitis B vaccination is anti-HBs (anti-HBsAg).

Quick Tip: The vaccine carries only surface antigen, so the antibody formed is against it.

78. The chance that a health worker gets HIV from an accidental needle prick is:

- (A) 1%
- (B) 10%
- (C) 95%
- (D) 100%

Correct Answer: (A) 1%

Solution:

Step 1: Concept. The risk of HIV transmission after a single percutaneous exposure (needle-stick injury) from a known HIV-positive source is low because the inoculum of virus delivered is very small compared with a transfusion.

Step 2: The number. The average risk of HIV seroconversion after a needle-stick or sharps injury in a health worker is about 0.3%, which is commonly quoted as roughly 1% in exam tables of relative risk per exposure. Hence option (a) is correct.

Step 3: Why the others are wrong. A 90% or more risk applies to transfusion of HIV-infected blood or blood products, not a needle prick. Values of 95% and 100% are far too high for percutaneous occupational exposure and do not match any real transmission category.

Step 4: Note. The risk rises with deep injury, visibly blood-stained device, hollow-bore needle in a vein or artery, and high source viral load. Post-exposure prophylaxis further lowers the chance of seroconversion.

Quick Tip: Percutaneous occupational HIV risk is roughly 0.3 percent, charted as about 1 percent.

79. The band form of *Plasmodium malariae* is which stage?

- (A) Schizont stage
- (B) Trophozoite stage
- (C) Merozoite stage
- (D) Gametocyte stage

Correct Answer: (B) Trophozoite stage

Solution:

Step 1: Concept. Plasmodium malariae has a characteristic morphology on the peripheral blood smear. As the parasite grows inside the red cell it stretches across the cell to form a so-called band form.

Step 2: Which stage. The band form is a growing (developing) trophozoite of P. malariae. The cytoplasm spreads as a band across the diameter of the infected red blood cell, with golden-brown malarial pigment. So option (b), the trophozoite stage, is correct.

Step 3: Why the others are wrong. The schizont stage of P. malariae shows segmentation into about 8 merozoites, often arranged as a rosette or daisy-head, not a band. A merozoite is the small invasive form released when the schizont ruptures. The gametocyte is the rounded sexual stage that fills the red cell and is taken up by the mosquito.

Step 4: Note. The band-form trophozoite is a useful clue to distinguish P. malariae from P. vivax and P. falciparum on a blood film.

Quick Tip: A cytoplasmic band stretching across the red cell is the growing form of the parasite.

80. Which of the following fungi is difficult to isolate in culture?

- (A) Candida
- (B) Dermatophytes
- (C) Cryptococcus
- (D) Malassezia furfur

Correct Answer: (D) Malassezia furfur

Solution:

Step 1: Concept. Most medically important fungi grow on routine Sabouraud dextrose agar. A fungus that needs special enriched media is harder to culture in the laboratory.

Step 2: Why *Malassezia furfur*. *Malassezia furfur* is a lipid-dependent (lipophilic) yeast. It does not grow on plain Sabouraud medium. The medium must be overlaid with a lipid source, classically 1% emulsified olive oil, or a specialised medium such as Dixon or modified Dixon medium must be used. This makes it the difficult one to isolate, so option (d) is correct.

Step 3: Why the others are wrong. *Candida*, dermatophytes, and *Cryptococcus* all grow readily on ordinary Sabouraud dextrose agar without a lipid supplement, so they are easy to recover in culture.

Step 4: Note. *Malassezia furfur* causes pityriasis versicolor, and the lipid requirement reflects its natural habitat on sebum-rich skin.

Quick Tip: Look for the lipid-dependent yeast that will not grow on plain Sabouraud agar.

81. Heat labile liquids are sterilized by which method?

- (A) Hot air oven
- (B) Autoclaving
- (C) Membrane filter
- (D) Moist heat

Correct Answer: (C) Membrane filter

Solution:

Step 1: Concept. Heat-labile means the substance is damaged or destroyed by heat. So any heat-based sterilization method is unsuitable, and a non-thermal method must be chosen.

Step 2: Why membrane filter. Filtration physically removes microbes from a fluid without heating it. Membrane filters (pore size about 0.22 micrometres) are used to sterilize heat-sensitive liquids such as serum, antibiotic solutions, ophthalmic preparations, oils, and liquid culture media. Hence option (c) is correct.

Step 3: Why the others are wrong. The hot air oven uses dry heat at about 160 degrees C and would denature the liquid. Autoclaving and moist heat both use steam under pressure at about 121 degrees C, again destroying a heat-labile substance.

Step 4: Note. Standard membrane filters remove bacteria and fungi but not viruses, so they sterilize only with respect to those larger organisms.

Quick Tip: If heat would wreck the liquid, you must physically strain out the microbes instead.



82. Blood agar is an example of which type of medium?

- (A) Enriched media
- (B) Indicator media
- (C) Enrichment media
- (D) Selective media

Correct Answer: (A) Enriched media

Solution:

Step 1: Concept. Culture media are classified by their function. An enriched medium has extra nutrients, such as blood, serum, or egg, added to a basal medium so that fastidious organisms can grow.

Step 2: Why enriched media. Blood agar is nutrient agar enriched with 5 to 10 percent sheep or human blood. The added blood supplies growth factors that

allow demanding bacteria to multiply, so it is the classic example of an enriched medium. Option (a) is correct.

Step 3: Why the others are wrong. An indicator medium contains a dye or chemical that changes colour to show a metabolic reaction, for example MacConkey agar for lactose fermentation. An enrichment medium (a liquid) favours one organism while suppressing others, for example selenite F or tetrathionate broth for Salmonella. A selective medium contains inhibitors that stop unwanted organisms, for example DCA or TCBS. Blood agar fits none of these.

Step 4: Note. Blood agar also reveals haemolysis patterns (alpha, beta, gamma), which adds differential value, but its primary classification is enriched.

Quick Tip: Adding blood as an extra nutrient to a basal agar defines this category.



83. Which of the following is true about endotoxin?

- (A) Protein
- (B) Highly antigenic
- (C) No enzymatic activity
- (D) Produced by gram positive bacteria

Correct Answer: (C) No enzymatic activity

Solution:

Step 1: Concept. Endotoxin is the lipopolysaccharide (LPS) of the outer membrane of gram-negative bacteria. Its toxic part is lipid A. Unlike exotoxins, which are secreted proteins with specific actions, endotoxin is a structural cell-wall component.

Step 2: Why no enzymatic activity. Because endotoxin is a lipopolysaccharide and not a protein enzyme, it has no enzymatic activity. It acts by triggering

cytokine release, fever, and shock rather than by catalysing a reaction. So option (c) is correct.

Step 3: Why the others are wrong. Endotoxin is lipopolysaccharide, not protein, so (a) is false. It is weakly antigenic (poorly immunogenic) compared with the highly antigenic exotoxins, so (b) is false. It is produced by gram-negative, not gram-positive, bacteria, so (d) is false.

Step 4: Note. Endotoxin is heat stable and cannot be converted into a toxoid, again contrasting with heat-labile, toxoid-forming exotoxins.

Quick Tip: Endotoxin is a lipopolysaccharide, not a protein enzyme, and is poorly antigenic.

84. Which of the following staphylococcal infections is not toxin mediated?

- (A) Toxic shock syndrome
- (B) Scalded skin syndrome
- (C) Food poisoning
- (D) Septic shock

Correct Answer: (D) Septic shock

Solution:

Step 1: Concept. Some staphylococcal diseases are caused by specific toxins, while others result from direct bacterial invasion and the bloodstream presence of organisms.

Step 2: Which one is not toxin mediated. Septic shock from *Staphylococcus aureus* is due to bacteraemia, that is, live bacteria circulating in the blood and provoking a systemic inflammatory response. It is not produced by a single named exotoxin. So septic shock is the answer, option (d).

Step 3: Why the others are toxin mediated. Toxic shock syndrome is caused by toxic shock syndrome toxin-1 (TSST-1). Staphylococcal scalded skin syndrome is caused by exfoliative (epidermolytic) toxin. Staphylococcal food poisoning is caused by a preformed heat-stable enterotoxin. All three are clearly toxin-driven.

Step 4: Note. Septic shock represents invasive infection with cytokine-driven shock, distinct from the toxin-mediated staphylococcal syndromes.

Quick Tip: Three options map to a named staphylococcal toxin; one is caused by bacteraemia.

85. Primordial prevention is done to prevent the development of?

- (A) Disease
- (B) Risk factors
- (C) Impairment
- (D) Disability

Correct Answer: (B) Risk factors

Solution:

Step 1: There are four levels of prevention: primordial, primary, secondary and tertiary. They are arranged according to how early in the natural history of disease they act.

Step 2: Primordial prevention is the earliest level. It aims to prevent the very emergence or development of risk factors in a population or country where those risk factors have not yet appeared. So the target here is the risk factor itself, not the disease.

Step 3: The main modes of intervention are individual education and mass education that discourage unhealthy lifestyles before they become established. This makes primordial prevention the best level of prevention for non-

communicable diseases.

Step 4: Disease (option a) is prevented by primary prevention, while impairment and disability (options c and d) are addressed by secondary and tertiary prevention. Hence the correct answer is risk factors.

Quick Tip: It is the earliest level, acting before the disease itself even has its drivers.

86. Which of the following is NOT true about vaccines?

- (A) Two live vaccines can be given at the same time at different sites
- (B) Two live vaccines at the same site should be given at least 3 weeks apart
- (C) In a vaccine vial monitor, if the colour of the inner square is the same as the outer background, the vaccine is good for use
- (D) Live and killed vaccines can be given together

Correct Answer: (C) In a vaccine vial monitor, if the colour of the inner square is the same as the outer background, the vaccine is good for use

Solution:

Step 1: The question asks for the false statement. A vaccine vial monitor (VVM) is a colour label put on vaccine bottles to show cumulative heat exposure. It has a deep blue outer circle with a white inner square.

Step 2: The inner square gradually turns blue as the vaccine is exposed to higher temperatures. When the inner square becomes as dark as the surrounding circle, that means the vaccine has had too much heat and must be discarded. So matching colour means the vaccine is BAD, not good. Option c is therefore false and is the answer.

Step 3: The other options are correct rules of vaccine administration. Two live vaccines can be given on the same day at different sites (option a is true). If not given together, two live vaccines should be spaced at least 3 to 4 weeks apart

(option b is true). Live and killed vaccines can be given together (option d is true).

Step 4: Hence the statement that does not hold is option c.

Quick Tip: On a VVM, the vial is usable only while the inner square stays lighter than the ring.

87. Missing cases are detected by?

- (A) Active surveillance
- (B) Passive surveillance
- (C) Sentinel surveillance
- (D) Prevalence rate

Correct Answer: (C) Sentinel surveillance

Solution:

Step 1: Surveillance is the continuous scrutiny of the factors that determine the occurrence and distribution of disease and other conditions of ill health. It is carried out in several forms.

Step 2: Sentinel surveillance is used to identify missing cases and to supplement notified cases. It samples selected reporting sites to estimate the true burden when routine notification is incomplete. In India it is used in the National AIDS Control Programme. So option c is correct.

Step 3: In passive surveillance the data report themselves to the health system, for example when a patient comes to a doctor; most national health programmes in India rely on it. In active surveillance the health system actively collects data, for example house-to-house blood slide collection for malaria. Neither is specifically meant to pick up missing cases.

Step 4: Prevalence rate (option d) is a measure of disease frequency, not a method of case detection. Hence the answer is sentinel surveillance.

Quick Tip: It uses select reporting sites to supplement notified cases and pick up the ones that slip through.

88. Matching is not required in which epidemiological study?

- (A) Case control study
- (B) Cohort study
- (C) Case report
- (D) Randomized control trial

Correct Answer: (C) Case report

Solution:

Step 1: Matching is a technique used to make two comparison groups similar for all characteristics except the exposure or factor being studied, so as to reduce confounding and bias.

Step 2: A case report is a detailed account of the symptoms, signs, diagnosis, treatment and follow-up of a single patient, usually describing an unusual or novel occurrence. Because it deals with only one case and has no comparison group, there is nothing to match. So matching is not required, and option c is correct.

Step 3: Case control studies, cohort studies and randomized controlled trials all compare two groups. To keep the groups comparable and limit bias, they must be matched for the relevant characteristics other than the one under study.

Step 4: Since only the case report lacks a comparison group, matching is not required only in a case report.

Quick Tip: Matching balances two groups, so a study with only a single patient needs none.

89. Which of the following is a positive mortality indicator?

- (A) IMR (Infant mortality rate)
- (B) Child mortality rate
- (C) MMR (Maternal mortality rate)
- (D) Life expectancy

Correct Answer: (D) Life expectancy

Solution:

Step 1: Mortality indicators include crude death rate, maternal mortality rate, life expectancy, disease-specific mortality, infant mortality rate, age-specific death rate, child mortality rate, adult mortality rate, under-5 proportional mortality and years of potential life lost.

Step 2: A positive health indicator is one where a higher value means better health. Among mortality indicators, only life expectancy (expectation of life) is positive, because a rise in life expectancy reflects improvement in the health of the community. So option d is correct.

Step 3: Infant mortality rate, child mortality rate and maternal mortality rate are negative indicators. When their values increase, it implies poorer health of the population. Hence they cannot be the answer.

Step 4: Therefore the only positive mortality indicator is life expectancy.

Quick Tip: Look for the one indicator where a higher value means a healthier population.

90. The STOP TB Strategy was launched in?

- (A) 2002
- (B) 2006
- (C) 2010
- (D) 2013

Correct Answer: (B) 2006

Solution:

Step 1: In 2006 the WHO launched the new Stop TB Strategy, whose core component is DOTS. So the correct answer is 2006 (option b).

Step 2: The strategy was designed to be implemented over the following 10 years, as set out in the Global Plan to Stop TB 2006 to 2015, with targets and indicators framed within the Millennium Development Goals.

Step 3: Its five principal indicators are case detection, treatment success, incidence, prevalence and deaths. The plan aimed by 2015 to halve TB prevalence and death rates relative to 1990 levels, and by 2050 to bring global incidence below one case per million per year.

Step 4: The other years listed do not correspond to the launch of this strategy, so the answer is 2006.

Quick Tip: Its Global Plan covered 2006 to 2015, which gives away the launch year.

91. The incubation period of measles is?

- (A) 18 to 72 hours
- (B) 10 to 14 days
- (C) 3 to 4 days
- (D) 20 to 25 days

Correct Answer: (B) 10 to 14 days

Solution:

Step 1: Measles is an acute viral infection caused by an RNA paramyxovirus, spread mainly by respiratory droplets.

Step 2: The incubation period of measles is about 10 to 14 days, counted from exposure to the onset of fever and the prodrome. This is roughly 10 days to fever and about 14 days to the appearance of the rash. So option b is correct.

Step 3: The very short 18 to 72 hour range fits illnesses like influenza or cholera, while 3 to 4 days is too brief and 20 to 25 days is too long for measles. These do not match measles.

Step 4: Hence the incubation period of measles is 10 to 14 days.

Quick Tip: Think roughly 10 days to fever and 14 days to the rash.

92. In the National AIDS Control Programme, for the syndromic management of STDs, the blue colored kit is used for the treatment of which condition?

- (A) Urethral discharge
- (B) Scrotal swelling
- (C) Genital ulcers
- (D) Ano-rectal discharge

Correct Answer: (C) Genital ulcers

Solution:

Step 1: NACO uses colour-coded pre-packed kits for the syndromic management of Sexually Transmitted Infections and Reproductive Tract Infections (STI/RTI). Each colour targets a specific clinical syndrome so that a peripheral health worker can treat without laboratory confirmation.

Step 2: The Blue kit (Kit 2) is used for genital ulcer disease of the non-herpetic type, mainly syphilis and chancroid. It contains injection Benzathine penicillin plus tablet Azithromycin.

Step 3: This makes option c, Genital ulcers, the correct answer.

Step 4: The other syndromes use different kits. Grey kit (Kit 1) is for urethral discharge, cervicitis and ano-rectal discharge. Scrotal swelling and inguinal bubo have their own kits. Hence urethral discharge, scrotal swelling and ano-rectal discharge are not the blue kit indications.

Quick Tip: NACO colour kits: blue equals genital ulcer disease, treated with benzathine penicillin and azithromycin.



93. What is the recommended dose of equine rabies immunoglobulin for post-exposure prophylaxis?

- (A) 10 IU/kg
- (B) 20 IU/kg
- (C) 30 IU/kg
- (D) 40 IU/kg

Correct Answer: (B) 20 IU/kg

Solution:

Step 1: Rabies immunoglobulin (RIG) gives passive immunity and is infiltrated into and around wounds in category III exposures, buying time until the active vaccine generates antibodies.

Step 2: Two preparations exist. Equine RIG (purified horse serum) is given at a dose of **20 IU/kg** body weight, while human RIG is given at **20 IU/kg** as well. The question specifies rabies immunoglobulin at 20 IU/kg, matching option b.

Step 3: Note that F(ab)₂ fragment products are dosed at **40 IU/kg**, which is why 40 IU/kg is a tempting distractor but refers to a different product.

Step 4: Therefore the correct dose is 20 IU/kg, option b.

Quick Tip: Equine RIG is 20 IU/kg; F(ab)₂ products are 40 IU/kg.

94. Endemic typhus is transmitted by which vector?

- (A) Louse
- (B) Flea
- (C) Tick
- (D) Mite

Correct Answer: (B) Flea

Solution:

Step 1: The typhus group of rickettsial diseases is split by vector. Knowing the vector of each form lets you separate endemic from epidemic typhus.

Step 2: Endemic (murine) typhus is caused by *Rickettsia typhi* and is transmitted by the rat flea (*Xenopsylla cheopis*). This matches option b, Flea.

Step 3: The distractors map to other diseases. Epidemic typhus (*Rickettsia prowazekii*) is spread by the body louse, tick typhus or spotted fever by ticks, and scrub typhus (*Orientia tsutsugamushi*) by the trombiculid mite.

Step 4: Therefore the vector of endemic typhus is the flea, option b.

Quick Tip: Endemic (murine) typhus equals *Rickettsia typhi*, spread by the rat flea.

95. Among the following states, the least neonatal mortality rate is seen in which state?

- (A) Delhi
- (B) Tamil Nadu
- (C) Karnataka
- (D) Maharashtra

Correct Answer: (B) Tamil Nadu

Solution:

Step 1: Neonatal mortality rate (NMR) is the number of deaths under 28 days of age per 1000 live births, a sensitive marker of perinatal and newborn care quality.

Step 2: Across India the lowest NMR is recorded in Kerala. Kerala is not in the option list, so we choose the best among those given.

Step 3: Among Delhi, Tamil Nadu, Karnataka and Maharashtra, Tamil Nadu has the lowest neonatal mortality, reflecting its strong maternal and child health services. This makes option b correct.

Step 4: The remaining states, though performing well, report higher NMR than Tamil Nadu, so they are not the answer.

Quick Tip: Kerala has lowest NMR overall; among the options Tamil Nadu is lowest.

96. What is the recommended level of fluoride in drinking water?

- (A) 0.2 to 0.5 mg/L
- (B) 0.5 to 0.8 mg/L
- (C) 0.8 to 1.2 mg/L
- (D) 1.2 to 2.0 mg/L

Correct Answer: (B) 0.5 to 0.8 mg/L

Solution:

Step 1: Fluoride in drinking water needs a narrow balance. Too little leads to dental caries, while too much causes dental and skeletal fluorosis.

Step 2: As per Park, the level of fluoride recommended in drinking water in India is **0.5 to 0.8 mg/L**. This corresponds to option b.

Step 3: The maximum permissible upper limit is **1.5 mg/L**; beyond this, fluorosis risk rises sharply, which is why higher ranges in the options are incorrect.

Step 4: Therefore the recommended range is 0.5 to 0.8 mg/L, option b.

Quick Tip: Recommended fluoride is 0.5 to 0.8 mg/L; maximum permissible is 1.5 mg/L.

97. Which of the following is true about the Indian reference female?

- (A) Height 161 cm
- (B) Weight 60 kg
- (C) BMI 22
- (D) None of the above

Correct Answer: (A) Height 161 cm

Solution:

Step 1: Nutritional energy requirements in India are defined for a standard reference man and reference woman, against whom adjustments are made for individuals who deviate from the standard.

Step 2: The Indian reference woman is aged 18 to 29 years, weighs **55 kg**, has a height of **1.61 m** (161 cm) and a BMI of about **21.2**.

Step 3: Comparing with the options, height 161 cm matches exactly, so option a is correct. Weight 60 kg belongs to the reference man, not the woman. BMI is 21.2, not 22.

Step 4: Hence the only true statement is height 161 cm, option a.

Quick Tip: Indian reference woman: 55 kg, 161 cm, BMI 21.2 (60 kg is the reference man).



98. Which disease is associated with an excessive maize diet?

- (A) Wernicke's encephalopathy
- (B) Pellagra
- (C) Beri-Beri
- (D) Scurvy

Correct Answer: (B) Pellagra

Solution:

Step 1: Pellagra is a niacin (vitamin B3) deficiency disorder, classically remembered by the 4 D's: dermatitis, diarrhoea, dementia and death.

Step 2: Maize is poor in available niacin and, more importantly, is rich in the amino acid leucine. Excess leucine interferes with the conversion of tryptophan to niacin, aggravating the pellagrigenic effect of a maize-based diet.

Step 3: This explains why pellagra is reported in maize and jowar (sorghum) eating populations, such as the Telangana region. Hence option b is correct.

Step 4: The distractors relate to other vitamins. Wernicke's encephalopathy and beri-beri are thiamine (B1) deficiencies, and scurvy is vitamin C deficiency, so none links to maize.

Quick Tip: Maize is high in leucine which blocks tryptophan to niacin, causing pellagra.



99. Which virus is NOT sensitive to disinfection of water by chlorination?

- (A) Rotavirus
- (B) Norwalk virus
- (C) Poliovirus
- (D) None

Correct Answer: (C) Poliovirus

Solution:

Step 1: Chlorination is one of the most important advances in water purification. It reliably kills most pathogenic bacteria, but in the usual doses it does NOT inactivate bacterial spores, cysts of *Entamoeba histolytica*, and certain resistant viruses.

Step 2: The classically chlorine-resistant viruses are poliovirus and the viral hepatitis (hepatitis A and E) viruses. These survive routine chlorination because of their compact, stable capsid structure. Hence poliovirus is the correct answer.

Step 3: Mechanism: when chlorine is added to water it forms hypochlorous acid, which ionises into hydrogen and hypochlorite ions. The disinfecting action is mainly due to hypochlorous acid, which is about 70 to 80 times more effective than the hypochlorite ion.

Step 4: Rotavirus and Norwalk virus are comparatively more susceptible than the enteroviruses, so among the options poliovirus is the one resistant to chlorination.

Quick Tip: Think of the non-enveloped enteric viruses that survive routine chlorine,

the same group as hepatitis A and E.



100. The most common cause of blindness in India is?

- (A) Cataract
- (B) Trachoma
- (C) Refractive errors
- (D) Vitamin A deficiency

Correct Answer: (A) Cataract

Solution:

Step 1: Blindness in India is graded by national surveys, and cataract is consistently the leading cause, accounting for roughly 60 to 70 percent of all blindness in the country.

Step 2: Cataract leads because it is an age-related lens opacity that is highly prevalent in an ageing population and accumulates as a backlog where surgical coverage is incomplete.

Step 3: Refractive error is the second most common cause and is the leading cause of avoidable visual impairment, but it ranks below cataract for blindness. Trachoma and vitamin A deficiency are now much less common causes after control programmes, so they are not the answer.

Quick Tip: It is an age-related lens opacity that contributes about two-thirds of blindness in India.



101. The "3 by 5" initiative in the AIDS control programme refers to?

- (A) Providing 3 million people treatment by the end of 2005
- (B) Providing treatment to 3 out of 5 patients
- (C) Reducing the incidence of AIDS by 3% by 2005
- (D) All of the above

Correct Answer: (A) Providing 3 million people treatment by the end of 2005

Solution:

Step 1: On 1st December 2003, WHO and UNAIDS launched the "3 by 5" initiative. The numbers stand for 3 million people and the year 2005.

Step 2: The goal was to provide antiretroviral treatment (ART) to 3 million people living with HIV/AIDS in developing countries by the end of 2005. The ultimate aim was universal access to ART for everyone who needs it.

Step 3: The other options misread the numbers. It was not a 3-out-of-5 patient ratio and not a 3 percent incidence reduction, so option a is correct.

Quick Tip: Decode the two numbers literally: 3 million people, by the year 2005.

102. The most common cause of trigeminal neuralgia is?

- (A) Infection
- (B) Trauma
- (C) Vascular compression
- (D) Iatrogenic

Correct Answer: (C) Vascular compression

Solution:

Step 1: Trigeminal neuralgia (tic douloureux) is characterised by intermittent, severe, shooting pain in the face due to involvement of the trigeminal nerve.

Step 2: In about 95 percent of cases the cause is pressure on the trigeminal nerve close to where it enters the brainstem, just past the Gasserian ganglion. This compression is most often by an aberrant artery or vein, hence vascular compression is the most common cause.

Step 3: Other less frequent causes include tumour, cysts, arteriovenous malformation, and multiple sclerosis. Infection, trauma, and iatrogenic injury are not the usual cause, so option c is correct.

Step 4: First-line medical therapy is carbamazepine; gabapentin and valproate are alternatives.

Quick Tip: Think of a neurovascular conflict at the nerve root entry zone near the brainstem.



103. Potato tumor (rhinophyma) is due to?

- (A) Hypotrophy of sebaceous glands of the nose
- (B) Hypertrophy of sebaceous glands of the nose
- (C) Hypotrophy of sweat glands of the nose
- (D) Hypertrophy of sweat glands of the nose

Correct Answer: (B) Hypertrophy of sebaceous glands of the nose

Solution:

Step 1: Rhinophyma, also called potato tumor, is a large, bulb-shaped, ruddy red nose. It is a slow-growing benign lesion.

Step 2: It results from hypertrophy of the sebaceous glands of the nose, so the correct option is hypertrophy of sebaceous glands. It is regarded as a severe end-stage form of acne rosacea, and the cause is otherwise unknown.

Step 3: The red or pink colour is due to engorgement of the superficial vessels. It

mainly affects men past middle age. The old belief that it is caused by heavy alcohol intake is incorrect; it occurs equally in non-drinkers.

Step 4: Treatment is surgical reshaping using carbon dioxide laser, scalpel, or dermabrasion, allowing the area to re-epithelialise; sometimes the tumour is excised and skin-grafted. The sweat-gland options are therefore wrong.

Quick Tip: The bulbous red nose of end-stage rosacea is an overgrowth of oil-producing glands.

104. Acute suppurative otitis media in children is most commonly caused by?

- (A) Streptococcus pneumoniae
- (B) Staphylococcus epidermidis
- (C) Staphylococcus aureus
- (D) Pseudomonas

Correct Answer: (A) Streptococcus pneumoniae

Solution:

Step 1: Acute suppurative otitis media (ASOM) is an acute bacterial infection of the middle ear cleft, especially common in infants and children.

Step 2: It usually follows an upper respiratory tract infection such as acute tonsillitis, common cold, or influenza, and the organisms ascend through the eustachian tube into the middle ear.

Step 3: The single most common causative organism is Streptococcus pneumoniae, followed by Haemophilus influenzae and Moraxella catarrhalis. Hence option a is correct.

Step 4: Staphylococcus epidermidis, Staphylococcus aureus, and Pseudomonas are not the usual primary cause of ASOM in children, so they are incorrect.

Quick Tip: The leading middle-ear pathogen in children, ahead of *H. influenzae* and *M. catarrhalis*.

105. The length of the external auditory meatus is?

- (A) 12 mm
- (B) 16 mm
- (C) 20 mm
- (D) 24 mm

Correct Answer: (D) 24 mm

Solution:

Step 1: The external auditory canal (external acoustic meatus) is an S-shaped canal extending from the concha to the tympanic membrane, with a length of about 24 to 25 mm. Hence the answer is 24 mm.

Step 2: It is divided into two parts. The outer or lateral one-third is the cartilaginous part, about 8 mm long, lined by thick skin with hair, ceruminous glands, and pilosebaceous glands; furuncles occur only here.

Step 3: The inner or medial two-thirds is the bony part, about 16 mm long, lined by thin skin without hair or ceruminous glands. The isthmus, the narrowest part, lies about 5 mm lateral to the tympanic membrane.

Step 4: Adding the cartilaginous 8 mm and bony 16 mm gives roughly 24 mm, so the shorter options are incorrect.

Quick Tip: Add the cartilaginous outer third (about 8 mm) to the bony inner two-thirds (about 16 mm).

106. When a patient fails to understand normal speech but can understand shouted or amplified speech, the hearing loss is termed?

- (A) Mild hearing loss
- (B) Moderate hearing loss
- (C) Severe hearing loss
- (D) Profound hearing loss

Correct Answer: (C) Severe hearing loss

Solution:

Step 1: Hearing loss is graded by the average hearing threshold (the quietest sound the better ear can detect, measured in decibels).

Step 2: In severe hearing loss the thresholds in the better ear lie roughly between 70 and 95 dB. At this level normal conversational speech (around 50 to 60 dB) is inaudible, so the patient cannot follow ordinary speech but can still respond to shouted or amplified sound.

Step 3: Such patients benefit from powerful hearing aids but rely heavily on lip-reading, and some use sign language. This matches the clinical picture described, so the answer is severe hearing loss.

Step 4: Mild (about 26 to 40 dB) and moderate (about 41 to 70 dB) losses still allow some understanding of normal speech, while profound loss (greater than 95 dB) leaves the patient unable to hear even shouted speech. Hence those options do not fit.

Quick Tip: Threshold of 70 to 95 dB in the better ear; normal speech missed but loud speech heard.

107. The most common bacterial cause of rhinitis is?

- (A) *Haemophilus influenzae*
- (B) *Streptococcus haemolyticus*
- (C) *Pasteurella multocida*
- (D) *Corynebacterium diphtheriae*

Correct Answer: (A) *Haemophilus influenzae*

Solution:

Step 1: Most rhinitis is viral in origin (viral rhinitis or the common cold). The question asks specifically for the most common bacterial cause, so we focus on the bacterial flora that cause acute bacterial rhinitis or its complications.

Step 2: Acute bacterial rhinitis is seen mainly in children, and in adults after nasal trauma, a viral upper respiratory infection, or surgery. Its clinical picture can be identical to the common cold.

Step 3: The common bacterial organisms are *Streptococcus pneumoniae*, *Haemophilus influenzae* and *Moraxella catarrhalis*. Among the listed options, *Haemophilus influenzae* is the recognised answer.

Step 4: *Streptococcus haemolyticus*, *Pasteurella multocida* (animal bite flora) and *Corynebacterium diphtheriae* are not typical causes of ordinary bacterial rhinitis, so they are excluded.

Quick Tip: Think of the respiratory trio that also causes otitis media and sinusitis.

108. Turban epiglottitis is seen in?

- (A) TB
- (B) Leprosy
- (C) Laryngeal papilloma
- (D) Epiglottitis

Correct Answer: (A) TB

Solution:

Step 1: Turban epiglottis refers to a diffuse, pale, pseudoedematous swelling of the epiglottis. It is a classic sign of tubercular laryngitis.

Step 2: Tubercular laryngitis is almost always secondary to pulmonary tuberculosis and usually affects the posterior third of the larynx. The structures involved in descending order are the interarytenoid fold, ventricular bands, vocal cords and epiglottis.

Step 3: Characteristic findings include hyperaemia of the cords, a mammilated interarytenoid swelling, mouse-nibbled ulceration of the cords, marked pallor of surrounding mucosa, and the pseudoedema of the epiglottis called turban epiglottis. Early symptoms are weakness of voice with periods of aphonia, followed by hoarseness, cough, odynophagia and referred otalgia.

Step 4: Leprosy, laryngeal papilloma and acute epiglottitis do not produce this specific pale turban-like epiglottic swelling, so TB is the answer.

Quick Tip: A pale pseudoedematous epiglottis in chronic posterior laryngitis.

109. Amsler's grid is used to evaluate?

- (A) Central 10 degrees of vision
- (B) Central 20 degrees of vision
- (C) Peripheral vision
- (D) Lens opacity

Correct Answer: (A) Central 10 degrees of vision

Solution:

Step 1: The Amsler grid is a square chart of fine lines with a central fixation dot, used to test the function of the macula, which subtends the central 10 degrees of the visual field.

Step 2: The patient fixes on the central dot and reports any distortion. Bent, wavy or missing lines indicate metamorphopsia, in which objects appear to have an altered or irregular contour.

Step 3: Because it maps the central 10 degrees, the grid detects and follows macular disease such as central serous chorioretinopathy, age-related macular degeneration, diabetic macular oedema and macular hole. Hence the answer is central 10 degrees of vision.

Step 4: Peripheral vision is assessed by perimetry, and lens opacity by slit-lamp examination, so those options are incorrect.

Quick Tip: A wavy-line chart that tests the macula.

110. Munson's sign is a feature of?

- (A) Keratoconus
- (B) Corneal ulcer
- (C) Pterygium
- (D) Posterior staphyloma

Correct Answer: (A) Keratoconus

Solution:

Step 1: Keratoconus is a progressive, non-inflammatory, bilateral ectatic disease in which the central or paracentral cornea thins and bulges forward into a cone, with the apex slightly below the centre.

Step 2: Munson's sign is the V-shaped indentation or bulging of the lower eyelid

produced by the conical cornea when the patient looks downward. It is a classic clinical sign of keratoconus.

Step 3: Other characteristic findings of keratoconus include a distorted corneal (window) reflex, the Fleischer ring of iron at the cone base, the oil-drop reflex and a scissoring retinoscopic reflex. The condition starts at puberty, causes progressive irregular astigmatism and myopia, and gives glare and halos.

Step 4: Corneal ulcer, pterygium and posterior staphyloma do not produce the conical lower-lid indentation, so keratoconus is the answer.

Quick Tip: Lower-lid bulges into a V on downgaze due to a cone-shaped cornea.

111. The gland of Moll opens in or on the?

- (A) Skin
- (B) Hair follicle
- (C) Tarsal plate
- (D) Ducts of Meibomian glands

Correct Answer: (B) Hair follicle

Solution:

Step 1: The glands of Moll are modified, apocrine sweat glands situated near the roots of the eyelashes at the lid margin.

Step 2: They do not open directly onto the skin surface. Instead they open into the hair follicles of the eyelashes, or sometimes into the ducts of the adjacent glands of Zeis.

Step 3: This distinguishes them from the Meibomian (tarsal) glands, which are modified sebaceous glands in the tarsal plate opening at the lid margin, and from the glands of Zeis, which are sebaceous glands opening into the lash follicles. So

the correct answer is the hair follicle.

Step 4: They open neither directly onto the skin nor into the tarsal plate nor into the Meibomian ducts, so those options are wrong.

Quick Tip: Modified sweat gland near the lash roots, draining into the lash follicle.

112. Diabetic ischemic maculopathy is characterized by all of the following except?

- (A) It occurs due to microvascular blockage
- (B) Mild visual loss
- (C) Areas of non-perfusion are evident on fluorescein angiography
- (D) Microaneurysms and hemorrhages are seen

Correct Answer: (B) Mild visual loss

Solution:

Step 1: Ischaemic diabetic maculopathy results from microvascular blockage at the macula, so retinal capillaries that should perfuse the fovea are occluded.

Step 2: Clinically it causes marked, not mild, visual loss because the central retina is deprived of blood. It is accompanied by microaneurysms, haemorrhages, only mild or no macular oedema, and a few hard exudates.

Step 3: Fluorescein angiography is diagnostic and shows areas of non-perfusion, beginning as enlargement of the foveal avascular zone, then capillary dropout, and in advanced cases blocked precapillary arterioles.

Step 4: The question asks for the false statement. Microvascular blockage, non-perfusion on angiography, and microaneurysms with haemorrhages are all true. The visual loss is marked, not mild, so mild visual loss is the exception and the correct answer.

Quick Tip: Foveal ischaemia kills central acuity, so visual loss is marked, not mild.

113. Bilateral inferior dislocation of the lens (ectopia lentis) is characteristically seen in which condition?

- (A) Marfan's syndrome
- (B) Homocystinuria
- (C) Weil-Marchesani syndrome
- (D) Trauma

Correct Answer: (B) Homocystinuria

Solution:

Step 1: Ectopia lentis means displacement of the crystalline lens from its normal position. When the lens lies completely outside the patellar fossa it is luxated (dislocated); when partially displaced but still within the lens space it is subluxated. The direction of displacement points to the cause.

Step 2: In homocystinuria the lens typically dislocates downward and inward (inferonasal), and the involvement is classically bilateral and symmetric. This makes bilateral inferior dislocation the hallmark of homocystinuria.

Step 3: In Marfan's syndrome the lens dislocates upward and outward (superotemporal), the opposite direction, so option a is wrong. In Weil-Marchesani syndrome the lens is small and spherical (microspherophakia) and usually dislocates anteriorly, so option c is wrong. Trauma causes unilateral, irregular displacement rather than a symmetric bilateral inferior pattern, so option d is wrong.

Step 4: Hence the answer is homocystinuria.

Quick Tip: Direction matters: down and in is metabolic, up and out is connective

tissue.



114. All of the following are clinical features of Leber's hereditary optic neuropathy EXCEPT:

- (A) Seen in the 2nd or 3rd decade of life
- (B) It is an example of gradual painless visual loss
- (C) Males can transmit the disease
- (D) No leak of dye is observed on fluorescein angiography

Correct Answer: (C) Males can transmit the disease

Solution:

Step 1: Leber's hereditary optic neuropathy (LHON) is caused by a point mutation in mitochondrial DNA. Mitochondrial DNA is inherited exclusively from the mother, so transmission follows a maternal pattern.

Step 2: Because affected and carrier males pass on only their nuclear DNA (not mitochondria), males CANNOT transmit the disease. The disease is transmitted only through carrier females. So option c is the false statement and is the answer.

Step 3: The other options are true. LHON presents in young adults (about 11-30 years, i.e. 2nd-3rd decade), so option a is correct. It causes painless, subacute, sequential visual failure, which fits gradual painless loss, so option b is correct. On fundus examination there is disc swelling and peripapillary telangiectasia, but characteristically there is NO leakage of dye on fluorescein angiography, so option d is correct.

Step 4: Therefore the exception is that males can transmit the disease.

Quick Tip: Mitochondrial DNA comes only from mother, so fathers never pass it on.



115. Retinoblastoma occurs bilaterally in what percentage of cases?

- (A) 10 - 15%
- (B) 15 - 20%
- (C) 20 - 25%
- (D) 25 - 30%

Correct Answer: (D) 25 - 30%

Solution:

Step 1: Retinoblastoma is the most common intraocular malignant tumour of childhood, usually affecting children aged 1-2 years, with no sex predilection.

Step 2: It is unilateral in about 70-75% of cases and bilateral in about 25-30% of cases. So the correct figure is 25-30%, which is option d.

Step 3: Bilateral disease is almost always hereditary. It follows Knudson's two-hit hypothesis with the RB tumour suppressor gene on chromosome 13 (13q14); the first hit is inherited in the germline and the second occurs postnatally. Sporadic unilateral disease requires both hits to occur postnatally.

Step 4: Hence the bilateral incidence is 25-30%.

Quick Tip: Unilateral is about 70-75%, so bilateral makes up the rest.

116. Bett's classification deals with which of the following?

- (A) Ocular trauma
- (B) Ocular foreign body
- (C) Squint
- (D) Maculopathy

Correct Answer: (A) Ocular trauma

Solution:

Step 1: BETTS stands for the Birmingham Eye Trauma Terminology System. As the name indicates, it is a standardized terminology for classifying eye injuries.

Step 2: It organizes ocular trauma into a logical framework (for example open-globe versus closed-globe injury, and subtypes such as rupture, penetrating, perforating, and intraocular foreign body injuries). Therefore Bett's classification deals with ocular trauma, which is option a.

Step 3: The other options refer to unrelated entities. Foreign body type is only a subcategory within trauma, not the overall scheme, so option b is too narrow. Squint and maculopathy have entirely separate classifications and are unrelated to BETTS, so options c and d are wrong.

Step 4: Hence Bett's classification is for ocular trauma.

Quick Tip: BETTS - Birmingham Eye TRAUMA Terminology System.

117. Which metabolic derangement is normally seen in pregnancy?

- (A) Metabolic acidosis
- (B) Metabolic alkalosis
- (C) Respiratory acidosis
- (D) Respiratory alkalosis

Correct Answer: (D) Respiratory alkalosis

Solution:

Step 1: During pregnancy, progesterone stimulates the respiratory centre, causing increased ventilation (hyperventilation) from early in gestation.

Step 2: Hyperventilation blows off carbon dioxide, lowering arterial pCO₂. A fall

in $p\text{CO}_2$ raises blood pH, producing a respiratory alkalosis. So the answer is option d.

Step 3: This is a chronic respiratory alkalosis, which the kidneys compensate for by excreting bicarbonate, keeping the pH near normal. The disturbance is respiratory (driven by the lungs/ CO_2), not metabolic, so options a and b are wrong. Because CO_2 is lowered rather than retained, there is no respiratory acidosis, so option c is wrong.

Step 4: Hence the normal acid-base picture of pregnancy is a compensated respiratory alkalosis.

Quick Tip: Progesterone drives hyperventilation, so CO_2 falls.

118. High steppage gait is seen in which of the following?

- (A) Foot drop
- (B) Frontal lobe stroke
- (C) Tabes dorsalis
- (D) Leprosy

Correct Answer: (C) Tabes dorsalis

Solution:

Step 1: A high steppage (foot drop) gait occurs when the patient cannot dorsiflex the foot. To avoid the toes catching the ground, the leg is lifted higher than normal and the forefoot strikes the ground first instead of the heel.

Step 2: This pattern is produced by motor peripheral neuropathies involving the common peroneal nerve. Among the listed options, tabes dorsalis is the classic disease association the source highlights, so the marked answer is option c.

Step 3: It is worth noting that foot drop (option a) is the mechanism rather than

a separate disease, and leprosy (option d) can also cause foot drop through peroneal nerve involvement. A frontal lobe stroke (option b) produces an apraxic/magnetic gait, not a steppage gait, so it is clearly wrong. For this single-best-answer question the intended choice is tabes dorsalis.

Step 4: Hence the answer is tabes dorsalis.

Quick Tip: Cannot dorsiflex, so the leg lifts high and the forefoot lands first.

119. Aphasia which affects the arcuate fibres (arcuate fasciculus) is called:

- (A) Global aphasia
- (B) Anomic aphasia
- (C) Conduction aphasia
- (D) Broca's aphasia

Correct Answer: (C) Conduction aphasia

Solution:

Step 1: The arcuate fasciculus is a bundle of white matter fibres that connects Broca's area (motor speech) to Wernicke's area (language comprehension).

Step 2: Damage to this connecting tract disconnects the two speech centres while leaving them individually intact. This produces conduction aphasia, so the answer is option c.

Step 3: The hallmark of conduction aphasia is markedly impaired repetition, with relatively preserved comprehension and fluent (though paraphasic) speech. The other options localize differently: Broca's aphasia is from frontal (Broca's area) damage with non-fluent speech, global aphasia involves widespread damage to both centres, and anomic aphasia is mainly a naming difficulty, none of which specifically reflects an isolated arcuate fasciculus lesion.

Step 4: Hence an arcuate fasciculus lesion causes conduction aphasia.

Quick Tip: Cutting the cable between Broca and Wernicke wrecks repetition.

120. Interstitial nephritis is common with

- (A) NSAID
- (B) Black water fever
- (C) Rhabdomyolysis
- (D) Tumor lysis syndrome

Correct Answer: (A) NSAID

Solution:

Step 1: Acute interstitial nephritis (AIN) is an immune-mediated, allergic injury of the renal interstitium and tubules. Drugs are by far the commonest cause, accounting for the majority of cases.

Step 2: Among the listed options, NSAIDs are a classic and very common cause of drug-induced interstitial nephritis. Other frequent culprits include beta-lactam antibiotics (penicillins, cephalosporins), sulfonamides, rifampicin, quinolones, allopurinol, diuretics (thiazides, furosemide), proton pump inhibitors like omeprazole, and H2 blockers.

Step 3: Why the other options are wrong - Blackwater fever (severe falciparum malaria), rhabdomyolysis and tumor lysis syndrome all damage the kidney chiefly by pigment or crystal/urate load causing acute tubular necrosis or tubular obstruction, not a primary allergic interstitial nephritis.

Step 4: Hence the answer is NSAID.

Quick Tip: Think of the drug class notorious for allergic tubulointerstitial injury, not pigment-induced ATN.

121. A patient is diagnosed with HIV and tuberculosis together. How should anti-tubercular treatment (ATT) and combination antiretroviral therapy (cART) be started?

- (A) Start ATT first
- (B) Start cART first
- (C) Start both simultaneously
- (D) Start cART only

Correct Answer: (A) Start ATT first

Solution:

Step 1: When HIV and active tuberculosis are diagnosed at the same time, anti-tubercular treatment (ATT) is started first, and antiretroviral therapy is added a little later.

Step 2: The reason is to reduce the risk of Immune Reconstitution Inflammatory Syndrome (IRIS). If cART is begun first, the recovering CD4 count mounts an exaggerated inflammatory response against the existing TB infection, paradoxically worsening symptoms.

Step 3: Starting ATT roughly 2 weeks before cART has been shown to lower the incidence of IRIS while still controlling the mycobacterial load. cART is not delayed indefinitely; it is initiated within the first few weeks based on the CD4 count.

Step 4: Therefore the correct sequence is to start ATT first.

Quick Tip: Control the mycobacteria before reviving the immune system to avoid IRIS.

122. All of the following are true about Rheumatoid arthritis, except

- (A) PIP and DIP joints involved equally
- (B) Pathology limited to articular cartilage
- (C) Women are affected 3 times more commonly than men
- (D) 20% of patients have extra articular manifestations

Correct Answer: (B) Pathology limited to articular cartilage

Solution:

Step 1: This is a negative-lead question asking which statement is false about rheumatoid arthritis (RA). Note that the printed key marks option (d), but the medically false statement is option (b).

Step 2: RA is a chronic systemic autoimmune disease associated with HLA-DR4. The pathology is NOT limited to the articular cartilage - it begins in the synovium with pannus formation and progressively destroys cartilage, subchondral bone, ligaments and tendons. So option (b) is false.

Step 3: Checking the true statements - RA characteristically spares the DIP joints and involves MCP and PIP joints; women are affected about 3 times more than men; and extra-articular features (nodules, vasculitis, lung, eye involvement) do occur in a notable fraction of patients. The claim in option (a) that PIP and DIP are involved equally is actually wrong too, but the single best false answer chosen here is the pathology being limited to cartilage.

Step 4: Therefore the exception (false statement) is that pathology is limited to articular cartilage.

Quick Tip: RA destroys far more than cartilage; the synovial pannus erodes bone and soft tissue too.

123. The finger-in-glove sign is seen in

- (A) Pulmonary alveolar proteinosis
- (B) Pneumocystis carinii
- (C) Tuberculosis
- (D) Bronchocele

Correct Answer: (D) Bronchocele

Solution:

Step 1: The finger-in-glove sign is a radiological appearance of branching tubular or Y-shaped/V-shaped opacities produced by mucoid impaction within dilated bronchi - that is, a bronchocele (mucoid impaction of bronchi).

Step 2: A bronchocele forms when a bronchus is obstructed or dilated and fills with inspissated mucus. Causes include bronchial obstruction (hamartoma, lipoma, carcinoid, bronchogenic carcinoma, congenital atresia) and non-obstructive conditions such as asthma, allergic bronchopulmonary aspergillosis (ABPA) and cystic fibrosis.

Step 3: Why the other options are wrong - Pulmonary alveolar proteinosis gives a crazy-paving pattern; Pneumocystis pneumonia gives bilateral perihilar ground-glass opacities; tuberculosis shows cavitation and upper-lobe fibronodular disease. None of these produces the gloved-finger mucoid impaction shadow.

Step 4: Hence the finger-in-glove sign indicates a bronchocele.

Quick Tip: Branching mucus-filled dilated bronchi resembling gloved fingers - think mucoid impaction.

124. The vitamin B12 level in chronic myeloid leukemia is

- (A) Elevated
- (B) Decreased
- (C) Normal
- (D) Markedly decreased

Correct Answer: (A) Elevated

Solution:

Step 1: In chronic myeloid leukemia (CML) there is a huge proliferation of granulocytes and their precursors. These cells produce transcobalamin I and III (vitamin B12 binding proteins).

Step 2: The raised levels of these binding proteins increase the total serum vitamin B12 carrying capacity, so the measured serum vitamin B12 level is elevated.

Step 3: Other characteristic laboratory findings in CML include a raised lactate dehydrogenase (LDH) and a low leukocyte alkaline phosphatase (LAP/ALP) score, which helps distinguish CML from a leukemoid reaction.

Step 4: Hence the vitamin B12 level in CML is elevated.

Quick Tip: Excess granulocytes raise B12-binding transcobalamins, pushing serum B12 up.

125. Brock's syndrome is

- (A) Bronchiectasis sicca
- (B) Middle lobe bronchiectasis
- (C) Kartagener's syndrome
- (D) Sarcoidosis

Correct Answer: (B) Middle lobe bronchiectasis

Solution:

Step 1: Brock's syndrome, also called middle lobe syndrome, is right middle lobe bronchiectasis or recurrent atelectasis occurring as a late sequel of primary tuberculosis.

Step 2: It is believed to result from external pressure of enlarged hilar lymph nodes (from primary TB in childhood) on the right middle lobe bronchus. The narrow, long middle lobe bronchus is especially vulnerable to such compression.

Step 3: Repeated episodes of atelectasis and fibrosis lead to bronchiectasis of the middle lobe. The term is also used for recurrent middle lobe atelectasis even without overt endobronchial obstruction.

Step 4: Why the others are wrong - bronchiectasis sicca is dry bronchiectasis seen in upper lobes, Kartagener's syndrome is situs inversus with bronchiectasis and sinusitis, and sarcoidosis is a granulomatous disease. Hence Brock's syndrome equals middle lobe bronchiectasis.

Quick Tip: Named the middle lobe syndrome - hilar TB nodes compress the right middle lobe bronchus.

126. Which of the following is given to treat thrombocytopenia secondary to anti-cancer therapy and is known to stimulate progenitor megakaryocytes?

- (A) Filgrastim
- (B) Oprelvekin
- (C) Erythropoietin
- (D) Anagrelide

Correct Answer: (B) Oprelvekin

Solution:

Step 1: The agent needed must boost platelet production by stimulating megakaryocyte progenitors and is used for chemotherapy-induced thrombocytopenia.

Step 2: Oprelvekin is recombinant human interleukin-11 (IL-11). It stimulates the proliferation of megakaryocyte progenitors and megakaryocyte maturation, thereby raising the platelet count, and is approved to prevent and treat thrombocytopenia after myelosuppressive chemotherapy.

Step 3: Why the others are wrong - Filgrastim is recombinant G-CSF that raises neutrophils, not platelets; erythropoietin stimulates red cell production for anaemia; anagrelide actually lowers the platelet count and is used in thrombocythemia, the opposite of what is needed here.

Step 4: Hence the correct drug is oprelvekin.

Quick Tip: Recombinant IL-11 that boosts megakaryocyte progenitors and platelet output.



127. Which of the following is NOT true regarding obstructive sleep apnea (OSA)?

- (A) Nocturnal asphyxia
- (B) Alcoholism is a cofactor
- (C) Prone to hypertension
- (D) Overnight oximetry is diagnostic and can replace polysomnography

Correct Answer: (D) Overnight oximetry is diagnostic and can replace polysomnography

Solution:

Step 1: Obstructive sleep apnea (OSA) is caused by repeated collapse of the upper airway during sleep, producing apneas and hypopneas. This leads to recurrent episodes of nocturnal asphyxia (oxygen desaturation), so option a is

true.

Step 2: Alcohol relaxes the pharyngeal muscles and worsens airway collapse, so alcoholism is a recognised cofactor (option b true). The repeated hypoxia and sympathetic surges make patients prone to systemic hypertension (option c true).

Step 3: Full-night, attended, in-laboratory polysomnography is the gold-standard diagnostic test for OSA. Overnight oximetry alone is only a screening tool with limited sensitivity and specificity; it cannot replace polysomnography. Hence option d is the statement that is NOT true.

Step 4: Therefore the false statement, and the correct answer, is option d.

Quick Tip: The gold-standard diagnostic test for OSA is full-night polysomnography, not oximetry.



128. The most common acute complication of dialysis is:

- (A) Hypotension
- (B) Bleeding
- (C) Dementia
- (D) Muscle cramps

Correct Answer: (A) Hypotension

Solution:

Step 1: Hemodialysis removes fluid by ultrafiltration. The most common acute (intradialytic) complication is hypotension, especially in patients with diabetes mellitus.

Step 2: The drop in blood pressure occurs because of excessive ultrafiltration with inadequate compensatory vascular refilling, impaired vasoactive or autonomic responses, osmolar shifts, and overzealous use of antihypertensive

drugs.

Step 3: Muscle cramps and bleeding can occur with dialysis but are less frequent than hypotension. Dementia (dialysis dementia, an aluminium-related encephalopathy) is a chronic, now rare complication, not an acute one.

Step 4: Hence the single most common acute complication is hypotension, option a.

Quick Tip: Think of the immediate drop in blood pressure caused by rapid fluid removal during a dialysis session.

129. All of the following statements about the genetics of G6PD deficiency are true, EXCEPT:

- (A) X-linked inheritance
- (B) More severe in men
- (C) Contradicts the Lyon hypothesis
- (D) May affect heterozygous females

Correct Answer: (C) Contradicts the Lyon hypothesis

Solution:

Step 1: The G6PD gene is located on the X chromosome (Xq28), so the disorder shows X-linked recessive inheritance (option a true). Males have a single X chromosome, so a variant gene is fully expressed and the disease is more severe in men (option b true).

Step 2: According to the Lyon hypothesis, one of the two X chromosomes in each cell of a female is randomly inactivated. A heterozygous female therefore has two red-cell populations, one normal and one G6PD deficient. Depending on the degree of lyonization, she may show normal, moderately reduced, or grossly deficient enzyme activity. Thus heterozygous females CAN be affected (option d

true).

Step 3: This mosaic pattern in heterozygous females is exactly what the Lyon hypothesis predicts. G6PD deficiency therefore SUPPORTS the Lyon hypothesis, it does not contradict it.

Step 4: The false statement, and hence the answer, is option c.

Quick Tip: Variable enzyme levels in carrier females actually prove, not disprove, random X-inactivation.



130. Asbestosis causes all of the following EXCEPT:

- (A) Shaggy heart borders
- (B) Honeycombing
- (C) Hilar lymphadenopathy
- (D) Basal peribronchial fibrosis

Correct Answer: (C) Hilar lymphadenopathy

Solution:

Step 1: Asbestosis is a diffuse interstitial pulmonary fibrosis that characteristically affects the lower lobes of the lung. The basal fibrosis gives basal peribronchial fibrosis (option d true) and, in advanced disease, honeycombing (option b true).

Step 2: Pleural plaques and basal fibrosis blur the cardiac and diaphragmatic outlines, producing the classic shaggy heart border appearance (option a true).

Step 3: Hilar lymphadenopathy is NOT a feature of asbestosis. It is more typical of sarcoidosis and silicosis (the latter giving eggshell calcification of hilar nodes). Asbestosis spares the hila.

Step 4: Hence the feature NOT caused by asbestosis is hilar lymphadenopathy, option c.

Quick Tip: Asbestosis is a basal interstitial fibrosis; enlarged hilar nodes point instead to sarcoidosis or silicosis.



131. Gout can be precipitated by all of the following EXCEPT:

- (A) Thiazides
- (B) Furosemide
- (C) Cyclosporine
- (D) High dose salicylates

Correct Answer: (D) High dose salicylates

Solution:

Step 1: Gout is precipitated by drugs that raise serum uric acid (hyperuricemia). Diuretics such as thiazides and loop diuretics (furosemide) reduce renal urate excretion and cause hyperuricemia, so options a and b can precipitate gout.

Step 2: Cyclosporine (and tacrolimus) decrease renal urate clearance and are a well known cause of drug-induced hyperuricemia, so option c can also precipitate gout.

Step 3: Salicylates have a dose-dependent effect on urate. Low doses (0.3 to 3.0 g/day) cause urate retention and hyperuricemia, but HIGH doses (greater than 3.0 g/day) are uricosuric and actually lower serum uric acid. Therefore high-dose salicylates do NOT precipitate gout.

Step 4: The exception, and the correct answer, is high dose salicylates, option d.

Quick Tip: Salicylates are biphasic: low doses retain urate, but high doses are



132. Rupture of a berry aneurysm most commonly results in:

- (A) Subarachnoid hemorrhage
- (B) Subdural hemorrhage
- (C) Extradural hemorrhage
- (D) Intra-parenchymal hemorrhage

Correct Answer: (A) Subarachnoid hemorrhage

Solution:

Step 1: Berry (saccular) aneurysms arise at the branch points of the circle of Willis, which lie in the subarachnoid space at the base of the brain.

Step 2: When such an aneurysm ruptures, blood escapes directly into the subarachnoid space, producing subarachnoid hemorrhage. Rupture of a berry aneurysm is in fact the most common cause of spontaneous (non-traumatic) subarachnoid hemorrhage.

Step 3: Subdural hemorrhage results from torn bridging veins and extradural hemorrhage from a torn middle meningeal artery, both usually traumatic. Intraparenchymal hemorrhage is typically due to hypertension. None of these is the usual result of a ruptured berry aneurysm.

Step 4: Therefore the correct answer is subarachnoid hemorrhage, option a.

Quick Tip: Berry aneurysms sit in the subarachnoid space at the circle of Willis.



133. Round pneumonia is seen with:

- (A) Streptococcal pneumonia
- (B) Kerosene oil aspiration
- (C) Lung cancer
- (D) Mendelson syndrome

Correct Answer: (A) Streptococcal pneumonia

Solution:

Step 1: Round (spherical) pneumonia is a focal consolidation that appears rounded on imaging. It is seen mainly in children because the collateral channels of air drift (pores of Kohn and canals of Lambert) are poorly developed, so infection stays localised and spreads spherically.

Step 2: The most common organism responsible for round pneumonia is *Streptococcus pneumoniae* (pneumococcus). Hence option a is correct.

Step 3: Kerosene aspiration causes hydrocarbon pneumonitis (usually basal), Mendelson syndrome is chemical aspiration pneumonitis of gastric acid, and lung cancer gives a mass lesion, not infective round pneumonia. The clinical importance of round pneumonia is that it can mimic a tumour mass and must be differentiated from it.

Step 4: Therefore round pneumonia is associated with streptococcal (pneumococcal) pneumonia, option a.

Quick Tip: This spherical consolidation in children is most often caused by the pneumococcus.

134. HIV RNA detection by PCR (viral load assay) can reliably detect as low as how many copies of viral RNA per millilitre of blood?

- (A) 30 copies viral RNA/ml of blood
- (B) 40 copies viral RNA/ml of blood
- (C) 50 copies of viral RNA/ml of blood
- (D) 60 copies of viral RNA/ml of blood

Correct Answer: (B) 40 copies viral RNA/ml of blood

Solution:

Step 1: The HIV viral load assay uses RT-PCR to quantify HIV RNA, reporting the result as the number of copies of HIV RNA per millilitre of serum or plasma. It is used to monitor disease progression and the response to antiretroviral therapy.

Step 2: Standard commercial viral load assays can reliably detect as few as 40 copies of HIV RNA per millilitre of plasma. Hence the correct answer is option b (40 copies/ml).

Step 3: The other figures (30, 50, 60 copies) are not the validated lower limit of standard assays. Note that specialised research based assays can push detection down to a single copy per millilitre, but that is not the routine clinical cut-off being asked here.

Quick Tip: Routine clinical viral load assays bottom out at this two-digit copy number.

135. What is the most common cause of complete heart block (acquired complete AV block) in infants and children?

- (A) SLE
- (B) Surgery for congenital heart disease
- (C) Viral myocarditis
- (D) Rheumatic fever

Correct Answer: (B) Surgery for congenital heart disease

Solution:

Step 1: In children the most common cause of permanent acquired complete atrioventricular (AV) block is surgery for congenital heart disease. So the answer is option b.

Step 2: Post-surgical complete AV block results from trauma to the AV node during the operation, through haemorrhage, ischaemia, necrosis, inflammation or direct traumatic disruption of conducting tissue.

Step 3: The second most common cause is congenital heart disease itself associated with congenital complete AV block (for example in maternal SLE, but that is a congenital, not acquired, mechanism).

Step 4: Viral myocarditis and acute rheumatic fever can both cause AV block, but these are usually reversible and far less common than surgical injury, so they are not the single most common cause.

Quick Tip: Think of trauma to the AV node during cardiac repair in children.

136. A banana shaped left ventricle on imaging is characteristically seen in which condition?

- (A) HOCM (hypertrophic obstructive cardiomyopathy)
- (B) DCM (dilated cardiomyopathy)
- (C) RCM (restrictive cardiomyopathy)
- (D) Takotsubo cardiomyopathy

Correct Answer: (A) HOCM (hypertrophic obstructive cardiomyopathy)

Solution:

Step 1: The shape of the left ventricular cavity gives a clue to the underlying cardiomyopathy. A banana shaped left ventricle is seen in hypertrophic

cardiomyopathy (HOCM), where asymmetrical septal hypertrophy narrows and deforms the cavity into a curved, banana-like outline. So the answer is option a.

Step 2: Contrast this with the other patterns: a spherical (globular) ventricle is typical of dilated cardiomyopathy.

Step 3: Apical ballooning of the ventricle is the hallmark of Takotsubo (stress) cardiomyopathy, while a spade-shaped ventricle is seen in apical hypertrophic cardiomyopathy.

Step 4: Distortion or aneurysmal bulging of the ventricle follows myocardial infarction and adverse remodelling. None of these gives the banana shape, confirming HOCM as the correct choice.

Quick Tip: Asymmetric septal thickening curves the cavity into a banana shape.



137. Which of the following is NOT a high pitched heart sound?

- (A) Mid systolic click
- (B) Pericardial shudder
- (C) Opening snap
- (D) Tumor plop sound

Correct Answer: (D) Tumor plop sound

Solution:

Step 1: Heart sounds can be grouped by pitch. High pitched sounds include the mid systolic click, opening snap and the pericardial knock or shudder. Low pitched sounds include S3, S4 and the tumour plop. The question asks for the one that is NOT high pitched, so the answer is option d, the tumour plop.

Step 2: The mid systolic click is heard in mitral valve prolapse during systole and is a high pitched sound.

Step 3: The opening snap and the pericardial knock are high pitched and occur in early diastole; the pericardial knock is a feature of constrictive pericarditis.

Step 4: A tumour plop is a lower pitched diastolic sound heard in atrial myxoma, arising as the tumour prolapses across the mitral valve. Being low pitched, it is the correct exception.

Quick Tip: Group these by pitch; the atrial myxoma sound is the low pitched one.



138. A Mentzer index of more than 13 suggests a diagnosis of which condition?

- (A) Iron deficiency anemia
- (B) Thalassemia
- (C) Hereditary spherocytosis
- (D) Autoimmune hemolytic anemia

Correct Answer: (A) Iron deficiency anemia

Solution:

Step 1: The Mentzer index helps differentiate iron deficiency anaemia from beta thalassaemia. It is calculated as the mean corpuscular volume (MCV in fL) divided by the red blood cell count (RBC in millions per microlitre).

Step 2: The rule: if the index is greater than 13, iron deficiency anaemia is more likely; if it is less than 13, thalassaemia is more likely. Since the value here is more than 13, the answer is option a, iron deficiency anaemia.

Step 3: Reason: in iron deficiency the marrow makes fewer and smaller red cells, so both the RBC count and MCV fall, and dividing a low MCV by a low RBC count gives a value above 13.

Step 4: In thalassaemia the globin synthesis defect produces a normal or high

RBC count with a low MCV, so dividing a low MCV by a normal/high RBC count gives a value below 13. Hereditary spherocytosis and autoimmune haemolytic anaemia are not differentiated by this index.

Quick Tip: MCV divided by RBC count above 13 points away from thalassaemia.

139. Chronic non-spherocytic hemolytic anemia is seen in which WHO class of G6PD deficiency?

- (A) Class I
- (B) Class II
- (C) Class III
- (D) Class IV

Correct Answer: (A) Class I

Solution:

Step 1: The WHO classifies G6PD variants by the degree of enzyme deficiency and the severity of haemolysis. Chronic (congenital) non-spherocytic haemolytic anaemia is the feature of Class I variants, so the answer is option a.

Step 2: Class I variants have such severe G6PD deficiency (less than 10 percent of normal activity) that lifelong haemolysis occurs even without infection or drug exposure. Mutations often lie at the glucose-6-phosphate or NADP binding sites, crippling enzyme function.

Step 3: Class II variants (for example G6PD Mediterranean) also have severe deficiency but cause only intermittent acute haemolysis triggered by infection, drugs or fava beans. Class III variants (for example G6PD A-) have moderate deficiency with episodic haemolysis.

Step 4: Class IV variants have normal activity with no haemolysis and Class V

have increased activity, so neither produces chronic haemolytic anaemia. This confirms Class I as the correct answer.

Quick Tip: The most severe variant with under 10 percent activity haemolyses for life.

140. All of the following are markers of acute kidney injury EXCEPT?

- (A) Clusterin
- (B) Osteopontin
- (C) Alanine aminopeptidase
- (D) Acid phosphatase

Correct Answer: (D) Acid phosphatase

Solution:

Step 1: Acute kidney injury (AKI) is detected early by a panel of urinary and serum biomarkers that reflect tubular damage and stress. The question asks which listed substance is NOT a recognised AKI marker.

Step 2: Clusterin, osteopontin and alanine aminopeptidase are all established markers of acute kidney injury, along with KIM-1, NGAL, IL-18, cystatin C, NAG and others.

Step 3: Acid phosphatase is not part of the AKI biomarker panel, so the exception is option d, acid phosphatase.

Step 4: Note that alkaline phosphatase (not acid phosphatase) is among the tubular enzyme markers, which is the likely source of the distractor.

Quick Tip: One option is the acid form of an enzyme; the alkaline form is the real renal marker.

141. Sialolithiasis is most commonly seen in which gland?

- (A) Parotid
- (B) Sublingual
- (C) Submandibular
- (D) Minor salivary gland

Correct Answer: (C) Submandibular

Solution:

Step 1: Sialolithiasis means stone formation within a salivary gland or its duct.

Step 2: The submandibular gland is the commonest site. About 80 percent of salivary calculi form here, most often within the submandibular (Wharton's) duct.

Step 3: This predilection is explained by the gland's secretion. Submandibular saliva is more alkaline, more mucinous and richer in calcium, and the long, upward-running Wharton's duct drains against gravity, all of which favour stasis and stone formation.

Step 4: The parotid (serous saliva) and sublingual gland are far less commonly affected, and minor salivary gland stones are rare. Hence the answer is the submandibular gland.

Quick Tip: Think of the gland with thick mucinous saliva and a long uphill duct.

142. ABPI (ankle brachial pressure index) of imminent necrosis is

- (A) < 0.3
- (B) 0.3
- (C) < 0.6
- (D) > 0.6

Correct Answer: (A) < 0.3

Solution:

Step 1: The ankle brachial pressure index (ABPI) compares systolic pressure at the ankle with that in the arm and grades the severity of peripheral arterial disease.

Step 2: Normal ABPI is about 0.9 to 1.3. Values fall as arterial disease worsens. Claudication appears around 0.5 to 0.9, and rest pain or critical limb ischaemia at roughly 0.4 or below.

Step 3: When the index drops very low, near **0.3** or less, tissue perfusion is so poor that ulceration, gangrene and imminent necrosis are expected. So **< 0.3** signals impending tissue death.

Step 4: The higher cut-offs of 0.3, 0.6 and > 0.6 reflect less severe disease and do not indicate imminent necrosis, so the answer is **< 0.3** .

Quick Tip: The lower the index, the worse the perfusion; necrosis looms at the very lowest value.

143. Indications of liver transplantation are all EXCEPT

- (A) Biliary atresia
- (B) Sclerosing cholangitis
- (C) Hepatitis A
- (D) Cirrhosis

Correct Answer: (C) Hepatitis A

Solution:

Step 1: Liver transplantation is reserved for severe, irreversible liver disease when other medical or surgical options have been exhausted.

Step 2: Recognised indications include biliary atresia (the commonest indication in children), primary sclerosing cholangitis, and cirrhosis of any cause (the commonest indication group in adults, chiefly chronic hepatitis C and alcoholic liver disease).

Step 3: Hepatitis A is an acute, self-limiting viral infection that resolves without causing chronic liver damage. It does not lead to irreversible disease and so is NOT an indication for transplant. Only the rare fulminant form would even be considered, which is not what plain hepatitis A implies.

Step 4: Therefore the exception is hepatitis A; the other three are accepted indications.

Quick Tip: Which listed liver disease is acute and self-limiting rather than chronic and irreversible?

144. Commonest type of anal canal carcinoma is?

- (A) Squamous cell carcinoma
- (B) Adenocarcinoma
- (C) Adenoacanthoma
- (D) Papillary type

Correct Answer: (A) Squamous cell carcinoma

Solution:

Step 1: The anal canal is lined predominantly by squamous epithelium below the dentate line, so tumours arising here tend to be squamous in origin.

Step 2: Squamous cell carcinoma is therefore the commonest malignancy of the anal canal, beginning in the outer squamous lining.

Step 3: The next most frequent anal canal tumours are basal cell carcinoma and melanoma, but these are far less common than squamous cell carcinoma.

Step 4: Adenocarcinoma is the commonest cancer of the rectum and colon, not the anal canal, and adenoacanthoma and papillary types are not the usual anal canal cancers. Hence the answer is squamous cell carcinoma.

Quick Tip: Match the tumour type to the squamous lining of the anal canal.

145. True about blind loop syndrome, all EXCEPT

- (A) Syndrome of bacterial overgrowth
- (B) Steatorrhea, megaloblastic anemia and deficiency of fat soluble vitamins
- (C) Surgery is not indicated
- (D) ^{14}C -xylose or ^{14}C -cholyglycine breath tests are indirect tests for bacterial overgrowth

Correct Answer: (C) Surgery is not indicated

Solution:

Step 1: Blind loop syndrome is a syndrome of small-bowel bacterial overgrowth, usually caused by stasis in a surgically created or anatomically blind segment of bowel.

Step 2: The overgrowing bacteria deconjugate bile salts and consume vitamin B12, producing steatorrhea, megaloblastic (vitamin B12 deficiency) anaemia, weight loss and deficiency of fat-soluble vitamins. So options a and b are true.

Step 3: Diagnosis uses indirect breath tests such as the ^{14}C -xylose or ^{14}C -cholyglycine (glycocholate) breath test, which detect bacterial metabolism. So option d is true.

Step 4: Treatment is broad-spectrum antibiotics and nutritional support, but

when an anatomical cause (blind loop, stricture, fistula) is responsible, corrective surgery IS indicated. Therefore the false statement is that surgery is not indicated, making option c the exception.

Quick Tip: If an anatomical blind loop causes the stasis, what definitive treatment may still be required?

146. Prophylactic thyroidectomy is indicated in

- (A) Hashimoto thyroiditis
- (B) MEN type 2
- (C) Riedel thyroiditis
- (D) De-Quervain's thyroiditis

Correct Answer: (B) MEN type 2

Solution:

Step 1: Prophylactic thyroidectomy means removing a clinically normal thyroid to pre-empt an almost certain future cancer.

Step 2: MEN type 2 (multiple endocrine neoplasia 2A and 2B) is driven by a germline RET proto-oncogene mutation and carries a near-inevitable risk of medullary thyroid carcinoma at a young age.

Step 3: Because the cancer risk is so high, once the RET mutation is confirmed, prophylactic total thyroidectomy is advised early in childhood (by about age 5 in MEN 2A, and even earlier in the more aggressive MEN 2B).

Step 4: Hashimoto, Riedel and De Quervain thyroiditis are inflammatory conditions managed medically and do not justify prophylactic gland removal. Hence the answer is MEN type 2.

Quick Tip: Which RET-linked syndrome makes medullary thyroid cancer nearly inevitable?

147. Choledochal cyst is dilatation of

- (A) Gall bladder
- (B) CBD
- (C) Hepatic duct
- (D) Bile duct

Correct Answer: (D) Bile duct

Solution:

Step 1: A choledochal cyst is a congenital cystic dilatation of the biliary tree.

Step 2: The dilatation can involve the extrahepatic or intrahepatic bile ducts, alone or in combination, as classified by the Todani system.

Step 3: Because the abnormality is of the bile duct in general (not confined to the common bile duct), the best and most complete answer is bile duct.

Step 4: The gall bladder and hepatic duct alone do not define the lesion, and CBD is only the most frequent single segment involved (type I), so the broader term bile duct is the intended answer.

Quick Tip: It is a congenital cystic dilatation of the biliary tree, not just one named duct.

148. A gum tumor measures 5 cm in dimension with contralateral lymph node enlargement of 2 cm. There is no distant metastasis. What is the stage of the tumor?

- (A) T3N2M0
- (B) T2N2M0
- (C) T1N2M0
- (D) T3N3M0

Correct Answer: (A) T3N2M0

Solution:

Step 1: Score the primary tumor (T). In the TNM staging of oral carcinoma, T1 is a tumor 2 cm or smaller, T2 is 4 cm or smaller, and T3 is larger than 4 cm. The gum tumor here is 5 cm, which is larger than 4 cm, so it is **T3**.

Step 2: Score the nodes (N). N2c is defined as metastasis in bilateral or contralateral lymph nodes, none larger than 6 cm in greatest dimension. The node here is contralateral and 2 cm (under 6 cm), so it falls under the N2 group, giving **N2**.

Step 3: Score metastasis (M). There is no distant metastasis, so it is **M0**.

Step 4: Combine. T3 + N2 + M0 = **T3N2M0**. This is the correct answer. T2N2M0 is wrong because 5 cm exceeds the 4 cm cut-off for T2. T1N2M0 is wrong for the same size reason. T3N3M0 is wrong because N3 requires a node larger than 6 cm, but this node is only 2 cm.

Quick Tip: A 5 cm primary is T3; a contralateral node under 6 cm is N2c; no distant spread is M0.

149. What is the appropriate management of a carcinoma of the lateral border of the tongue after primary resection?

- (A) Chemotherapy
- (B) Radiotherapy
- (C) Observation
- (D) Neck dissection

Correct Answer: (D) Neck dissection

Solution:

Step 1: Understand the lymphatic risk. Carcinoma of the lateral border of the tongue has a high propensity for spreading to the cervical lymph nodes, even when nodes are not clinically obvious. So management cannot stop at resecting the primary tumor.

Step 2: Address the neck. When regional lymph nodes are involved, the neck must be treated surgically. This is done by a modified radical neck dissection or a selective neck dissection. Therefore **neck dissection** is the correct next step after resection.

Step 3: Why the other options are wrong. Observation risks leaving occult nodal disease untreated. Postoperative radiotherapy is added only for specific high-risk features such as perineural or angiolymphatic spread or positive nodes, not as the standalone next step. Chemotherapy alone is not the primary locoregional treatment here.

Quick Tip: Tongue cancers spread early to neck nodes, so the neck must be cleared surgically.

150. Over the last decade, the morbidity of duodenal ulcer has been reduced mainly due to which of the following?

- (A) Life style modification
- (B) Eradication of H. pylori
- (C) Proton pump inhibitors
- (D) None

Correct Answer: (C) Proton pump inhibitors

Solution:

Step 1: Identify the key advance. The single biggest pharmacological change in peptic ulcer disease over recent decades has been the introduction of proton pump inhibitors (PPIs). These potently suppress gastric acid and allow rapid ulcer healing.

Step 2: Link to reduced morbidity. With PPIs and a better understanding of perforated peptic ulcer aetiology, the incidence and morbidity of complications such as perforation have reportedly fallen in Western countries. So **proton pump inhibitors** is the correct answer.

Step 3: Why H. pylori eradication is not the chosen answer here. While eradicating H. pylori is central to preventing recurrence, this recall question credits the marked reduction in duodenal ulcer morbidity to acid suppression with PPIs. Life style modification has a much smaller role for duodenal ulcer specifically.

Quick Tip: Potent acid suppression that heals ulcers fast is the key advance.

151. What is the most common conotruncal anomaly?

- (A) TGA
- (B) Tetralogy of Fallot
- (C) Truncus arteriosus
- (D) Double outlet right ventricle

Correct Answer: (A) TGA

Solution:

Step 1: Define conotruncal defects. Conotruncal anomalies are abnormalities of outflow tract septation or of ectomesenchymal (neural crest) tissue migration. They include transposition of the great arteries, tetralogy of Fallot, truncus arteriosus, and double outlet right ventricle.

Step 2: Identify the commonest. Among the conotruncal group, the most common defect is transposition of the great arteries (TGA). So **TGA** is the answer.

Step 3: Why the other options are not chosen. Tetralogy of Fallot, truncus arteriosus, and double outlet right ventricle are all conotruncal anomalies, but in this recall TGA is listed as the most common one.

Quick Tip: Among outflow-tract (conotruncal) defects, transposition of the great arteries tops this list.

152. Which of the following is NOT included as a major criterion in the modified Jones criteria?

- (A) Polyarthralgia
- (B) Carditis
- (C) Chorea
- (D) Erythema marginatum

Correct Answer: (A) Polyarthralgia

Solution:

Step 1: Recall the Jones major criteria. The major criteria for acute rheumatic fever are carditis, polyarthritides, chorea, erythema marginatum, and subcutaneous nodules.

Step 2: Spot the odd one out. Carditis, chorea, and erythema marginatum are all major criteria. Polyarthralgia (joint pain without inflammation) is a **minor** criterion, not a major one. So **polyarthralgia** is the answer to a question asking what is not a major criterion.

Step 3: Note the distinction. Polyarthritis (with inflammation) is a major criterion, but polyarthralgia (pain only) is minor. The wording therefore separates option a from the other three, which are major manifestations.

Quick Tip: Polyarthritis is major; mere joint pain (arthralgia) is a minor criterion.

153. The swallowing and breathing reflex is NOT seen in the fetus until which gestational age?

- (A) 14 weeks
- (B) 12 weeks
- (C) 16 weeks
- (D) Appears in all the above periods

Correct Answer: (B) 12 weeks

Solution:

Step 1: Trace fetal behavioural milestones. Muscle contractions first appear around 8 weeks, followed soon by lateral flexion movements. Breathing and swallowing motions then appear by about 13 to 14 weeks of gestation.

Step 2: Apply to the question. Since breathing and swallowing motions begin at roughly 13 to 14 weeks, they are NOT yet present at 12 weeks. So the reflex is not seen at **12 weeks**, which is the answer.

Step 3: Why the others are wrong. By 14 weeks and 16 weeks these motions have already appeared, so they cannot be the period in which the reflex is absent.

The grasp reflex appears at 17 weeks and eye opening around 26 weeks, confirming the timeline.

Quick Tip: Breathing and swallowing motions begin by 13 to 14 weeks, so they are absent at 12 weeks.

154. What is the most common bacterial cause of lower respiratory tract infection in a 3 year old child?

- (A) Klebsiella
- (B) H. influenzae
- (C) Streptococcus pneumoniae
- (D) Staphylococcus aureus

Correct Answer: (C) Streptococcus pneumoniae

Solution:

Step 1: Separate viral from bacterial causes. Overall, the most common cause of pneumonia in children is respiratory syncytial virus (RSV), with influenza virus, adenovirus, rhinovirus, and parainfluenza virus also contributing.

Step 2: Identify the leading bacterium. Among bacterial causes, the most common cause of pediatric pneumonia is Streptococcus pneumoniae (pneumococcus). The options here are all bacteria, so the question is asking for the leading bacterial pathogen.

Step 3: Choose and justify. Therefore **Streptococcus pneumoniae** is the answer. Klebsiella, H. influenzae, and Staphylococcus aureus can all cause pneumonia in children but are less common than pneumococcus. Mycoplasma and chlamydia are the usual causes of atypical pneumonia.

Quick Tip: Among bacteria, pneumococcus leads childhood pneumonia (RSV leads

overall).



155. Pedigree analysis chart is used for which of the following purposes?

- (A) Used for growth monitoring
- (B) To assess side effect during chemotherapy
- (C) Used to see genetic transmission
- (D) To assess developmental delay in infant

Correct Answer: (C) Used to see genetic transmission

Solution:

Step 1: A pedigree is a diagram that gives a graphic depiction of a family structure and its medical history across generations. The person who first brings the family to medical attention and provides the information is called the proband.

Step 2: Standardised symbols are used for each individual and relationship, and a three generation pedigree is usually drawn. This allows the doctor to track how a trait or disease is passed from parents to children, so the main use is to study genetic transmission of a condition within a family.

Step 3: The closer the relationship of the proband to an affected person, the greater the chance of a shared genetic component. Growth monitoring, chemotherapy side effects and developmental delay are clinical or follow up assessments and have nothing to do with charting inheritance, so options a, b and d are incorrect.

Quick Tip: Think about a family tree drawn with symbols to track inheritance.



156. In exchange blood transfusion, which blood product is used?

- (A) Whole blood
- (B) EPP
- (C) Serum
- (D) Packed cells

Correct Answer: (A) Whole blood

Solution:

Step 1: Exchange transfusion is a procedure in which the patient's blood is slowly removed in small aliquots and replaced with fresh donor blood. It is most often performed in neonates with severe hyperbilirubinaemia or haemolytic disease of the newborn.

Step 2: Because the goal is to replace both the cellular and plasma components and to remove bilirubin, antibodies and toxins, fresh whole blood is the product of choice. Whole blood supplies red cells together with plasma in physiological proportions.

Step 3: Serum and EPP do not provide red cells, so they cannot correct anaemia or remove sensitised cells. Packed cells lack plasma and would not replace the volume or remove plasma bound substances effectively, so option a, whole blood, is correct.

Quick Tip: You need to replace both red cells and plasma at the same time.

157. The most common organ involved in congenital tuberculosis is?

- (A) Liver
- (B) Pancreas
- (C) Kidney
- (D) Lung

Correct Answer: (A) Liver

Solution:

Step 1: Congenital tuberculosis is rare because a mother with active tuberculosis often presents with infertility. When transmission does occur, the route of spread determines the organ that is first affected.

Step 2: The tubercle bacilli most commonly pass from the mother through the umbilical vein to the fetus. The umbilical vein drains into the fetal liver, so a primary focus, the hepatic primary complex, develops there. This makes the liver the most commonly involved organ.

Step 3: If the neonate instead aspirates amniotic fluid containing bacilli, gastrointestinal or lung infection can follow, which is why lung is a tempting but less common answer. Neonates usually present with respiratory distress, hepatosplenomegaly and lymphadenopathy. The pancreas and kidney are not characteristic primary sites, so the correct answer is liver.

Quick Tip: Follow the umbilical vein and ask which organ it drains into first.

158. Antiendomysial antibody is used in the screening of?

- (A) Myasthenia gravis
- (B) Autoimmune hepatitis
- (C) Coeliac disease
- (D) Graves disease

Correct Answer: (C) Coeliac disease

Solution:

Step 1: Coeliac disease is a permanent intolerance of the small intestine to gluten. It is marked by gluten dependent changes in villous morphology and signs of immune activation in the lamina propria of the intestinal mucosa.

Step 2: Serum anti-endomysial antibodies (EMA) are highly suggestive of coeliac disease because they have both high sensitivity and high specificity. They are therefore a key serological screening test for this condition.

Step 3: Other supportive antibodies are tissue transglutaminase and antigliadin antibodies, and the treatment is a strict gluten free diet. Myasthenia gravis is screened with anti-acetylcholine receptor antibodies, autoimmune hepatitis with anti smooth muscle and ANA, and Graves disease with TSH receptor antibodies, so none of those match. The correct answer is coeliac disease.

Quick Tip: Link this antibody to a gluten driven small bowel disorder.

159. Regarding Reye syndrome, the FALSE statement is?

- (A) Associated with salicylate ingestion
- (B) Hepatomegaly
- (C) Jaundice
- (D) Hypoglycemia

Correct Answer: (C) Jaundice

Solution:

Step 1: Reye syndrome is an acute, non inflammatory encephalopathy with fatty change in the liver. It damages many organs, especially the brain and liver, and characteristically causes low blood sugar (hypoglycaemia) and a raised ammonia level. So hypoglycaemia (option d) is a true feature.

Step 2: It typically follows a viral illness such as influenza or varicella, classically when salicylates (aspirin) have been given. This confirms that salicylate ingestion (option a) is a true association. The fatty infiltration enlarges the liver, so hepatomegaly (option b) is also true.

Step 3: The liver shows fatty change with minimal inflammation, and liver

enzymes rise while bilirubin stays normal. Because bilirubin is not elevated, jaundice is usually absent. Therefore the false statement is jaundice, making option c the answer.

Quick Tip: Enzymes rise but bilirubin stays normal, so what skin sign is missing?

160. A child has a lesion on the buttocks for 2 years, spreading peripherally with central scarring and non symptomatic. The diagnosis is?

- (A) Erythema annulare centrifugum
- (B) Erythema migrans
- (C) Erythema marginatum
- (D) Erythema gyratum repens

Correct Answer: (A) Erythema annulare centrifugum

Solution:

Step 1: Erythema annulare centrifugum is an asymptomatic or mildly itchy eruption of variable duration. It begins as red papules that spread peripherally while clearing centrally, producing annular, arcuate or polycyclic plaques that enlarge slowly at about 2 to 5 mm per day. The chronic, asymptomatic, centrifugally spreading lesion described here fits it best.

Step 2: Erythema migrans is usually less numerous, less circinate and is linked to a tick bite (Lyme disease), so the long standing buttock lesion without a tick history does not fit. Erythema gyratum repens spreads much faster, has a bizarre wood grain pattern and is almost always tied to an underlying malignancy.

Step 3: Erythema marginatum is a non scaling gyrate erythema seen specifically with rheumatic fever, which would present acutely with carditis or arthritis rather than a slow chronic plaque. By exclusion and by the classic centrifugal spread with central clearing, the diagnosis is erythema annulare centrifugum.

Quick Tip: A chronic, asymptomatic ring that clears centrally and creeps outward slowly.

161. The most common cause of amputation in India is?

- (A) Diabetic gangrene
- (B) Gas gangrene
- (C) Road traffic accident
- (D) Tumors

Correct Answer: (C) Road traffic accident

Solution:

Step 1: Amputation is a procedure in which part of a limb is removed through one or more bones, whereas disarticulation removes the limb through a joint. Indications are grouped as absolute, such as gas gangrene, diabetic gangrene and irreparable loss of blood supply, and relative, such as trauma, tumours and severe loss of limb function.

Step 2: Overall, the single most common cause of amputation is trauma to a limb. Trauma may damage the blood supply or destroy the limb beyond repair, leading to amputation.

Step 3: The most common cause of such limb trauma in India is a road traffic accident. Diabetic and gas gangrene are important but less frequent overall, and tumours are an uncommon relative indication. Therefore the correct answer is road traffic accident.

Quick Tip: Trauma leads overall, and what causes most limb trauma in India?

162. Which of the following is NOT a complication of fracture of the neck of the femur?

- (A) Non-union
- (B) Malunion
- (C) Avascular necrosis (AVN)
- (D) Osteoarthritis

Correct Answer: (B) Malunion

Solution:

Step 1: Fractures of the neck of the femur are intracapsular and disrupt the precarious blood supply to the femoral head. They are far more prone to serious complications than almost any other fracture, especially when there is displacement.

Step 2: The three classic complications are avascular necrosis of the femoral head (the most common, seen in 15-35% of displaced fractures), non-union (the second most common, 10-30% of displaced fractures), and secondary osteoarthritis of the hip that follows collapse of the femoral head.

Step 3: Malunion is NOT a recognised complication here. The fracture is intracapsular and bathed in synovial fluid, with very little surrounding periosteum and soft tissue, so healing in a deformed position (malunion) does not occur. Instead the bone either fails to unite (non-union) or the head dies (AVN).

Step 4: Hence the odd one out is malunion. AVN, non-union and osteoarthritis are all genuine sequelae.

Quick Tip: An intracapsular fracture with a poor blood supply dies or fails to unite, it does not heal in a deformed position.

163. The most common site for an osteoporotic vertebral fracture is?

- (A) Dorsolumbar spine
- (B) Cervical spine
- (C) Lumbosacral spine
- (D) Dorsal spine

Correct Answer: (A) Dorsolumbar spine

Solution:

Step 1: Osteoporosis is usually silent until a fragility fracture occurs. The commonest symptom is back pain caused by a vertebral compression fracture.

Step 2: Vertebral compression fractures cluster at the thoracolumbar junction, that is the dorsolumbar spine, because this is the transition zone between the relatively rigid thoracic spine and the mobile lumbar spine, where mechanical stress is greatest.

Step 3: Other typical fragility fracture sites are the lower end of the radius (Colles fracture) and the neck of the femur, but among vertebrae the dorsolumbar region is the most frequent.

Step 4: Remember that serum calcium, phosphate and alkaline phosphatase are all normal in osteoporosis, which helps separate it from osteomalacia. The answer is the dorsolumbar spine.

Quick Tip: Think of the thoracolumbar junction, where the stiff thoracic spine meets the mobile lumbar spine.

164. Froment's sign is positive in cases of weakness of?

- (A) Thumb adduction
- (B) Thumb abduction
- (C) Thumb flexion
- (D) Thumb extension

Correct Answer: (A) Thumb adduction

Solution:

Step 1: Froment's sign tests the integrity of the adductor pollicis, a muscle supplied by the ulnar nerve. It therefore detects ulnar nerve palsy through weakness of thumb adduction.

Step 2: The patient is asked to hold a book or sheet of paper between the thumb and the side of the index finger. Normally the adductor pollicis and first dorsal interosseous grip firmly with the thumb kept extended.

Step 3: If the ulnar nerve is injured, the adductor pollicis is paralysed, so the patient compensates by using the flexor pollicis longus (median nerve). This flexes the thumb at the interphalangeal joint to pinch the object.

Step 4: The visible flexion of the terminal thumb joint when trying to hold the object, made more obvious when the examiner pulls it away, is a positive Froment's sign, also called the book test. The defect is weakness of thumb adduction.

Quick Tip: It signals ulnar nerve palsy, paralysing the adductor pollicis so the thumb flexes to grip.

165. CASPAR criteria are used in the diagnosis of?

- (A) Psoriatic arthritis
- (B) Rheumatoid arthritis
- (C) Ankylosing spondylitis
- (D) Reactive synovitis

Correct Answer: (A) Psoriatic arthritis

Solution:

Step 1: CASPAR stands for ClASsification criteria for Psoriatic ARthritis, so by its very name it is used to classify and diagnose psoriatic arthritis.

Step 2: To meet CASPAR, a patient must have inflammatory articular disease (joint, spine or enthesal) plus a score of at least 3 points from five categories.

Step 3: The five categories are current psoriasis or a personal or family history of psoriasis, typical psoriatic nail dystrophy, a negative rheumatoid factor, current or past dactylitis, and radiographic juxta-articular new bone formation in the hand or foot.

Step 4: Rheumatoid arthritis uses ACR/EULAR criteria, ankylosing spondylitis uses the modified New York or ASAS criteria, so the correct answer is psoriatic arthritis.

Quick Tip: Read the acronym - Classification criteria for Psoriatic ARthritis.

166. The development of chondrosarcoma is related with?

- (A) Maffucci syndrome
- (B) Felty syndrome
- (C) Both a and b
- (D) None of the above

Correct Answer: (A) Maffucci syndrome

Solution:

Step 1: Chondrosarcoma arising in a pre-existing benign cartilage lesion is called secondary chondrosarcoma. Its main precursors are osteochondroma and enchondroma.

Step 2: Maffucci syndrome (multiple enchondromas plus soft tissue haemangiomas) and Ollier disease (multiple enchondromatosis) both carry a high risk of malignant change, roughly a 25-30% lifetime risk of developing chondrosarcoma.

Step 3: Compare this with the small 2% risk in a solitary osteochondroma and 5-25% in multiple hereditary osteochondromatosis. So Maffucci syndrome is strongly linked to chondrosarcoma.

Step 4: Felty syndrome is the triad of rheumatoid arthritis, splenomegaly and neutropenia, and has nothing to do with cartilage tumours. Hence the answer is Maffucci syndrome alone.

Quick Tip: Multiple enchondromas with haemangiomas carry a high malignant change risk; Felty is an RA triad.

167. Infection of the index finger spreads to which space?

- (A) Thenar space
- (B) Mid palmar space
- (C) Hypothenar space
- (D) Flexion space

Correct Answer: (A) Thenar space

Solution:

Step 1: The palm has two potential fascial spaces deep to the long flexor tendons, separated by a fibrous septum running to the third metacarpal: the thenar space (radial) and the mid palmar space (ulnar).

Step 2: The thenar space communicates with the index finger through the lumbrical canal, while the mid palmar space communicates with the middle, ring and little fingers.

Step 3: Therefore a deep infection of the index finger tracks into and collects in the thenar space, whereas infection of the middle, ring or little finger spreads to the mid palmar space.

Step 4: The hypothenar space is small and not a usual site of spread, so the correct answer is the thenar space.

Quick Tip: Index finger drains radially to the thenar space; middle, ring and little drain to mid palmar.

168. The amniotic fluid volume is maximum at?

- (A) 32 weeks
- (B) 34 weeks
- (C) 36 weeks
- (D) 40 weeks

Correct Answer: (C) 36 weeks

Solution:

Step 1: Amniotic fluid volume rises steadily through pregnancy, reaching its peak in the early third trimester and then declining toward term.

Step 2: The maximum volume, about 1000 mL, is reached at around 36 weeks of gestation.

Step 3: After 36 weeks the volume falls, dropping to roughly 600-800 mL at 40 weeks (term) and falling further in post-term pregnancy, which is why oligohydramnios is a concern in prolonged pregnancy.

Step 4: So the correct answer is 36 weeks, not 40 weeks where the volume is already declining.

Quick Tip: The liquor peaks near 1 litre in the early third trimester, then falls before term.

169. Beta hCG is detected earliest in serum by which day of conception?

- (A) 8 days
- (B) 15 days
- (C) 21 days
- (D) 30 days

Correct Answer: (A) 8 days

Solution:

Step 1: Beta hCG (human chorionic gonadotropin) is a glycoprotein hormone secreted by the syncytiotrophoblast of the developing placenta. Its detection forms the basis of all pregnancy tests.

Step 2: hCG begins to be produced very soon after implantation. It can be detected in serum as early as 8 to 9 days after conception, which is the earliest reliable window.

Step 3: In urine, hCG appears later, by about 14 days after conception, because serum levels rise before urinary concentrations become measurable. This is why serum testing is more sensitive and earlier than urine testing.

Step 4: Options of 15, 21 and 30 days are too late for the earliest serum detection, so the correct answer is 8 days.

Quick Tip: Serum picks up hCG before urine does - think roughly a week after conception.



170. Which of the following is true about the expected date of delivery (EDD)?

- (A) Less than 10% of deliveries occur on EDD
- (B) Less than 20% of deliveries occur on EDD
- (C) 80% of the deliveries occur on EDD
- (D) 90% of the deliveries occur on EDD

Correct Answer: (A) Less than 10% of deliveries occur on EDD

Solution:

Step 1: The EDD is calculated using Naegele's rule, which estimates a date 40 weeks from the last menstrual period. It is only an estimate, not a fixed delivery day.

Step 2: In reality, fewer than 5% of all pregnancies actually deliver exactly on the EDD. The majority deliver within about 7 days on either side of the EDD.

Step 3: Around 13% of births are preterm and about 5 to 7% are post term. Because the true figure is under 5%, the statement that less than 10% of deliveries occur on EDD is correct.

Step 4: The other options (80% or 90% on EDD) are grossly overstated and medically false.

Quick Tip: The EDD is just an estimate - very few babies actually arrive on that exact day.

171. What is the most common cause of tenth day (secondary) postpartum bleeding?

- (A) Retained bits of membrane or placenta
- (B) Infection
- (C) Endometritis
- (D) Blood coagulopathy

Correct Answer: (A) Retained bits of membrane or placenta

Solution:

Step 1: Bleeding around the tenth postpartum day is secondary (late) postpartum hemorrhage, defined as abnormal bleeding from the genital tract occurring between 24 hours and 6 weeks after delivery.

Step 2: The single most common cause is retained bits of placental tissue or membranes. These retained products prevent proper uterine involution and act as a focus for bleeding.

Step 3: Infection and endometritis often coexist with retained products and can aggravate the bleeding, but they are secondary contributors rather than the leading primary cause.

Step 4: Coagulopathy is an uncommon cause of late PPH. Hence retained bits of membrane is the best answer.

Quick Tip: Late postpartum bleeding usually means something was left behind in the uterus.

172. Which of the following is true about luteoma of pregnancy?

- (A) Usually bilateral
- (B) It is a benign self limiting condition
- (C) It consists of luteinized cells
- (D) All of the above

Correct Answer: (D) All of the above

Solution:

Step 1: Luteoma of pregnancy is a non-neoplastic, tumour-like enlargement of the ovary that develops during pregnancy in response to high levels of human chorionic gonadotropin.

Step 2: It typically appears as bilateral, multinodular, solid ovarian masses, so the statement that it is usually bilateral is correct.

Step 3: Histologically, normal ovarian tissue is replaced by a solid proliferation of luteinized stromal cells, so the option about luteinized cells is also correct.

Step 4: It is a benign, self-limiting condition that regresses spontaneously after delivery and needs no treatment. Since all three statements are true, the answer is all of the above.

Quick Tip: Bilateral, luteinized, and resolves after delivery - all the listed features apply.

173. The investigation of choice for the diagnosis of pelvic inflammatory disease (PID) is?

- (A) Laparoscopy
- (B) Colposcopy
- (C) Hysteroscopy
- (D) Ultrasonography

Correct Answer: (A) Laparoscopy

Solution:

Step 1: Pelvic inflammatory disease is infection and inflammation of the upper genital tract. A clinical diagnosis alone has limited accuracy, so a confirmatory investigation is often needed.

Step 2: Laparoscopy is the gold standard or investigation of choice for diagnosing PID. It allows direct visualization of the tubes and pelvis, can confirm tubal hyperemia, edema and exudate, and permits sampling.

Step 3: Colposcopy examines the cervix for neoplasia and hysteroscopy views the uterine cavity, so neither is suited to confirming upper tract or tubal inflammation.

Step 4: Ultrasonography is useful for tubo-ovarian abscess but is less definitive overall, making laparoscopy the best answer.

Quick Tip: Think of the test that lets you directly see the inflamed tubes.

174. All of the following are true regarding the management of intrauterine fetal death EXCEPT?

- (A) In 50% of cases spontaneous expulsion occurs in 2 weeks
- (B) Fibrinogen levels should be checked weekly
- (C) Delivery by medical induction is preferred if spontaneous expulsion does not occur
- (D) Caesarean section has a limited place in the management of intrauterine fetal death

Correct Answer: (A) In 50% of cases spontaneous expulsion occurs in 2 weeks

Solution:

Step 1: This is a negative (EXCEPT) question, so we must find the one statement that is false about managing intrauterine fetal death (IUFD).

Step 2: After fetal death, spontaneous labour occurs in the large majority of women, with about 80% delivering spontaneously within 2 to 3 weeks. The claim that only 50% expel spontaneously in 2 weeks understates this and is therefore the false statement.

Step 3: The other statements are correct. With prolonged retention there is a risk of disseminated intravascular coagulation, so fibrinogen and coagulation should be monitored periodically. If spontaneous expulsion does not occur, medical induction (for example with prostaglandins) is the preferred route.

Step 4: Caesarean section is reserved for special indications only and has a limited role, so option D is also true. The exception is option A.

Quick Tip: Most women (about 80%) go into labour spontaneously within a couple of weeks - the 50% figure is the wrong one.



175. Mediolateral episiotomy is preferred because it?

- (A) Reduces damage to the anal sphincter and anal canal
- (B) Causes less blood loss
- (C) Is easy to suture
- (D) Is an easy technique

Correct Answer: (A) Reduces damage to the anal sphincter and anal canal

Solution:

Step 1: An episiotomy is a surgical incision of the perineum to enlarge the vaginal outlet during delivery. The two main types are midline and mediolateral.

Step 2: A mediolateral incision is directed away from the anus and rectum, so its

key advantage is a much lower risk of the cut extending into the anal sphincter and anal canal (third and fourth degree tears).

Step 3: The midline type is easier to suture and bleeds less, but it carries a higher risk of extension into the sphincter. Mediolateral actually tends to bleed slightly more and is a little harder to repair.

Step 4: Therefore the reason mediolateral is preferred is the reduced damage to the anal sphincter and canal, not less bleeding or ease of suturing.

Quick Tip: The angled cut keeps the incision away from the anus, protecting the sphincter.

176. The following are causes of oligohydramnios EXCEPT?

- (A) IUGR
- (B) Postmaturity
- (C) Maternal dehydration
- (D) Labetolol

Correct Answer: (D) Labetolol

Solution:

Step 1: Oligohydramnios means a reduced volume of amniotic fluid (amniotic fluid index below 5 cm or a single deepest pocket below 2 cm). The fluid late in pregnancy is largely fetal urine, so anything that lowers fetal urine output or causes loss of fluid leads to oligohydramnios.

Step 2: IUGR (intrauterine growth restriction) is associated with placental insufficiency. Reduced placental perfusion shunts fetal blood away from the kidneys, lowering urine production and causing oligohydramnios.

Step 3: In postmaturity (post-term pregnancy) placental function declines and

amniotic fluid falls progressively, so it is a classic cause. Maternal dehydration reduces maternal plasma volume and is a recognised, reversible cause of low fluid.

Step 4: Labetalol is a combined alpha and beta blocker used to treat hypertension in pregnancy. It is NOT a cause of oligohydramnios. The drugs that classically cause oligohydramnios are ACE inhibitors, ARBs and prostaglandin synthase inhibitors (NSAIDs like indomethacin), which reduce fetal renal output. Hence Labetalol is the exception.

Quick Tip: Think which antihypertensive is safe in pregnancy versus ACE inhibitors and NSAIDs that drop fetal urine.

177. The most common breech presentation in a primigravida is?

- (A) Flexed breech
- (B) Frank breech
- (C) Footling presentation
- (D) Incomplete breech

Correct Answer: (B) Frank breech

Solution:

Step 1: In breech presentation the buttocks or lower limbs occupy the lower pole of the uterus. The subtypes are frank (extended legs), complete (flexed hips and knees) and footling or incomplete (one or both feet presenting).

Step 2: Frank breech, also called breech with extended legs, has the hips flexed and the knees extended so the legs lie straight up against the body. This is the most common breech variety overall and especially in the primigravida.

Step 3: A primigravida has a firm, well-toned uterus and tight abdominal wall. This firm tone splints the fetal legs in the extended position, favouring the frank

breech presentation.

Step 4: Complete (flexed) and footling presentations are commoner in multigravidae with lax uterine tone. Hence in a primigravida the answer is frank breech.

Quick Tip: A firm primigravid uterus keeps the fetal legs extended.

178. The most common malignant ovarian tumor is?

- (A) Serous cystadenocarcinoma
- (B) Mucinous cystadenocarcinoma
- (C) Malignant teratoma
- (D) Sarcoma

Correct Answer: (A) Serous cystadenocarcinoma

Solution:

Step 1: Ovarian tumors are grouped into surface epithelial, germ cell and sex cord stromal tumors. The surface epithelial group accounts for the majority of all ovarian cancers.

Step 2: Within the epithelial group, serous tumors are the commonest, and the malignant form, serous cystadenocarcinoma, is the single most common malignant ovarian tumor. It is often bilateral and may show psammoma bodies on histology.

Step 3: Mucinous cystadenocarcinoma is less frequent than the serous type. Malignant teratoma is a germ cell tumor and is much rarer, while ovarian sarcoma is uncommon.

Step 4: Therefore the most common malignant ovarian tumor is serous cystadenocarcinoma.

Quick Tip: Surface epithelial serous type leads, often bilateral with psammoma bodies.

179. The first line treatment of infertility in PCOS is?

- (A) Clomiphene
- (B) FSH
- (C) GnRH
- (D) Assisted reproductive techniques

Correct Answer: (A) Clomiphene

Solution:

Step 1: Infertility in polycystic ovary syndrome (PCOS) is mainly due to anovulation. The goal of treatment is therefore to induce ovulation.

Step 2: Clomiphene citrate is a selective estrogen receptor modulator. It blocks estrogen receptors at the hypothalamus, removing negative feedback so that FSH and LH rise and stimulate follicular growth and ovulation.

Step 3: Because it is oral, inexpensive and effective, clomiphene citrate is the first line agent for ovulation induction in an anovulatory PCOS woman, usually after lifestyle and weight measures.

Step 4: Injectable gonadotropins (FSH), GnRH and assisted reproductive techniques such as IVF are reserved for women who fail to ovulate or conceive on first line therapy. Hence the answer is clomiphene.

Quick Tip: An oral anti-estrogen that induces ovulation comes first.

180. A patient presents with discharge per urethra and microscopy shows the presence of intracytoplasmic gram negative cocci. What is the most probable diagnosis?

- (A) Gonorrhea
- (B) Donovanosis
- (C) Bacterial vaginosis
- (D) Syphilis

Correct Answer: (A) Gonorrhea

Solution:

Step 1: The clinical clue is purulent urethral discharge in a male together with a microscopy finding of intracytoplasmic gram negative cocci, that is, gram negative diplococci inside neutrophils.

Step 2: *Neisseria gonorrhoeae* is a gram negative diplococcus that is classically seen inside polymorphonuclear leukocytes (intracytoplasmic). Acute urethritis with a purulent discharge is its most common presentation. This combination is essentially diagnostic of gonorrhea.

Step 3: The distractors do not fit. Donovanosis (*Klebsiella granulomatis*) shows intracellular Donovan bodies in tissue smears, not urethral diplococci. Bacterial vaginosis is a vaginal condition with clue cells, not a male urethral discharge. Syphilis is caused by *Treponema pallidum*, a spirochete, and presents with a painless chancre.

Step 4: The intracellular gram negative cocci with purulent urethral discharge therefore point to gonorrhea.

Quick Tip: Gram negative diplococci inside neutrophils plus urethral pus.

181. The treatment of nodulocystic acne is?

- (A) Erythromycin
- (B) Tetracycline
- (C) Isotretinoin (Retinoic acid)
- (D) Steroids

Correct Answer: (C) Isotretinoin (Retinoic acid)

Solution:

Step 1: Nodulocystic acne is the most severe form of acne, with deep inflamed nodules and cysts that scar. It often resists conventional topical and oral antibiotics.

Step 2: Oral isotretinoin (13-cis-retinoic acid) is the drug of choice for severe nodulocystic acne. It shrinks the sebaceous glands, reduces sebum production, normalises follicular keratinisation and reduces Cutibacterium acnes, addressing every step of acne pathogenesis.

Step 3: Erythromycin and tetracycline are oral antibiotics useful for mild to moderate inflammatory acne but are not adequate alone for severe nodulocystic disease. Steroids are not a primary treatment for ordinary acne.

Step 4: Hence the treatment of nodulocystic acne is isotretinoin. It needs strict contraception monitoring as it is teratogenic.

Quick Tip: Severe scarring acne needs the oral vitamin A derivative.

182. Alopecia areata is a/an?

- (A) Autoimmune disorder
- (B) Allergic disorder
- (C) Anaphylactic disorder
- (D) Bacterial infection

Correct Answer: (A) Autoimmune disorder

Solution:

Step 1: Alopecia areata presents as sudden, well-defined, round or oval patches of non-scarring hair loss, often with exclamation mark hairs at the margins.

Step 2: It is an organ-specific autoimmune disorder. T lymphocytes attack the anagen hair follicles after a loss of their normal immune privilege, halting hair growth in the affected areas.

Step 3: It is associated with other autoimmune conditions such as thyroid disease, vitiligo and pernicious anemia, supporting its autoimmune basis. The hair loss is reversible because the follicles are not destroyed.

Step 4: It is not an allergic, anaphylactic or bacterial process. Therefore alopecia areata is an autoimmune disorder.

Quick Tip: T cells attack hair follicles, linked with thyroid disease and vitiligo.

183. Groove sign of Greenblatt is seen in?

- (A) LGV
- (B) Donovanosis
- (C) Chancroid
- (D) Genital Herpes

Correct Answer: (A) LGV

Solution:

Step 1: The groove sign of Greenblatt is a classic clinical feature of Lymphogranuloma venereum (LGV), caused by *Chlamydia trachomatis* serovars L1, L2 and L3.

Step 2: In LGV, the inguinal and femoral lymph nodes enlarge and become matted (buboes). Poupart's ligament (the inguinal ligament) runs across this region and separates the enlarged inguinal nodes from the femoral nodes, producing a visible linear depression known as the groove sign. It is considered pathognomonic of LGV.

Step 3: The other options do not produce this sign. Donovanosis (granuloma inguinale) causes beefy-red ulcers and pseudobuboes; chancroid produces painful suppurating buboes without the groove; genital herpes produces grouped painful vesicles. Hence the answer is LGV.

Quick Tip: Think of the STI where Poupart's ligament splits enlarged inguinal nodes.

184. Post spinal headache lasts for?

- (A) 10 min
- (B) 1 hr
- (C) 10 days
- (D) 1 week

Correct Answer: (C) 10 days

Solution:

Step 1: Post dural puncture headache (PDPH), also called post spinal headache, is caused by leakage of CSF through the dural puncture, which lowers intracranial pressure and produces traction on the meninges and supporting structures.

Step 2: It is positional - it worsens on sitting or standing upright and is relieved by lying down. Onset is usually 12 to 72 hours after the procedure. In most cases it lasts about 7 to 10 days, so 10 days is the best fit among the options.

Step 3: Risk factors include young age, female sex, pregnancy, large bore needle

and multiple punctures. Conservative treatment includes bed rest, fluids, analgesics, caffeine and sumatriptan; if it fails, an epidural blood patch is given. The shorter options (10 min, 1 hr, 1 week) are too brief for the typical self-limiting course, hence 10 days is correct.

Quick Tip: Self-limiting postural headache from a CSF leak resolves in about a week to ten days.

185. False about local anesthetics?

- (A) Prilocaine is less toxic than lignocaine
- (B) Lignocaine is used as an antiarrhythmic
- (C) Mixture of lignocaine and prilocaine is known as eutectic
- (D) Lidocaine is shorter acting than bupivacaine

Correct Answer: (A) Prilocaine is less toxic than lignocaine

Solution:

Step 1: The question asks for the FALSE statement about local anaesthetics. Prilocaine is actually the least toxic of the amide local anaesthetics in terms of CNS and cardiac toxicity, so calling it more toxic than lignocaine would be wrong - but the listed statement claims prilocaine is less toxic, which seems true at first glance.

Step 2: The catch is that prilocaine carries a unique toxicity not shared by lignocaine - its metabolite ortho-toluidine causes methaemoglobinaemia. Because of this distinctive and clinically important toxic effect, the blanket statement that prilocaine is simply less toxic than lignocaine is considered false in this question; prilocaine's methaemoglobinaemia risk makes it the odd one out.

Step 3: The other statements are true. Lignocaine is a class Ib antiarrhythmic. EMLA is a eutectic mixture of lignocaine and prilocaine. Lidocaine is indeed

shorter acting than the long acting bupivacaine. Hence the false option is that prilocaine is less toxic than lignocaine.

Quick Tip: Which amide local anaesthetic uniquely causes methaemoglobinaemia?

186. Anesthetic agent or agents which have a tocolytic effect are?

- (A) Halothane
- (B) Enflurane
- (C) Isoflurane
- (D) All the above

Correct Answer: (D) All the above

Solution:

Step 1: Tocolysis means relaxation of the uterus (decrease in uterine tone). The volatile halogenated inhalational anaesthetics relax smooth muscle, including uterine smooth muscle, in a dose dependent fashion.

Step 2: Halothane, enflurane and isoflurane all produce a dose dependent fall in uterine tone. They reduce both the frequency of uterine contractions and the interval between them, so all three are tocolytic.

Step 3: Since each of the three agents listed has this effect, the correct answer is All the above. This property is why deep volatile anaesthesia can be used for uterine relaxation during procedures like manual removal of a retained placenta.

Quick Tip: Halogenated volatile anaesthetics share a dose-dependent uterine relaxing effect.

187. Shortest acting non depolarizing muscle relaxant is?

- (A) Mivacurium
- (B) Doxacurium
- (C) Pipecuronium
- (D) Vecuronium

Correct Answer: (A) Mivacurium

Solution:

Step 1: Non depolarizing (competitive) neuromuscular blockers act by competing with acetylcholine at the nicotinic receptor of the motor end plate. They are classified by duration of action.

Step 2: Mivacurium is the shortest acting non depolarizing muscle relaxant because it is rapidly hydrolysed by plasma pseudocholinesterase, giving it a brief duration of action.

Step 3: The distractors are longer acting. Doxacurium and pipecuronium are long acting agents, and vecuronium is an intermediate acting agent. Note that suxamethonium (succinylcholine) is the shortest acting skeletal muscle relaxant overall, but it is depolarizing, so among non depolarizing agents the answer is mivacurium.

Quick Tip: Which competitive blocker is metabolised by plasma pseudocholinesterase?

188. Henry's law states that?

- (A) At a constant temperature gas dissolves in solution in proportion to its partial pressure
- (B) At a constant pressure gas dissolves in solution in proportion to its temperature
- (C) At a constant temperature gas dissolves in solution in proportion to its fat solubility
- (D) At a constant pressure gas dissolves in solution in proportion to its fat solubility

Correct Answer: (A) At a constant temperature gas dissolves in solution in proportion to its partial pressure

Solution:

Step 1: Henry's law describes how much of a gas dissolves in a liquid. It states that at a constant temperature, the amount of a gas that dissolves in a given liquid is directly proportional to the partial pressure of that gas in equilibrium with the liquid.

Step 2: This can be written as $C = k \times P$, where C is the concentration of dissolved gas, P is the partial pressure of the gas and k is the solubility constant for that gas and temperature. Higher partial pressure forces more gas into solution.

Step 3: Option a matches this exactly. The other options are wrong because solubility relates to partial pressure (not temperature directly) at constant temperature, and fat solubility is a separate property and not the defining variable in Henry's law. Hence the answer is option a.

Quick Tip: At fixed temperature, dissolved gas tracks its partial pressure.

189. Radiocontrast is contraindicated in all of the following conditions except?

- (A) Renal failure
- (B) Patient on metformin
- (C) Dehydration
- (D) Obesity

Correct Answer: (D) Obesity

Solution:

Step 1: Iodinated radiocontrast agents are nephrotoxic and can precipitate contrast induced nephropathy, so several conditions are relative or absolute

contraindications. The question asks which option is NOT a contraindication.

Step 2: Renal failure is a contraindication because impaired kidneys are at high risk of contrast nephropathy. Dehydration worsens renal perfusion and increases this risk, so it is also a contraindication. Metformin must be withheld around contrast administration because of the risk of lactic acidosis if renal function declines, making it a caution or contraindication.

Step 3: Obesity is not a contraindication to radiocontrast administration. It does not increase the risk of contrast nephropathy or lactic acidosis in the way the other factors do. Therefore obesity is the exception and is the correct answer.

Quick Tip: Which option carries no renal or lactic-acidosis risk with contrast?

190. Radiological features of coarctation of aorta is/are -

- (A) Reverse figure of 3 sign
- (B) Rock sign
- (C) Double aortic knuckle
- (D) All of the above

Correct Answer: (D) All of the above

Solution:

Step 1: Coarctation of aorta (COA) is a narrowing of the aorta, classically at or near the site of the ligamentum arteriosum, just distal to the left subclavian artery. The narrowing alters blood flow and produces several characteristic plain-film signs.

Step 2: The reverse figure of 3 sign (also called the double bulge or E sign) is seen on a frontal chest radiograph. The upper bulge is the dilated aorta above the coarctation, the indentation is the coarctation itself, and the lower bulge is post-stenotic dilatation.

Step 3: The double aortic knuckle reflects the same two bulges of the aortic contour above and below the constriction. Rib notching (Dock's sign) results from enlarged collateral intercostal arteries eroding the inferior rib margins.

Step 4: Since all three features - reverse figure of 3 sign, the rib notching collateral sign, and the double aortic knuckle - are recognised radiological features of COA, the correct answer is All of the above.

Quick Tip: Think of the aortic contour, rib changes, and the reversed 3 outline together.

191. Best view for sphenoid sinus is -

- (A) Water's view
- (B) Caldwell view
- (C) Basal view
- (D) Towne's view

Correct Answer: (C) Basal view

Solution:

Step 1: Each paranasal sinus is best demonstrated by a specific radiographic projection that throws the sinus clear of overlapping bony structures.

Step 2: The sphenoid sinus lies deep at the skull base. The basal view, also called the submentovertical view, projects the X-ray beam from below the chin to the vertex and shows the sphenoid sinus best.

Step 3: The distractors map to other sinuses: Water's view (occipito-mental) is best for the maxillary sinus, Caldwell view (occipito-frontal) is best for the frontal and ethmoid sinuses, and Towne's view is used for the occiput and condyles, not the sphenoid.

Step 4: Therefore the basal (submentovertical) view is the correct answer for the sphenoid sinus.

Quick Tip: Submentovertical projection reaches the deep skull-base sinus.

192. Which of the following is/are radiological features of fluorosis?

- (A) Osteosclerosis
- (B) Cortical thickening
- (C) Enthesopathy
- (D) All the above

Correct Answer: (D) All the above

Solution:

Step 1: Skeletal fluorosis results from chronic excessive fluoride intake, which stimulates osteoblasts and causes abnormal new bone formation and mineralisation.

Step 2: The increased dense bone is seen radiologically as osteosclerosis, particularly involving the axial skeleton (spine and pelvis).

Step 3: Excess bone also deposits in the cortex, producing cortical thickening that encroaches on the medullary cavity. At tendon and ligament attachments fluoride drives ossification, giving enthesopathy and ligamentous calcification, along with large spinal osteophytes.

Step 4: Because osteosclerosis, cortical thickening, and enthesopathy are all genuine features of fluorosis, the answer is All the above.

Quick Tip: Dense bone, thick cortex, and ligament ossification all coexist.

193. Best imaging modality in patients with breast implants is:

- (A) MRI scan
- (B) CT scan
- (C) Mammography
- (D) Radionuclide scan

Correct Answer: (A) MRI scan

Solution:

Step 1: Breast implants are radiopaque and physically obscure overlying and underlying breast tissue, so they limit conventional X-ray based imaging.

Step 2: MRI gives excellent soft tissue contrast without ionising radiation and is not blocked by the implant, so it is the best modality for women with implants. It also reliably detects implant rupture.

Step 3: MRI is additionally valuable to distinguish post-treatment scar tissue from tumour recurrence in women who have had breast conservation therapy for cancer.

Step 4: Mammography is limited because the implant hides tissue, CT is not a routine breast modality, and radionuclide scanning lacks the anatomical detail required. Hence MRI scan is the answer.

Quick Tip: No radiation, superb soft-tissue contrast, not blocked by the prosthesis.

194. Obsessive attention by an individual towards another person is called?

- (A) Stalking
- (B) Perceiving
- (C) Following
- (D) Pressurizing

Correct Answer: (A) Stalking

Solution:

Step 1: The question describes unwanted, persistent, obsessive attention directed by one person (or group) toward another individual.

Step 2: This pattern of behaviour is termed stalking. It is characterised by harassment and intimidation and may include physically following the victim or otherwise monitoring them.

Step 3: The term stalking is used in psychology and psychiatry and is also defined as a criminal offence in many legal jurisdictions.

Step 4: Perceiving, following, and pressurizing each describe only an isolated act and do not capture the obsessive, repeated, attention-seeking pattern; stalking is the correct term.

Quick Tip: Persistent, unwanted, harassing fixation on another person.

195. A woman firmly and persistently feels her husband is cheating on her and she refuses to accept any proof given in the husband's support. The other family members do not support her belief. This is an example of -

- (A) Illusion
- (B) Delusion
- (C) Hallucination
- (D) Perversion

Correct Answer: (B) Delusion

Solution:

Step 1: A delusion is a fixed, false belief that is firmly held despite clear evidence to the contrary and is not shared by others of the same culture or background.

Step 2: Here the woman holds an unshakable false belief that her husband is unfaithful and rejects all proof and reassurance from relatives. This jealous, evidence-resistant conviction is a delusion (specifically the delusion of infidelity, or Othello syndrome).

Step 3: The distractors are different: an illusion is a misperception of a real external stimulus, a hallucination is a perception without any external stimulus, and perversion refers to deviant behaviour, not a false belief.

Step 4: Because the core abnormality is a false fixed belief held against evidence, the answer is delusion.

Quick Tip: Fixed false belief held despite contrary proof equals a disorder of thought content.

196. All the following are true regarding pseudocyesis, EXCEPT:

- (A) The patients usually have an intense desire to have children
- (B) Change in the breast may be present
- (C) There may be considerable increase in the size of the abdomen
- (D) The labour pain invariably continues to persist even if she is told that she is not pregnant

Correct Answer: (D) The labour pain invariably continues to persist even if she is told that she is not pregnant

Solution:

Step 1: Pseudocyesis (phantom or false pregnancy) is a condition in which a non-pregnant woman develops signs and symptoms of pregnancy. It is typically seen in women nearing menopause or in young women who intensely desire children, and most have an underlying psychic or hormonal disturbance. So option a is true.

Step 2: Real pregnancy-like changes can appear, including breast changes and apparent abdominal enlargement (the abdomen distends mainly due to deposition of fat). So options b and c are true.

Step 3: The symptoms are psychologically driven, so the perceived labour pains tend to stop abruptly once the woman is convincingly told she is not pregnant. The claim that labour pain invariably persists even after she is told she is not pregnant is therefore false.

Step 4: Since the question asks for the EXCEPT (the false statement), option d is the answer.

Quick Tip: The phantom labour pains usually stop once she is convinced she is not pregnant.

197. Which is the most common type of persistent delusional disorder?

- (A) Delusion of persecution
- (B) Somatic delusion
- (C) Delusion of jealousy
- (D) Delusion of grandeur

Correct Answer: (A) Delusion of persecution

Solution:

Step 1: Persistent delusional disorder is a psychotic condition in which one or more non-bizarre, fixed false beliefs persist for at least 1 to 3 months, with

relatively well-preserved functioning and no prominent hallucinations.

Step 2: Among the recognised subtypes, the persecutory (paranoid) type is the most common. Here the patient firmly believes that they are being conspired against, cheated, spied on, followed, poisoned, or otherwise harassed.

Step 3: The other options are valid but rarer subtypes. Somatic delusion involves a false belief about bodily function or a physical defect, the jealous type involves a fixed belief that the partner is unfaithful, and the grandiose type involves an inflated belief about one's own worth, power, or identity. None of these is as frequent as the persecutory type.

Step 4: Therefore the most common type of persistent delusional disorder is the delusion of persecution.

Quick Tip: Think of the paranoid theme - being targeted or conspired against.

198. Following are the major symptoms of obsessive compulsive disorder?

- (A) Contamination
- (B) Pathological doubts
- (C) Intrusive thoughts
- (D) All the above

Correct Answer: (D) All the above

Solution:

Step 1: Obsessive compulsive disorder (OCD) is characterised by recurrent obsessions (intrusive, unwanted thoughts) and compulsions (repetitive behaviours done to reduce anxiety). It has four classic major symptom patterns.

Step 2: Contamination is the most common pattern, an obsession about dirt or germs, usually followed by a washing compulsion (the washer).

Step 3: Pathological doubt is the second most common pattern, an obsession about whether a task was done correctly, followed by compulsive checking (the checker). Intrusive thoughts (pure obsessions) are the third pattern, recurrent obsessional thoughts, often of a sexual or aggressive nature, without a clear compulsion. Symmetry or precision is the fourth pattern.

Step 4: Since contamination, pathological doubts, and intrusive thoughts are all recognised major symptom patterns of OCD, the correct answer is All the above.

Quick Tip: Each option is a recognised major OCD symptom pattern.

199. Mania is characterized by:

- (A) Paranoid delusion
- (B) Loss of orientation
- (C) High self esteem
- (D) All

Correct Answer: (C) High self esteem

Solution:

Step 1: Mania is a mood episode marked by elevated, expansive, or irritable mood lasting at least one week, with increased energy and activity. A core feature is inflated self-esteem or grandiosity.

Step 2: The manic patient typically shows high self-esteem, reduced need for sleep, pressured speech, flight of ideas, distractibility, and excessive involvement in risky pleasurable activities.

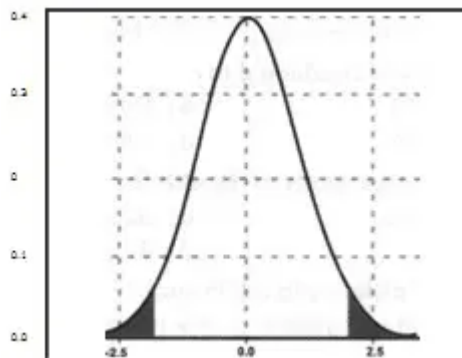
Step 3: Paranoid delusion can appear in severe mania with psychotic features, but it is not a defining or universal characteristic. Loss of orientation (disorientation in time, place, or person) is a feature of delirium or organic

states, not of uncomplicated mania, where sensorium remains clear.

Step 4: The single feature that best and consistently characterises mania is high self-esteem (grandiosity), so option C is correct, and "All" is wrong because loss of orientation does not belong.

Quick Tip: Grandiosity is core; disorientation points to delirium, not mania.

200.



What percentage of the normal distribution curve is represented by the shaded area?

- (A) 1%
- (B) 3%
- (C) 5%
- (D) 7%

Correct Answer: (C) 5%

Solution:

Step 1: In a normal (Gaussian) distribution, the data are symmetrically spread around the mean, and known percentages of the area fall within fixed multiples of the standard deviation (SD).

Step 2: The empirical rule states that about 68% of values lie within mean plus or minus 1 SD, about 95% lie within mean plus or minus 2 SD, and about 99.7%

lie within mean plus or minus 3 SD.

Step 3: Since 95% of the area lies within mean plus or minus 2 SD, the remaining area outside these limits (both tails together) is 100% minus 95%, which equals 5%. The shaded tails in the figure correspond to this region beyond 2 SD.

Step 4: Therefore the shaded area represents 5% of the normal distribution curve.

Quick Tip: Area outside mean plus or minus 2 SD equals 100% minus 95%.